



UNIVERSIDAD DE CHILE
FACULTAD DE CIENCIAS FÍSICAS Y MATEMÁTICAS
DEPARTAMENTO DE CIENCIAS DE LA COMPUTACIÓN

**COMPUTATIONAL APPROACHES TO LEXICAL SEMANTIC CHANGE
DETECTION WITH WIC-BASED AND LLM-BASED SENSE INDUCTION**

TESIS PARA OBTENER AL GRADO DE
DOCTOR EN COMPUTACIÓN

FRANK D. ZAMORA-REINA

PROFESORES GUÍAS:

Felipe Bravo-Marquez

Dominik Schlechtweg

Nikolay Arefyev

MIEMBROS DE LA COMISIÓN:

Richard Johansson

Marcelo Mendoza Rocha

Andrés Abeliuk

Este trabajo ha sido parcialmente financiado por
ANID y el Instituto Milenio Fundamento de los Datos.

SANTIAGO DE CHILE

2026

RESUMEN

La Detección de Cambio Semántico Léxico (LSCD, por sus siglas en inglés) tiene como objetivo identificar y cuantificar cambios en el significado de las palabras a lo largo del tiempo. A pesar de los importantes avances empíricos alcanzados recientemente, los enfoques actuales suelen depender de supuestos implícitos sobre la estructura de los sentidos léxicos, y la relación precisa entre LSCD y tareas orientadas al sentido, tales como la Inducción de Sentidos de Palabras (WSI), la Desambiguación de Sentidos de Palabras (WSD) y Word-in-Context (WiC), sigue siendo insuficientemente comprendida. Asimismo, aún no está claro si los modelos de lenguaje de gran escala pueden reemplazar clasificadores especializados dentro de los pipelines de LSCD, particularmente en distintos idiomas y condiciones de recursos.

Esta tesis aborda estas preguntas mediante tres estudios interrelacionados, situando la evaluación multilingüe, especialmente para el español, en el centro de la investigación. En primer lugar, presentamos el primer conjunto de datos y tarea compartida de LSCD para español, estableciendo la base empírica necesaria para investigar el cambio semántico más allá de escenarios de evaluación predominantemente centrados en el inglés. El conjunto de datos en español no solo constituye un recurso de referencia, sino también un medio para evaluar si los supuestos de modelado desarrollados principalmente para inglés generalizan adecuadamente a otros idiomas. El análisis de los resultados de la tarea compartida revela que los sistemas con mejor desempeño dependen consistentemente de representaciones relacionadas con sentidos léxicos, lo que motiva un examen más profundo del papel estructural del modelado de sentidos en LSCD.

En segundo lugar, a partir de esta observación, investigamos la relación entre la Inducción de Sentidos de Palabras y LSCD mediante la introducción de un conjunto de datos diacrónico anotado a nivel de sentidos siguiendo un esquema de anotación inspirado en WSD. A través de experimentos sistemáticos de validación cruzada, mostramos que las configuraciones de hiperparámetros optimizadas para WSI producen consistentemente un mejor desempeño en LSCD, mientras que las configuraciones optimizadas directamente para LSCD no se transfieren de manera confiable hacia WSI. Esta transferencia asimétrica resalta la importancia de la granularidad de optimización: dado que las puntuaciones de LSCD se derivan de distribuciones sobre sentidos, los modelos ajustados para recuperar distinciones finas de sentido se benefician de una supervisión más rica y de una estimación de parámetros más estable. Los resultados sugieren, por tanto, que optimizar modelos computacionales a nivel de sentidos constituye una estrategia más confiable para LSCD que ajustarlos únicamente respecto de métricas agregadas de cambio semántico. Complementamos además estos hallazgos con un análisis cualitativo de las estructuras semánticas inducidas, mostrando que un buen desempeño cuantitativo no garantiza necesariamente la obtención de clusters semánticamente coherentes o fácilmente interpretables desde una perspectiva lingüística.

Finalmente, investigamos si los modelos de lenguaje de gran escala pueden reemplazar clasificadores basados en WiC que estiman proximidad semántica a nivel de sentido entre usos contextualizados de palabras dentro de pipelines de LSCD. Mostramos que, utilizando prompts optimizados automáticamente, los LLMs pueden generar juicios confiables de proximidad semántica y sustituir efectivamente el componente WiC. Bajo este enfoque, los LLMs superan a sistemas basados en WiC de estado del arte para inglés, mientras que los modelos especializados basados en WiC continúan siendo superiores para español. Este contraste demuestra además que el modelado de proximidad semántica y el desempeño en LSCD siguen

siendo fuertemente dependientes del idioma. En conjunto, estos hallazgos evidencian tanto el potencial como las limitaciones actuales de los LLMs para LSCD multilingüe.

ABSTRACT

Lexical Semantic Change Detection (LSCD) aims to identify and quantify changes in word meaning over time. Despite strong empirical progress, current approaches often rely on implicit assumptions about word senses, and the precise relationship between LSCD and sense-oriented tasks such as Word Sense Induction (WSI), Word Sense Disambiguation (WSD), and Word-in-Context (WiC) remains insufficiently understood. In addition, it is unclear whether recent large language models can replace task-specific classifiers within LSCD pipelines, particularly across different languages and resource conditions.

This thesis addresses these questions through three interconnected studies, placing cross-linguistic evaluation, particularly for Spanish, at the center of the investigation. First, we introduce the first LSCD dataset and shared task for Spanish, establishing the empirical foundation necessary for systematic investigation of semantic change beyond predominantly English-centered evaluation settings. The Spanish dataset serves not only as a benchmark resource, but also as a means of testing whether modeling assumptions developed primarily for English generalize across languages. Analysis of the shared task results reveals that the best-performing systems consistently rely on sense-related representations, motivating a deeper examination of the structural role of sense modeling in LSCD.

Second, building on this observation, we investigate the relationship between Word Sense Induction and LSCD by introducing a diachronic sense-annotated dataset following a WSD-style annotation scheme. Through systematic cross-validation experiments, we show that hyperparameter configurations optimized for WSI consistently yield superior LSCD performance, whereas configurations optimized directly for LSCD do not reliably transfer back to WSI. This asymmetric transfer highlights the importance of optimization granularity: because LSCD scores are derived from distributions over senses, models tuned to recover fine-grained sense distinctions benefit from richer supervision and more stable parameter estimation. The results therefore suggest that optimizing computational models at the sense level constitutes a more reliable strategy for LSCD than tuning them solely against aggregated change metrics. We further complement these findings with a qualitative analysis of the induced semantic structures, revealing that strong quantitative performance does not necessarily guarantee linguistically coherent or easily interpretable sense clusters.

Finally, we investigate whether large language models can replace WiC-based classifiers that estimate sense-level semantic proximity between contextualized word usages in LSCD pipelines. We show that, when using automatically optimized prompts, LLMs can generate reliable semantic proximity judgments and effectively substitute the WiC component. Using this approach, LLMs outperform state-of-the-art WiC-based systems for English, while specialized WiC models remain superior for Spanish. This contrast further demonstrates that semantic proximity modeling and LSCD performance remain strongly language-dependent. Overall, these findings highlight both the potential and the current limitations of LLMs for multilingual LSCD.

A mi mamá, lo más grande que tengo.

Acknowledgments

First, I would like to express my deepest gratitude to the people who welcomed me upon my arrival in this country. Without their support, I would not have been able to begin this journey.

I am especially grateful to Felipe; without your guidance, this work would not have been possible. Thank you for your timely advice and for your trust in me. To Dominik, whose work in the LSCD field served as a primary inspiration for this research, I offer my sincere thanks; your constant support and patience were crucial to the development of this thesis. To Nikolay, your insightful comments throughout the process not only strengthened the quality of this work but also contributed significantly to my growth as a researcher.

To my family, thank you for your unconditional love. To my friends, who have been witnesses of this path, thank you for your companionship and encouragement.

I would also like to extend my thanks to Sandra, whose constant help with the administrative procedures of the PhD program was invaluable. Finally, I would like to thank the Department of Computer Science (DCC) for making this dream possible, and Chile—a remarkable country—for welcoming me and giving me the opportunity to pursue this path.

Thank you.

Table of content

1. Introduction	1
1.1. Research problem	2
1.2. Research Approach	4
1.2.1. Optimization Transfer between WSI and LSCD	4
1.2.2. General-Purpose LLMs vs. Small Specialized Models in LSCD	5
1.3. Main Results	5
1.4. Publications	6
1.5. Thesis outline	7
2. Background and Foundations of LSCD	9
2.1. Word senses	9
2.2. Word meaning annotation	12
2.3. Task definitions	13
2.3.1. WiC	13
2.3.2. WSI	14
2.3.3. LSCD definitions	14
2.3.3.1. Graded change	15
2.3.3.2. Evaluation Metrics	17
2.3.4. How tasks relate to each other	18
2.4. LSCD datasets	18
2.4.1. DUREL style	19
2.4.2. WUG	20
2.4.3. Sense-based style	22
2.5. Computational approaches	26
2.5.1. Type-based LSCD models	28
2.5.1.1. Semantic representation	29
2.5.2. Token-based LSCD models	29
2.5.2.1. Semantic representation	30
2.5.2.2. Clustering methods	30
2.5.2.3. Change measure	32
2.5.3. WiC-based LSCD models	33
3. Building a High-Quality Spanish LSCD Dataset and Evaluating State-of-the-Art Approaches	35
3.1. Introduction	35
3.2. Background	35
3.3. Task description	36

3.3.1.	Phase 1: Graded Change Discovery	37
3.3.2.	Phase 2: Binary Change Detection	37
3.3.3.	Optional tasks	37
3.4.	Data	38
3.4.1.	Corpora	38
3.4.2.	Target words	38
3.4.2.1.	Phase 1 (Graded Discovery)	38
3.4.2.2.	Phase 2 (Binary Detection)	39
3.4.3.	Word usages	39
3.4.4.	Annotation	39
3.5.	Analysis of Inter-Annotator Disagreement	41
3.6.	Systems	43
3.6.1.	Baselines	43
3.6.2.	Participating systems	44
3.6.3.	Summary	46
3.7.	Results	47
3.7.1.	Discussion	49
3.8.	Conclusion	51
4.	Optimizing WSI for LSCD	52
4.1.	Introduction	52
4.2.	Background	53
4.3.	Datasets	53
4.4.	Tasks and Evaluation	54
4.5.	Models	54
4.5.1.	Word-in-Context	55
4.5.1.1.	The architecture of DeepMistake	56
4.5.1.2.	WiC training and model variants	56
4.5.2.	Building the Word Usage Graph	58
4.5.3.	Graph Clustering	59
4.5.3.1.	Chinese Whispers	59
4.5.3.2.	Correlation Clustering	60
4.5.3.3.	Weighted Stochastic Block Model	60
4.5.3.4.	Spectral Clustering	60
4.5.4.	Methodological Scope and Alternative Approaches	61
4.6.	Experiments	62
4.6.1.	Analysis of the German Dataset	63
4.6.2.	Analysis of the Spanish Dataset	64
4.6.3.	Analysis of the English Dataset	65
4.6.4.	Comparison with APD	66
4.6.5.	Summary of Findings	68
4.7.	Conclusions	68
5.	Enhancing LSCD predictions through LLMs	70
5.1.	Introduction	70
5.2.	Background	71
5.3.	Data	73

5.4.	Experimental setup	74
5.4.1.	WUG	75
5.4.2.	Clustering methods	75
5.5.	Prompts and their optimizations	75
5.6.	Experiments for graded change detection	77
5.6.1.	APD	78
5.6.2.	Cross validation with clustering	79
5.7.	Experiments at WiC annotation level	80
5.8.	Sensitive and Qualitative analysis	81
5.8.1.	Experimental Configuration for Qualitative Analysis	82
5.8.2.	Spectral Clustering Analysis	82
5.8.2.1.	Qualitative Analysis of Induced Clusters	83
5.8.3.	WSBM Analysis	85
5.8.3.1.	Qualitative Analysis of Induced Clusters	86
5.9.	Conclusion	89
6.	Conclusion	90
Appendices		93
A.	Lemmatization	93
B.	Target indices of annotated usages	93
C.	Type-based models	94
C.1.	Semantic representation	94
C.2.	Alignment methods	95
C.3.	Change measures	96
D.	Experiments with models based on substitution	97
E.	Cross-Validation Results and Hyperparameter Configurations	97
E.1.	Results per fold	97
F.	Annotation Guidelines	99
Bibliography		108

List of Tables

2.1.	Example of a Word Sense Disambiguation task instance using a predefined sense inventory. The system must select the appropriate sense of the target word from the inventory based on its context.	9
2.2.	Brown’s scale of sense distinction.	10
2.3.	DURel relatedness scale.	12
2.4.	Example usages of the target word <i>banco</i> across two time periods.	16
2.5.	Overview of datasets. LGS = languages, Id = identifier of the dataset, n = number of target words, C1 = old period of time, C2 = modern period of time.	22
2.6.	Statistics for the latest version (2.3.0) of DWUG DE and the new DWUG DE Sense dataset. Both datasets contain German word usages from two historical corpora covering 1800–1899 and 1946–1990 respectively. n = no. of target words, N/V/A = no. of nouns/verbs/adjectives, $ U $ = no. usages per word ($t_1 + t_2$), AN = no. of annotators, $ J $ = avg. no. judgments per annotation instance, KRI = Krippendorff’s α , STYLE = annotation style.	23
2.7.	Agreement (Krippendorff’s α) between annotators on sense definition annotation. Left: Pairwise agreement between annotators on full data. Right: Overall agreement between annotators per time period.	24
2.8.	Statistics of cleaned datasets. $ U $ = average number of usages per time period, $ S $ = average number of senses (in all cases the median number of senses is 2).	24
3.1.	Sizes of both corpora.	38
3.2.	Overview target words. n = no. of target words, N/V/A = no. of nouns/verbs/adjectives+adverbs, $ U $ = avg. no. of usages per word, AN = no. of annotators, JUD = total no. of judged usage pairs, AV = avg. no. of judgments per usage pair, KRI = Krippendorff’s α , SPR = weighted mean of pairwise Spearman, UNC = avg. no. of uncomparing multi-cluster combinations, LOSS = avg. of normalized clustering loss * 10, $LSC_{B/G}$ = mean binary/Graded Change score.	41
3.3.	Annotation disagreement patterns for the word <i>planta</i>	42
3.4.	Summary of system performance in phase 1. Teams are ranked according to SPR score for the Graded Change subtask in decreasing order. The values corresponding to the three best systems are highlighted in bold type.	47
3.5.	Summary of the results of Phase 2 for subtasks Graded Change, COMPARE and Binary Change. Teams are ranked according to F1 score for subtask Change binary in decreasing order. The values corresponding to the three best systems are highlighted in bold type.	48
3.6.	Summary of the results of Phase 2 for subtasks Sense loss and Sense gain. The values corresponding to the three best systems are highlighted in bold type.	49

4.1.	Training data of the employed DeepMistake models. Each example is a pair of usages of the same word in the same language, except for en-nen* which are cross-lingual examples with one usage in English and another in French, Russian, Arabic or Chinese.	57
4.2.	Hyperparameters to build a WUG.	58
4.3.	Clustering method hyperparameters.	59
4.4.	Results on the German dataset. The table reports the rank (in parentheses) obtained in the cross-validation experiments together with the corresponding performance metrics (ARI for WSI and Spearman's ρ for LSCD) for each clustering method.	63
4.5.	Results on the Spanish dataset. The table reports the rank (in parentheses) obtained in the cross-validation experiments together with the corresponding performance metrics (ARI for WSI and Spearman's ρ for LSCD) for each clustering method.	64
4.6.	Results on the English dataset. The table reports the rank (in parentheses) obtained in the cross-validation experiments together with the corresponding performance metrics (ARI for WSI and Spearman's ρ for LSCD) for each clustering method.	65
4.7.	Comparison between the best clustering-based LSCD configuration and the APD baseline across datasets. Results are reported in terms of Spearman's rank correlation coefficient (ρ).	67
5.1.	Overview of datasets. LGS = languages, Id = identifier of the dataset, n = number of target words, C1 = old period of time, C2 = modern period of time.	73
5.2.	Class distribution across DWUG EN resampled and development version of DWUG ES datasets.	73
5.3.	Statistics for each partition of the DWUG EN (resampled) and the development version of the DWUG ES datasets used to optimize the prompts for annotating semantic proximity examples.	74
5.4.	Accuracy of non-optimized prompts (NOP) and optimized prompts (OP) for DWUG EN and DWUG ES datasets, using prompts in English (PrE) and Spanish (PrS). Results are reported for Mixtral, Llama 3.1, and Llama 3.3 as the underlying LLMs.	76
5.5.	Performance of various methods across cross-validation experiments on the DWUG ES and DWUG EN datasets. The table compares different clustering approaches: AC (Agglomerative Clustering), SC (Spectral Clustering), WSBM (Weighted Stochastic Block Model). APD is reported to evaluate the performance of the models in identifying the different senses of a word, as opposed to quantifying the semantic change of a word without discerning the senses of the target words. The random baseline assigns a score between 1 and 4 randomly.	79
5.6.	Best-performing Spectral clustering configurations used for qualitative analysis on DWUG-ES dataset using semantic proximity annotations generated by Llama3.3:70B and MCL→es. The table reports the hyperparameter configurations achieving the highest Spearman correlation with the gold graded change scores.	82

5.7.	Best-performing WSBM configurations for the DWUG-ES dataset using semantic proximity annotations generated by Llama3.3:70B and MCL→es. The table reports the hyperparameter configurations achieving the highest Spearman correlation with the gold graded change scores.	85
E.1.	Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the WSI task for the DWUG DE dataset. Each row corresponding to a method signifies a distinct fold.	101
E.2.	Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the LSCD task for the DWUG DE dataset. Each row corresponding to a method signifies a distinct fold.	102
E.3.	Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the WSI task for the DWUG ES dataset. Each row corresponding to a method signifies a distinct fold.	103
E.4.	Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the LSCD task for the DWUG ES dataset. Each row corresponding to a method signifies a distinct fold.	104
E.5.	Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the WSI task for the DWUG EN dataset. Each row corresponding to a method signifies a distinct fold.	105
E.6.	Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the LSCD task for the DWUG EN dataset. Each row corresponding to a method signifies a distinct fold.	106
F.1.	Annotationstabelle.	107

List of Figures

2.1.	Graph-based representation of semantic change for the target word <i>banco</i> . The top row shows the full graph and its clustered version. The bottom row presents the induced subgraphs for each time period and the cross-period (COMPARE) graph.	17
2.2.	Visual comparison of tasks involved in LSCD. u_1 , u_2 , and u_3 represent word use samples. Arrows represent use-sense assignments for WSD, cluster assignments for WSI, and pairwise judgments for WiC.	18
2.3.	Sense frequency distribution for <i>Ohrwurm</i> from cleaned dataset (maj_3).	26
2.4.	Sense frequency distribution for <i>Manschette</i> from cleaned dataset (maj_3).	27
2.5.	Number of senses per word for cleaned dataset (maj_3).	28
2.6.	Jensen-Shannon distance between sense probability distributions for cleaned dataset (maj_3).	29
2.7.	Comparison of the DeepMistake (left) and XL-LEXEME (right) architectures, illustrating their respective mechanisms for generating graded change predictions based on contextualized embeddings.	33
3.1.	Word Usage Graph <i>servidor</i> (left), subgraphs for old corpus G_1 (middle) and for modern corpus G_2 (right). The colors correspond to the clusters. black /gray lines indicate high /low edge weights.	40
3.2.	F1 scores over binarization thresholds based on percentiles on submitted Graded Change predictions for top four teams in evaluation phase 1.	50
4.1.	Overview of the architecture used to generate predictions for the WSI and LSCD tasks.	55
5.1.	Spearman correlation between human and model annotations for word usage pairs (SPR_WiC), and between gold graded change scores and model predictions using the APD-based approach (SPR_LSCD). “ZS” denotes Zero-Shot, “FS” stands for Few-Shot, and “CT” refers to Chain-of-Thought prompting.	77
5.2.	APD-based approach with various WiC models and LLMs, Spearman correlation with the gold graded change word scores is reported. “COMPARE” calculates APD across usage pairs consisting of usages from different time periods, while “ALL” employs usage pairs regardless of their time periods.	78
5.3.	Spearman correlation between the annotations provided by human annotators, LLMs, DeepMistake models, and XL-LEXEME.	81
5.4.	SFD for the word <i>planta</i> obtained from the annotations generated by Llama3.3:70B using Spectral Clustering. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.	84
5.5.	SFD for the word <i>planta</i> obtained from the annotations generated by MCL→es using Spectral Clustering. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.	84

5.6.	SFDs for the word <i>historia</i> obtained from the annotations generated by MCL→es and Llama3.3:70B using Spectral Clustering. Both models produced the same cluster distribution, where the bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.	85
5.7.	SFD for the word <i>planta</i> obtained from the annotations generated by Llama3.3:70B using WSBM. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.	86
5.8.	SFD for the word <i>planta</i> obtained from the annotations generated by MCL→es using WSBM. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.	87
5.9.	SFD for the word <i>historia</i> obtained from the annotations generated by Llama3.3:70B using WSBM. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.	87
5.10.	SFD for the word <i>historia</i> obtained from the annotations generated by MCL→es using WSBM. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.	88

Chapter 1

Introduction

Human language is constantly evolving, reflecting complex processes within both language and society [1]. For instance, the word *cell* originally referred to a small monastery cell and later to a prison room for those who violated the law. In 1977 [2], it gained a new meaning related to telecommunications, as people began using small devices to communicate. The study of changes in meaning at the word level is often called Lexical Semantic Change (LSC) [3].

Advancements in the digitization of historical texts and computational modeling have enabled large-scale analysis of LSC, giving rise to Lexical Semantic Change Detection (LSCD). LSCD is an area of research within computational linguistics that focuses on identifying and analyzing word meaning evolution over time, including meaning shifts, sense gain, and sense loss. This has been facilitated by the development of computational models that learn the semantic representation of words from their contextual usage, leveraging techniques such as distributional semantics, word embeddings, and contextualized language models. Additionally, the availability of human-annotated benchmark datasets, such as the one introduced in the SemEval-2020 Task 1 on LSCD [4], has played a crucial role in evaluating and advancing these models.

LSCD has diverse applications across several domains. In linguistics, it provides insights into the mechanisms of language evolution, helping researchers understand how social and cultural dynamics, cognitive processes, and lexical semantic structure interact in shaping patterns of word usage over time [3]. In lexicography, LSCD techniques assist in tracking the emergence of new word senses and refining historical dictionaries by identifying terms that have undergone significant semantic shifts. Beyond lexicography, LSCD has implications for natural language processing (NLP) tasks, such as Word Sense Disambiguation (WSD) [5, 6], information retrieval, and historical document analysis, where understanding diachronic meaning variations can improve model performance. Moreover, LSCD has increasingly been applied in the social sciences and political science to investigate cultural and ideological shifts, the evolution of political discourse [7], and semantic signals associated with societal phenomena such as armed conflict [8]. As language continues to evolve, LSCD remains essential not only for linguistic research but also for interdisciplinary work spanning digital humanities, cultural analytics, and computational social science.

This thesis investigates how integrating word sense information into LSCD can enhance the detection of semantic change. It builds on the observation that LSCD can be viewed as a modular task composed of interacting subtasks, notably a sense induction component and a sense comparison component. In this perspective, Word Sense Induction (WSI)—which aims to

automatically identify the different senses of a word from its contextual usage—approximates the discovery of sense inventories at different time periods, while Word-in-Context (WiC), a task that evaluates whether a target word conveys the same meaning across two contexts [9], models the comparison of meaning across specific usages. By explicitly modeling these modules and their interaction—particularly by strengthening the sense induction stage—this thesis injects structured sense information into the LSCD process, enabling more fine-grained and interpretable analyses of lexical meaning variation over time and leading to improved LSCD performance through a more accurate representation of underlying sense distributions. Furthermore, it examines the effectiveness of large language models (LLMs), particularly those based on GPT-3 [10], in capturing these changes. By leveraging both sense-based analysis and the contextual representations offered by LLMs, this thesis proposes and evaluates hybrid approaches that aim to improve computational modeling of lexical semantic change.

1.1. Research problem

Despite significant advancements in computational linguistics, the study of word meanings and their evolution over time remains a complex challenge. In particular, the role of word senses in LSCD has not been thoroughly investigated, leaving important questions unanswered. Before 2020, most LSCD research relied on static word embedding models [11, 12], which, although useful for identifying broad semantic shifts, suffered from a critical limitation known as meaning conflation deficiency [13] —the inability to represent multiple senses of a word distinctly within a single vector.

The introduction of a shared task on LSCD in 2020 [4] marked a significant shift in the field, prompting the development of models aimed at addressing limitations in earlier approaches. Interestingly, type-based models—which assign a single embedding to each word—outperformed token-based models that compute representations for each word occurrence individually. While this result was counterintuitive, it does not necessarily imply that fine-grained sense information is irrelevant for modeling lexical semantic change. Rather, the overall performance of all participating systems remained relatively modest, and subsequent analyses have suggested that the weaker performance of token-based models may be attributable to factors such as representation noise and instability rather than to their sensitivity to contextual variation [14]. A closer examination of the results indicates that token-based approaches are not as ineffective as initially suggested. Rather than demonstrating that contextualized encoders are inherently unsuitable for LSCD, the findings point to limitations in early implementations and to the need for more robust and structured modeling strategies that better control noise and more explicitly incorporate sense-level information.

Despite these conceptual advances, progress in LSCD has also been constrained by limitations at the data level. Most existing LSCD benchmarks focus on a small set of languages and provide limited support for systematically analyzing sense-related modeling decisions across tasks and linguistic settings. In particular, Spanish has remained underrepresented in LSCD research, restricting empirical investigation of whether observations reported for English and German generalize to other languages. The introduction of new graded LSCD benchmarks—such as the Spanish dataset considered in this thesis—not only addresses this imbalance but, crucially, introduces a modular annotation design in which sense-level information is explicitly represented and can be independently modeled and evaluated. This modular structure makes it possible to decompose LSCD into interacting components—such as sense induction, sense comparison, and change aggregation—thereby enabling controlled

cross-linguistic comparisons of modeling paradigms when sense information is made explicit or implicitly exploited. Beyond filling a resource gap, the availability of high-quality Spanish sense-annotated data has already demonstrated measurable downstream impact on related semantic tasks. In subsequent multilingual ordinal WiC experiments, models trained using DWUG-ES achieved the strongest reported performance for Spanish among the evaluated languages, illustrating how resource development for historically underrepresented languages can directly influence widely used contextual embedding models [15]. Crucially, such datasets make it possible to observe performance patterns—e.g., the strong effectiveness of WiC-based and clustering-based approaches—that raise fundamental questions about the role of word senses in LSCD and their relationship to closely related tasks such as WiC, WSI, and WSD.

As more advanced contextualized representations models emerged—such as BERT [16] and XLM-RoBERTa [17]—and empirical evidence accumulated showing that modern pre-trained models markedly outperform traditional distributional and count-based approaches on the LSCD task [18], researchers increasingly began to explore how to transfer knowledge from related semantic tasks into LSCD. A notable development in this direction was the introduction of models whose parameters were trained or fine-tuned on the WiC task [19, 20]. Given the conceptual proximity between WiC and LSCD (see Section 2.3.4), these models learn to determine whether two sentences containing the same target word express the same meaning. The key distinctions are that LSCD explicitly incorporates temporal information—each sentence is associated with a specific time period—and that its scoring scale captures finer semantic nuances than the binary WiC formulation. In parallel, models were also developed for the WSI task, aiming to induce the full range of senses associated with a given target word.

As researchers increasingly integrated WiC- and WSI-based models into LSCD systems, it became evident that these tasks are closely related to the way semantic change is operationalized in modern LSCD datasets. In particular, the annotation procedures underlying graded LSCD benchmarks rely on identifying and comparing word senses across contexts, which naturally connects LSCD to tasks such as WSI and WiC. In this sense, WiC and WSI form conceptual components of the LSCD annotation pipeline, as detecting semantic change ultimately depends on inferring and comparing senses across time periods. WSD, by contrast, relies on predefined sense inventories and therefore relates to LSCD in a different and less direct way. Despite these conceptual connections at the annotation level, it remains unclear to what extent models developed for WiC and WSI can be effectively leveraged for LSCD in practice, leaving open the question of how modeling advances in these tasks translate into improvements in semantic change detection.

At the same time, LSCD systems increasingly began to incorporate WiC-based classifiers as a crucial component for transferring sense-related information into the LSCD pipeline. These systems depend on WiC judgments—effectively treating them as an intermediate semantic signal—to approximate whether two word usages reflect the same sense. This reliance on WiC models introduced an important methodological question: Could the role of WiC-based systems be replaced by a more flexible and powerful mechanism, one capable of producing semantic proximity judgments without requiring task-specific fine-tuning? The emergence of LLMs has brought this question into focus. LLMs have demonstrated remarkable zero-shot [21] and few-shot [22] abilities across a wide range of semantic tasks, raising the possibility that they could generate the pairwise semantic proximity judgments traditionally supplied by WiC models (often represented as probabilities). If so, LLMs might serve as direct drop-in replacements for WiC classifiers within LSCD pipelines, potentially

simplifying the modeling process while providing richer and more context-aware assessments of meaning proximity.

In summary, this thesis investigates whether explicit modeling of word senses provides a principled and effective foundation for LSCD. To this end, it examines how sense-oriented computational approaches—including WiC-based semantic proximity modeling, WSI, and LLMs—can be used to represent, recover, and compare meanings across time. Through multilingual experimentation and resource creation, the thesis evaluates the extent to which sense-aware modeling improves semantic change detection and whether LLMs can serve as viable alternatives to specialized WiC-based components within LSCD pipelines.

1.2. Research Approach

To address the research challenges outlined in Section 1.1, this thesis proposes two complementary approaches. The first investigates the relationship between WSI and LSCD, leveraging the conceptual connections among sense-related tasks. The second examines the performance of large, general-purpose language models relative to smaller, specialized LSCD systems, assessing whether LLMs constitute a viable alternative for modeling semantic change. A central aspect of this second approach is evaluating whether LLM-based predictions—obtained through natural language prompting—can effectively replace the WiC component traditionally used in LSCD pipelines and serve as a direct source of semantic proximity judgments. Together, these approaches address the broader question of how LSCD can be modeled either through explicit sense representations or through the implicit semantic knowledge encoded in modern language models. The WSI-based approach evaluates whether optimization for WSI also leads to improved performance in LSCD, whereas the LLM-based approach tests whether large models can achieve similar or superior performance without relying on explicit sense modeling. Taken together, they offer complementary perspectives on the role of sense information in LSCD and enable a unified evaluation of both traditional sense-centric methods and contemporary prompt-based neural approaches.

1.2.1. Optimization Transfer between WSI and LSCD

As discussed in Section 1.1, we investigate whether optimizing the clustering parameters used for WSI—based on pairwise semantic proximity scores produced by a WiC model—can improve performance in the LSCD task, and conversely whether parameter choices that optimize LSCD evaluation also lead to improved WSI results. We propose the following approach: First, we manually annotate the senses of the extracted usages for a set of target words. Next, we employ a WiC-based model to generate semantic proximity scores for all usage pairs of each target word. For each word, we then construct a graph whose nodes represent individual usages and whose weighted edges encode the model’s estimated degree of semantic proximity between usage pairs. Applying clustering algorithms to this graph allows us to address the WSI task by inducing sense clusters.

To determine the most effective configuration, we conduct a systematic hyperparameter search using k -fold cross-validation [23], selecting the parameters that best reconstruct the reference sense inventory. Our central hypothesis is that hyperparameters optimized for WSI will also support strong performance on the LSCD task, enabling results comparable to state-of-the-art systems for the target language. Conversely, we also evaluate whether hyperparameters optimized directly for LSCD can generalize back to the WSI task, thereby

testing whether performance optimization transfers across the two tasks.

To guide this investigation, we formulate the following research question:

- Is solving the underlying WSI task sufficient for strong LSCD performance, and can LSCD models simultaneously generalize their representations to address WSI?

1.2.2. General-Purpose LLMs vs. Small Specialized Models in LSCD

LLMs introduce a fundamentally different modeling paradigm for LSCD: instead of being trained on specialized objectives, they generate answers directly through natural language prompting. However, LLM outputs are often highly sensitive to prompt formulation [24], making manual prompt design difficult and inconsistent. In this work, we investigate whether automatic prompt optimization can yield more stable and effective performance. This leads to our first research question:

- Can automatically optimized prompts yield better results for the LSCD task than manually crafted prompts?

We further examine whether LLMs are capable of solving the graded LSCD task by designing prompts inspired by the DUREL annotation guidelines [25]. This motivates our second research question:

- Can LLMs achieve strong performance on the graded LSCD task, potentially surpassing state-of-the-art WiC-based models?

Finally, because it remains unclear whether LLMs can emulate human annotation behavior, we assess how closely their usage-pair proximity judgments align with human ratings, and whether such alignment translates into improved LSCD performance. Accordingly, we pose a third research question:

- Can LLMs outperform state-of-the-art LSCD models at the WiC annotation level?

Through these experiments, this thesis seeks to determine whether general-purpose LLMs offer a viable and competitive alternative to specialized models for LSCD, and to examine the trade-offs between model scale, interpretability, and efficiency in modeling diachronic semantic change.

Beyond the quantitative evaluation of LSCD systems, this thesis also examines the linguistic interpretability of the sense structures induced by clustering methods. While evaluation metrics such as Spearman correlation [26] quantify how well predicted semantic change scores align with gold standards, they do not necessarily reveal whether the underlying sense inventories correspond to meaningful linguistic distinctions. To address this issue, we conduct a qualitative analysis of the induced clusters and investigate the following research question:

- What is the relationship between quantitative LSCD performance and the linguistic interpretability of the induced sense structures?

1.3. Main Results

This thesis yields three main results that advance the study of LSCD, spanning data creation, modeling methodology, and the evaluation of LLMs.

First, we introduce the first graded LSCD dataset and shared task for Spanish, addressing a major gap in existing LSCD benchmarks. The dataset provides human-annotated semantic proximity judgments across time periods and enables systematic evaluation of LSCD models in a language that has been largely underrepresented in prior work. Analysis of the shared task submissions showed that several of the top-performing systems relied on clustering-based or WiC-based approaches, suggesting that successful LSCD models often benefit from capturing sense-level distinctions. These empirical patterns raised fundamental questions about the role of word senses in LSCD and directly motivated the subsequent investigations of sense-oriented modeling strategies pursued in this thesis.

Second, building on these observations, we show that solving WSI—a more general and foundational sense-oriented task—can function as an effective auxiliary objective for LSCD. Through systematic cross-validation and hyperparameter optimization within the WSI framework, we observe that parameters selected for WSI consistently yield strong LSCD performance. In contrast, hyperparameters optimized directly for LSCD do not generalize as effectively to the WSI task. These findings suggest that tuning clustering parameters with respect to WSI provides a more reliable strategy for model selection than optimizing directly for LSCD, likely due to the limited size of available LSCD evaluation data.

Third, we conduct a detailed empirical comparison between LLMs and smaller, task-specific models developed explicitly for LSCD. Using both the newly introduced Spanish dataset and existing English benchmarks enriched with human- and model-generated sense annotations, we show that LLMs—when prompted using automatically optimized formulations—can match or surpass specialized LSCD systems. In particular, large general-purpose models such as Mixtral:8x7B, Llama3.1:8B, and Llama3.3:70B outperform state-of-the-art WiC-based approaches for English, while specialized WiC models remain superior for Spanish. These findings demonstrate both the potential and current limitations of LLMs for multilingual LSCD and indicate that their effectiveness is conditioned by language-specific and data-related factors.

Finally, qualitative analyses of the induced sense inventories show that strong performance on LSCD evaluation metrics does not necessarily correspond to linguistically interpretable sense structures. While some clustering configurations recover meaningful semantic distinctions, others produce highly fragmented clusters influenced by discourse or contextual variation. These findings suggest that quantitative LSCD evaluation and linguistic interpretability capture complementary aspects of model quality and should be considered jointly when assessing sense-based approaches.

Together, these findings underscore the value of integrating sense-based information into LSCD workflows, while also highlighting the strong potential of prompt-optimized LLMs as effective and scalable alternatives to traditional task-specific modeling strategies.

1.4. Publications

Conference papers:

- **Frank D. Zamora-Reina**, Felipe Bravo-Marquez, Dominik Schlechtweg and Nikolay Arefyev. 2025. Can Large Language Models Compete with Specialized Models in Lexical Semantic Change Detection? In *Frontiers in Artificial Intelligence and Applications*, Volume 413: ECAI 2025 (pp. 4201–4208). IOS Press. <https://doi.org/10.3233/FAIA251313>.

Journal papers:

- Dominik Schlechtweg, **Frank D. Zamora-Reina**, Felipe Bravo-Marquez, and Nikolay Arefyev. 2024. Sense through time: Diachronic word sense annotations for word sense induction and lexical semantic change detection. *Language Resources and Evaluation*.

Workshops:

- **Frank D. Zamora-Reina** and Felipe Bravo-Marquez. 2020. DCC-Uchile at SemEval-2020 Task 1: Temporal Referencing Word Embeddings. In *Proceedings of the 14th International Workshop on Semantic Evaluation*, Barcelona, Spain. Association for Computational Linguistics.
- **Frank D. Zamora-Reina**, Felipe Bravo-Marquez, and Dominik Schlechtweg. 2022. LSCDiscovery: A shared task on semantic change discovery and detection in Spanish. In *Proceedings of the 3rd International Workshop on Computational Approaches to Historical Language Change*, Dublin, Ireland. Association for Computational Linguistics.

1.5. Thesis outline

This thesis is organized as follows:

Chapter 2 offers an overview of the related work relevant to the research presented in this thesis. In this chapter, we introduce the fundamental concepts of the LSCD task. In addition, we review previous studies that focused on annotating LSCD examples. Furthermore, we analyze the primary models developed within the LSCD field and discuss the key metrics used to evaluate their performance. Finally, we examine research related to the use of general-purpose LLMs in the LSCD domain and provide a critical assessment of these works.

Chapter 3 describes the creation of a Spanish dataset for LSCD with the goal of extending LSCD research beyond the languages that have been predominantly studied so far. This dataset is then released in the context of a shared task, allowing an analysis of how different research groups approach the LSCD problem when applied to Spanish. The results obtained from this shared task provide insights into current modeling strategies and contribute additional evidence on the strengths and limitations of state-of-the-art LSCD approaches, which informs the experimental design of the subsequent chapters.

Chapter 4 builds on the findings of Chapter 3, where several of the strongest-performing systems relied on clustering-based representations of word senses. Motivated by this observation, we investigate whether improving the WSI component of LSCD pipelines can lead to better semantic change detection. We develop and evaluate a framework in which WSI is incorporated into an LSCD pipeline through clustering over pairwise semantic proximity scores. Through cross-validation and hyperparameter tuning, we demonstrate that optimization for WSI can also improve LSCD performance, providing empirical support for the importance of sense-aware modeling in semantic change detection.

Chapter 5 investigates whether the WiC component commonly used in modern LSCD pipelines can be replaced by LLMs. Building on the sense-oriented perspective developed throughout the thesis, we analyze models of varying sizes, including Mixtral-8x7B, Llama3.1:8B, and Llama3.3:70B, and benchmark their performance against specialized WiC-based and

LSCD models. This chapter examines the impact of prompt optimization and domain-specific training on the quality of the semantic proximity judgments produced by these models and evaluates whether LLMs constitute competitive alternatives to task-specific approaches for LSCD.

Chapter 6 presents the conclusions of this thesis, highlighting the main findings and outlining potential directions for future work.

Chapter 2

Background and Foundations of LSCD

In this chapter, we introduce the fundamental concepts relevant to the LSCD field and the scope of this thesis. We review prior research related to the annotation of semantic change data and examine the primary models developed within the LSCD domain, alongside the key evaluation metrics employed to assess their performance. Furthermore, we explore the foundational theories and historical context underlying the broader field of LSCD.

2.1. Word senses

WSD is the computational task of assigning a sense label $s \in S(w)$ to an occurrence of a target word w given its context c , where $S(w)$ denotes the predefined inventory of possible senses associated with w . A major limitation of WSD is its dependence on external knowledge sources such as dictionaries and lexical databases, whose quality and coverage vary according to the underlying corpora from which they are derived. Moreover, manual construction of these resources is time-consuming and labor-intensive, complicating the disambiguation process [6]. Table 2.1 illustrates a typical WSD task instance in which the system must choose the correct sense of the word *bank* based on its context.

Table 2.1: Example of a Word Sense Disambiguation task instance using a predefined sense inventory. The system must select the appropriate sense of the target word from the inventory based on its context.

Context	She deposited the money in the bank .
Candidate senses	1. bank#1: financial institution that accepts deposits 2. bank#2: sloping land beside a body of water
Correct sense	bank#1

Addressing these limitations, Kilgarrieff [27, 28] critically challenges the traditional notion of discrete word senses. He argues that word senses are not fixed entities but rather flexible abstractions derived from patterns of language use observed in corpora. According to this view, senses emerge through the grouping of word occurrences in real contexts (concordances), where similarity of use and meaning serves as the organizing principle. Importantly, this grouping is not purely objective but involves interpretative decisions influenced

by the practical goals and subjective criteria of the lexicographer. As a result, defining clear boundaries between word senses becomes inherently problematic, since there are no definitive criteria for determining where one sense ends and another begins.

Table 2.2: Brown’s scale of sense distinction.

	4: Same sense
↑	3: Closely Related
	2: Distantly Related
	1: Homonymy

The main conclusion of the article is that closely related word senses may be represented in the mental lexicon in a manner similar to a single sense, potentially sharing a central semantic core. This implies that grouping such closely related senses in WSD tasks could lead to minimal information loss. In contrast, responses to distantly related senses resembled those to homonyms, suggesting that distinguishing between them would be advantageous in WSD systems. The study also observed that distinctions between closely related senses often involved concrete meanings, whereas distinctions among more distantly related senses encompassed both concrete and figurative usages. Finally, the finding that response time and accuracy vary gradually with the degree of semantic relatedness supports the notion that the mental representation of word senses is not strictly discrete.

In their work, McCarthy and Navigli [29] advocate for a less discrete, more gradual view of word meanings. They highlight the lack of consensus within the WSD community regarding what constitutes the “correct” sense distinctions for evaluation. Concerns about the granularity of these distinctions—where overly fine-grained senses challenge both human annotators and computational models—suggest that sense boundaries are often not sharply defined. To address this, the authors introduce the lexical substitution task, which allows annotators to propose up to three equally valid substitutes for a given word in context. This approach implies that no single “correct” answer exists; rather, there is a range of plausible alternatives that vary in their semantic similarity to the original word. The observed low inter-annotator agreement further supports the notion that substitutability is not a binary issue, but instead reflects a spectrum—reinforcing the idea of a gradual approach to sense interpretation.

A pivotal advancement in the LSCD field came with the work of Erk et al. [30], who introduced innovative methods for representing and evaluating word meanings in context. Departing from the traditional WSD paradigm, which assigns a single “best” sense per occurrence, they proposed two novel annotation schemes—WSsim and Usim—that capture meaning from a gradual, rather than categorical, perspective.

- **WSsim (Word Sense Similarity):** Annotators evaluate the applicability of each dictionary sense (specifically from WordNet) to a word occurrence on a five-point ordinal scale. The goal is to capture the idea that senses may not apply in a binary manner, allowing for more nuanced judgments.
- **Usim (Usage Similarity):** Annotators directly assess the meaning similarity between pairs of usages of the same word (lemma), also using a five-point ordinal scale. This method aims to evaluate usage similarity without relying on a predefined inventory of senses.

The study involved two rounds of annotation (R1 and R2). In the second round (R2), traditional Word Sense Disambiguation tasks (WSbest) and lexical substitution tasks (SYNbest) were included within the same dataset, allowing for more direct comparisons across different annotation types. The authors analyzed inter-annotator agreement for both gradual annotation schemes and found significant correlations. Notably, annotators made full use of the ordinal scale and often provided nuanced judgments that “mixed and matched” senses across individual usages.

The article further investigates the relationship between the gradual annotation schemes (WSSim and Usim) and more traditional approaches, such as single-sense annotation and lexical substitution. A strong correlation was observed, validating the new annotation methods and suggesting that these diverse approaches capture the same underlying phenomenon of word meaning in context.

Brown [31] notes that while WordNet [32] is a widely used resource for WSD due to its extensive coverage and accessibility, concerns have been raised about whether its fine sense distinctions are truly necessary or even reflective of how humans represent word meanings. The article mentions that WSD systems tend to be more successful at distinguishing general senses rather than fine ones. It also highlights that human annotators struggle to reliably differentiate fine senses, which sets an upper limit on the performance of WSD systems.

The central experiment of the article directly investigated sense distinctions with varying degrees of semantic relatedness: homonymy, distantly related senses, closely related senses, and identical senses (see Table 2.2). The results from the semantic decision experiment showed that subjects were faster and more accurate when transitioning between usages of a word with the same meaning. However, a crucial finding was that subjects were also significantly faster and more accurate with pairs of closely related senses compared to distantly related and unrelated senses (homonyms). Additionally, a linear progression in response time and accuracy was observed as the relationship between senses decreased.

McCarthy et al. [33] challenge the traditional assumption in WSD and WSI—namely, that the occurrences of a lemma can always be partitioned into discrete senses. They argue instead for a spectrum of *partitionability*, where some lemmas are inherently more amenable to sense division than others. This notion is closely aligned with the gradual view of word senses proposed by Erk et al.

The study by McCarthy et al. centers on quantifying how easily the usages of a word can be grouped into meaningful clusters. Its novelty lies in being the first to explicitly measure partitionability and to bridge the machine learning literature on *clusterability* with the word usage data employed in computational linguistics.

To operationalize the notion of partitionability, the authors introduce the concept of *clusterability*, defined as a measure of how easily the occurrences of a lemma can be grouped into distinct clusters. In the context of a gradual perspective on word senses, this concept offers a way to quantify that graduality. Lemmas with high clusterability tend to align with the traditional view of discrete, well-defined senses, whereas those with low clusterability indicate greater semantic vagueness or a continuum of meaning, where sense boundaries are diffuse. The authors suggest that for such lemmas, more nuanced representations—such as soft clustering or non-strict partitioning methods—may be more appropriate.

Blank, in his work, distinguishes two main types of lexical semantic change [1, 25]:

- **innovative meaning change:** Emergence of a full-fledged additional meaning of a word; old and new meaning are related by polysemy.

- **reductive meaning change:** Loss of a full-fledged meaning of a word.

Building on prior work [1, 30, 31], Schlechtweg et al. [25] proposed a language-agnostic framework for constructing test sets to evaluate lexical semantic change based on the annotation schema developed by Erk et al. [30]. This framework is capable of detecting two main types of change: innovative and reductive meaning shifts. It was also used to annotate a German lexical semantic change test set, achieving high inter-annotator agreement, thereby validating the reliability of the annotations. The test set construction relied on a rating scale adapted from Brown’s work (see Table 2.2).

Table 2.3: DUREl relatedness scale.

	4: Identical
↑	3: Closely Related
	2: Distantly Related
	1: Unrelated

Furthermore, Schlechtweg et al. [4, 34] created a large-scale multilingual resource that constructs a Word Usage Graph (WUG) for each lemma. These WUGs include annotations of word usages from two distinct time periods, enabling fine-grained analysis of semantic evolution. The annotation process relies on graded pairwise semantic proximity judgments (Table 2.3), which capture degrees of semantic relatedness between usages without presupposing discrete sense boundaries. The resulting graphs can then be analyzed using clustering algorithms, which partition usages into groups that approximate word senses. While the underlying annotations are graded, this discretization step provides a practical—though imperfect—way to relate the induced clusters to definitions of semantic change that assume distinct senses, such as those proposed by Blank.

Throughout the thesis, we will mean by a **word usage** an occurrence of a word within an instance of text such as a sentence or a paragraph. By a **sense definition** we will mean a textual description of some meaning a target word can express. For shortness, we will sometimes refer to these concepts by ‘usage’ and ‘sense’ if the context sufficiently disambiguates what we mean.

2.2. Word meaning annotation

Word meaning annotated data falls into three main categories: (i) use-sense, (ii) use-use and (iii) lexical substitution annotation [cf. 30, 3, p. 20]. The first type, use-sense annotation has a long tradition within the task of WSD [5]. Annotators usually choose the best-fitting word sense definition for a word use as in this example:

use: [...] and taking a knife from her pocket, she opened a vein in her little **arm**, and dipping a feather in the blood, wrote something on a piece of white cloth, which was spread before her.

sense1: a human limb

sense2: weapon system

There has been extensive work on use-sense annotation and several large-scale annotation projects have been carried out, as e.g. SemCor and OntoNotes [35, 36]. Such data can also be used for the WSI task as it provides a mapping of usages to clusters, i.e., all usages annotated with the same sense definition receive the same cluster label [cf. 6, 37].

The (discrete) use-sense assignment approach was criticized to be empirically inadequate [27, 38–40]. This was (amongst others) supported by the observation that inter-annotator agreement for certain words was consistently low [36, 41]. Hence, the alternative annotation strategies for use-use pairs and lexical substitutions were developed. In the former approach annotators typically judge usages of a word for their semantic proximity [25, 30] while in the latter approach annotators find synonymous substitution words [42]. However, these alternative strategies do not solve the problem of low agreement [cf. 30, 42] and they do not directly provide data usable for WSD and WSI [33]. While word sense clusters for WSI can be obtained through clustering algorithms McCarthy et al. (2013) [33], Schlechtweg et al. (2021) [34, 43], such approaches are currently not thoroughly validated.

There is a growing body of work with a diachronic focus on word meaning [e.g. 44–46], tackling tasks such as LSCD [3]. This development has brought a number of diachronic word meaning datasets [e.g. 25, 34, 47–50]. In contrast to their synchronic counterparts, these datasets are mainly annotated within the use-use paradigm.¹ On several of these datasets word sense clusters have been inferred from the human annotation [e.g. 34, 50–52] which could be used for WSI tasks in a diachronic setting. However, the inferred clusters often suffer from ambiguity and sparsity of annotation [3, p. 57ff.]. While we deem this a promising approach, it is still ongoing research.

Despite the above-mentioned criticisms, the traditional use-sense annotation approach has several advantages over the use-use setting: (i) The number of annotations needed is much lower [3, p. 45]. (ii) The sense clusters follow directly, so no additional sense/cluster inference procedure introducing additional noise is needed. (iii) The annotation is richer as it provides a sense definition and thus allows to define more tasks on the data, as e.g. WSD. (iv) The agreement is higher than with other approaches [cf. 30, 42]. (v) The data can be cleaned easily as agreement is calculated on instances rather than pairs of instances, hence individual instances can easily be excluded. However, to our knowledge only a small fraction of diachronic datasets is use-sense annotated [53–56]. Of these only y McGillivray [55]’s Latin is publicly available. The sense example sentences from historical dictionaries [e.g. 57–59] can be seen as use-sense annotated data [cf. 60]. However, these sentences do not provide a realistic task scenario as they are few (often only one per sense) and not randomly sampled from a corpus.

2.3. Task definitions

2.3.1. WiC

The WiC task is to determine whether two occurrences of the same ambiguous target word in different sentences or text fragments express the same or different senses [9]. In our work, the annotated usage pairs obtained from this task constitute the first step of our framework, as they enable the direct quantification of semantic change and serve as the basis for subsequent

¹ This is probably an influence from the early DUREL annotation approach [25] relying on the insight that one does not have to annotate word senses in order to annotate (specific types of) word sense change.

modeling (see Section 4.5).

2.3.2. WSI

The WSI task is an unsupervised task that considers grouping usages corresponding to different senses of a given target word into clusters or groups. These clusters are interpreted as individual senses considering the context in which the given target word occurs [61, 62], and do not have to correspond to a predefined set of senses for evaluation. It is closely related to WSD, but with a crucial difference: WSD requires a predefined sense inventory for each word, making it a supervised task.

2.3.3. LSCD definitions

A general definition of Lexical Semantic Change Detection was proposed by Kutuzov et al. [45]. Given a sequence of corpora $[C_1, C_2, \dots, C_n]$ corresponding to different time periods $[t_1, t_2, \dots, t_n]$, the task is to identify words whose meanings differ across these time periods.

With the organization of the SemEval-2020 Task 1 shared task, a more concise and widely adopted definition of LSCD was introduced. It is typically framed as two sub-tasks:

Binary classification: Given a set of target words, determine which words have gained or lost senses between C_1 and C_2 , and which have remained semantically stable.

Ranking task: Rank a set of target words according to the degree of lexical semantic change between C_1 and C_2 by comparing their sense frequency distributions across the two corpora.

In addition, an auxiliary task known as COMPARE [25] is often used as a proxy for graded change LSCD [3]. This task is defined as follows:

- **COMPARE:** Given two usages of a target word, this task focuses on estimating semantic change by considering pairs of usages in which one occurrence comes from the earlier corpus and the other from the later one.

By restricting predictions to cross-period usage pairs, COMPARE provides an indirect measure of semantic shift. Schlechtweg [3, p. 66] show that this formulation serves as a meaningful approximation of the graded change LSCD task.

Most research in LSCD focuses primarily on the graded change task, where the degree of semantic change is defined as the divergence between the sense frequency distributions of a word across time periods [4]. Following this prevailing approach, this thesis concentrates exclusively on graded change LSCD. Under this formulation, the semantic change of a target word is quantified across corpora, even though the specific senses involved are not explicitly identified when aggregating over all its usages.

As outlined earlier, data for this task is annotated on a 4-point scale, where a score of 4 indicates that two usages of a target word share the same meaning, while a score of 1 indicates that their meaning are completely unrelated. Because LSCD datasets are typically annotated by multiple annotators, a given pair of usages may receive several judgments. In such cases, the final proximity score assigned to the pair is computed as the median of all annotations, ensuring a robust aggregate assessment of semantic proximity across annotators.

The annotated data is represented in a mathematical structure known as a graph, specifically referred to as a **Word Usage Graph** (WUG) in the context of semantic change [34].

A WUG $\mathbf{G} = (\mathbf{U}, \mathbf{E}, \mathbf{W})$ is a weighted, undirected graph where nodes $u \in U$ represent word usages, weights $w \in W$ represent the semantic proximity of a pair of usages $(u_1, u_2) \in E$ [33], and $W : E \mapsto \mathbb{R}$ where $Z \subset \mathbb{R}$ maps each edge to its weight [3, p. 33].

WUGs represent the semantic proximity of a set of word usages U . These usages U can be sampled from a single corpus or from multiple corpora that represent different time periods $t_1, t_2, \dots, t_n$. Typically, the usages U are sampled from two distinct time periods t_1, t_2 corresponding to the subsets $U_1 \in U$ and $U_2 \in U$. Additionally, for the two time periods t_1, t_2 , it can be defined two subgraphs G_1, G_2 containing only the usages from U_1 and U_2 , respectively, and the edges between them [3].

The WUGs obtained from the annotation process are weighted, undirected, and often noisy [4]. As explained by Schlechtweg et al. [4] and Schlechtweg [3, p. 34], this presents a specific challenge that requires a robust clustering algorithm. Schlechtweg et al. To address this, they employ a variation of correlation clustering to minimize the sum of cluster disagreements, which includes the sum of negative edge weights within a cluster and the sum of positive edge weights across clusters. The authors assume that usage pairs with judgments of 3 and 4 are more likely to belong to the same sense, whereas judgments of 1 and 2 are more likely to indicate different senses. They adjust the weight $W(e)$ of all edges $e \in E$ in a usage graph $\mathbf{G} = (U, E, W)$ by subtracting 2.5 from $W(e)$. The edges $e \in E$ with a weight $W(e) \geq 0$ are classified as positive edges P_E , while edges with weights $W(e) < 0$ are considered negative edges N_E . Let $C : U \mapsto L$ represent a clustering of U , $\phi_{E,C}$ denote the set of positive edges across any of the clusters in clustering C , and $\psi_{E,C}$ represent the set of negative edges within any of the clusters. Then, a clustering C is sought that minimizes the sum of weighted cluster disagreements:

$$L(C) = \sum_{e \in \phi_{E,C}} W'(e) + \sum_{e \in \psi_{E,C}} |W'(e)|$$

By applying the clustering method to the WUGs, a clustering $C : U \mapsto L$ is established, mapping each usage $u \in U$ to a cluster label $l \in L \subset \mathbb{N}$. Next, a cluster frequency distribution D is calculated, which encodes the size of each cluster as follows:

$$D = (f(L_1), f(L_2), \dots, f(L_i))$$

where $L_i < L_{i+1}$ and $f(L_i)$ represents the number of times any usage from U was mapped to the cluster label L_i . Subsequently, two distributions D_1 and D_2 are derived from C for the two time-specific usage sets U_1 and U_2 . The distributions D , D_1 , and D_2 are ordered and contain the frequencies for the complete set of cluster labels L , ensuring that the i -th index consistently corresponds to the same cluster label. From these sense frequency distributions (SFD), cluster sense probability distributions P_1 and P_2 are obtained by dividing D_1 and D_2 by their respective total frequencies.

2.3.3.1. Graded change

After the sense frequency distributions D and E are normalized to probability distributions P and Q . The graded change score of a word w is defined as Jensen-Shannon Distance (JSD) between the two probability distribution (normalized distributions) [3, p. 38]:

$$G(w) = G(P, Q) = JSD(P, Q)$$

Table 2.4: Example usages of the target word *banco* across two time periods.

ID	Year	Usage
A	1810	Fui al banco a retirar dinero.
B	1855	El banco me concedió un préstamo.
E	1906	El banco central emitió un comunicado oficial.
C	1994	Nos sentamos en el banco del parque a descansar.
D	2008	El banco estaba mojado después de la lluvia.
F	2020	Se quedó dormido en un banco de madera junto al río.

where the Jensen-Shannon distance is the symmetrized square root of the Kullback-Leibler Divergence [63, 64]:

$$JSD(P, Q) = \sqrt{\frac{KLD(P||M) + KLD(Q||M)}{2}}$$

where

$$KLD(P||Q) = \sum_i^k p_i \log_2\left(\frac{p_i}{q_i}\right),$$

$$M = \frac{P + Q}{2}$$

Schlechtweg explains that the Jensen-Shannon Distance is preferred over the Kullback-Leibler Divergence because it constitutes a true metric, satisfying properties such as symmetry, i.e., $JSD(P, Q) = JSD(Q, P)$ [3, p. 38]. This measure underlies the graded formulation of semantic change.

This graded formulation enables direct comparison between words in terms of the extent of their semantic evolution. In particular, it allows us to assess nuanced scenarios, such as whether the loss of a highly frequent sense results in greater change than the loss of a rare one, or whether gaining multiple senses leads to less change than losing several senses. Moreover, graded change mitigates several limitations associated with binary change. It is more robust to noise introduced by infrequent senses and occasional annotation or clustering errors, as such discrepancies do not necessarily lead to maximal error. Additionally, graded change is sensitive to shifts in sense frequency distributions, ensuring that even subtle shifts in usage probabilities are captured, rather than only complete sense gain or loss [3, p. 39].

Example

To illustrate the notion of graded semantic change, we consider the target word *banco*, which exhibits multiple senses in Spanish.

Consider the example shown in Figure 2.1. The WUG G represents the semantic proximity structure over the set of usages U of the Spanish word *banco* listed in Table 2.4. The usages are divided into two time periods, t_1 (1810–1906) and t_2 (1994–2020), with $U_1 = \{A, B, E\}$ and $U_2 = \{C, D, F\}$.

Applying a clustering procedure to G yields a partition $C : U \rightarrow L$ with two clusters of high internal similarity, which can be interpreted as distinct senses. One cluster groups

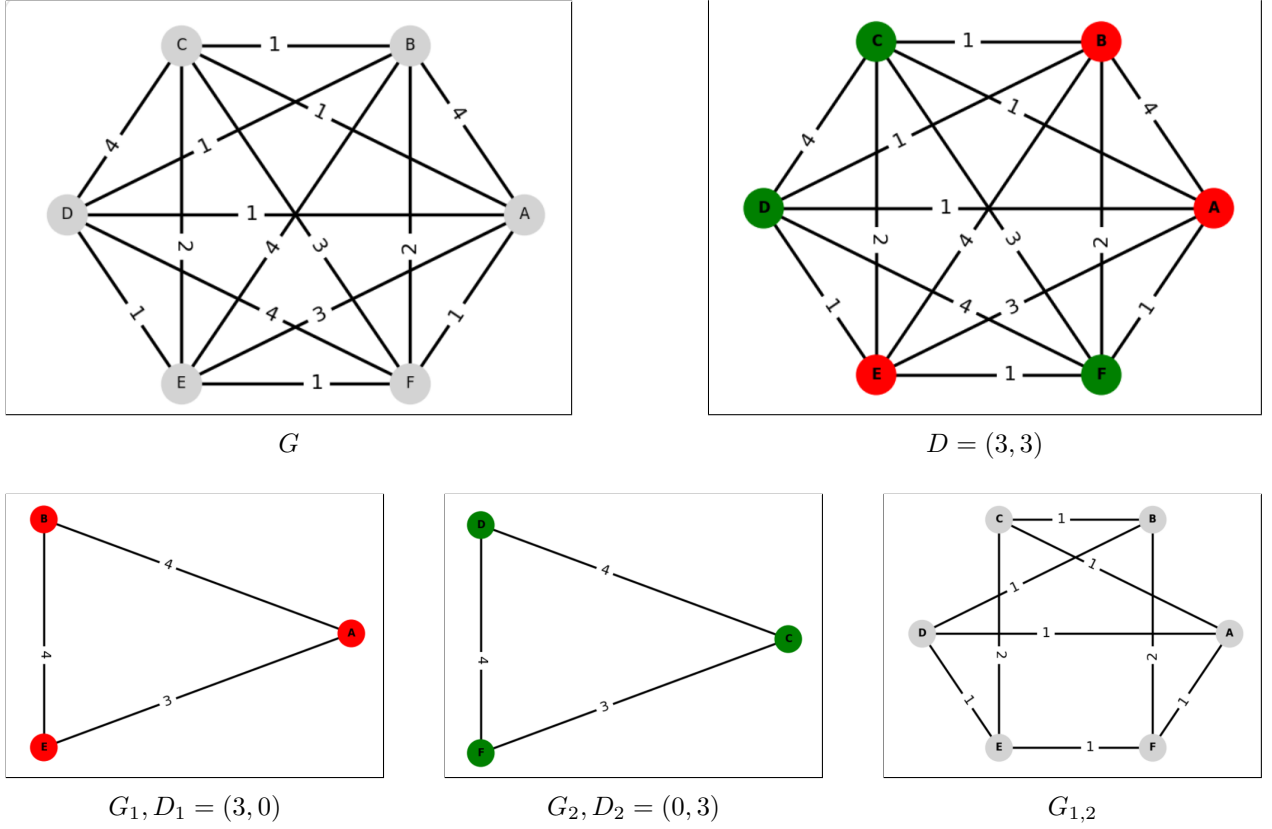


Figure 2.1: Graph-based representation of semantic change for the target word *banco*. The top row shows the full graph and its clustered version. The bottom row presents the induced subgraphs for each time period and the cross-period (COMPARE) graph.

usages related to the financial institution sense, while the other captures the physical location sense. These clusters define the semantic structure of G and allow us to derive time-specific subgraphs G_1 and G_2 , reflecting how usages are organized within each period.

Based on this partition, we compute sense frequency distributions over L for each period. Let P and Q denote the normalized distributions derived from G_1 and G_2 , respectively.

In this example, the clustering assigns all usages in t_1 (A, B, E) to the financial institution sense l_1 , and all usages in t_2 (C, D, F) to the physical object sense l_2 . This yields the following normalized distributions:

$$P = [1.0, 0.0], \quad Q = [0.0, 1.0].$$

In this case, the dominant sense in t_1 is entirely different from the dominant sense in t_2 , indicating a complete shift in sense prevalence over time. This change is quantified using the Jensen–Shannon Divergence between P and Q , which captures differences in the distribution of senses while remaining sensitive to gradual variations in their relative frequencies.

2.3.3.2. Evaluation Metrics

Evaluating LSCD systems requires task-specific metrics that capture different facets of how word meanings evolve over time; these metrics quantify how closely a model’s predictions match gold-standard scores derived from human annotations so systems can be compared

and ranked. LSCD is typically framed in two ways: as a binary classification task or as a graded ranking task, each with its own evaluation methodology.

In the graded ranking setting, the objective is to assign each target word the graded-change score defined earlier— $G(w) = \text{JSD}(P, Q)$, the JSD between the normalized sense-frequency distributions—so that system-generated continuous scores can be compared against gold-standard annotations (see Section 2.3.3.1). Evaluation uses Spearman’s rank correlation coefficient [26] to assess the monotonic relationship between system-generated $G(w)$ scores and gold-standard annotations. This measures how well a system recovers the relative ranking of words by semantic-change magnitude rather than exact numeric values. Spearman’s ρ ranges from -1 to 1, where $\rho = 1$ indicates perfect agreement between the model’s $G(w)$ ranking and the gold-standard ranking (identical order), $\rho = 0$ indicates no monotonic relationship (model rankings are uncorrelated with gold), and $\rho = -1$ indicates perfect inverse agreement (model ranks exactly opposite to gold).

2.3.4. How tasks relate to each other

Note that there is a close conceptual connection between WSD, WiC, WSI and the various definitions of LSCD [65] as the process to derive gold labels for LSCD requires at least one of the other tasks to be solved. For instance, Schlechtweg et al. [4] employ a WiC and a WSI step in their annotation process while Basile et al. [54] employ a single WSD step and Schlechtweg et al. [25] employ a single WiC step. All of these approaches have in common that they require some form of sense distinctions: WiC asks to distinguish senses of use pairs, WSI asks to distinguish senses within sets of usages, and WSD asks to assign senses to individual usages. Hence, because of this sense distinction information, they can all be applied to measure sense changes over time. We summarize the relations between the tasks in Figure 2.2.

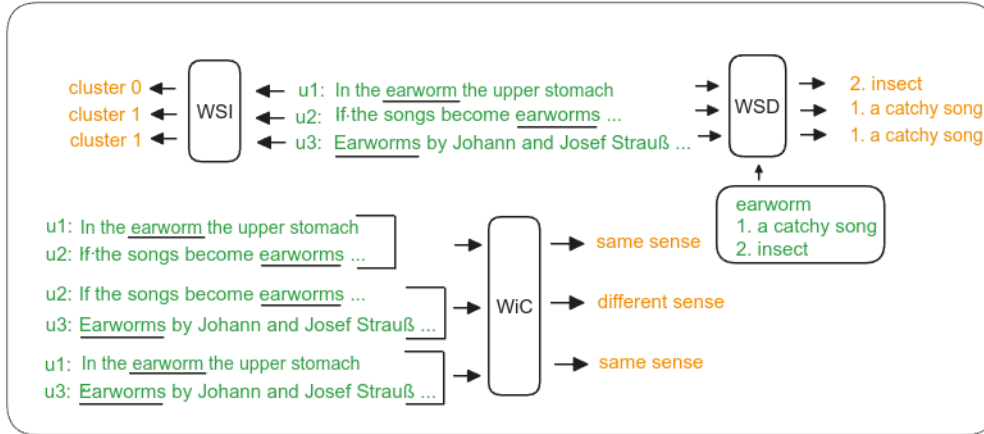


Figure 2.2: Visual comparison of tasks involved in LSCD. $u1$, $u2$, and $u3$ represent word use samples. Arrows represent use-sense assignments for WSD, cluster assignments for WSI, and pairwise judgments for WiC.

2.4. LSCD datasets

The field of LSCD has historically faced a significant challenge due to the absence of annotated test sets, which are essential for training models and evaluating their effectiveness on

diachronic annotated datasets. However, this trend began to shift in 2020 with the introduction of the first multilingual dataset [4], marking a pivotal moment for research in this area.

These datasets are annotated following several styles:

1. DUREl
2. WUG
3. Sense-based annotation style

The annotation styles used for LSCD datasets differ in their underlying methodology. The WUG approach constructs a connected graph of usage pairs annotated for semantic proximity and then applies clustering techniques to derive the gold data [4, 34]. The DUREl style, in contrast, organizes annotations into three groups—EARLIER, LATER, and COMPARE—computes the average judgment score for each group, and publishes the COMPARE scores as the gold standard [25]. Finally, the senses style identifies the senses of each target word based on its set of usages. Gold data is then derived by computing the frequency distribution of these senses across time periods and applying a suitable metric to the normalized distributions; at this stage the resulting per-period sense-frequency representations are formally equivalent to those obtained from the WUG approach, and the same downstream distance/divergence metrics and evaluation procedures can be applied, although methodological differences in sense induction or labeling can affect the precise form of those distributions.

2.4.1. DUREl style

RuSemShift [49] is a dataset designed for modeling historical lexical semantic change in Russian. This dataset provides a structured framework for researchers to analyze semantic shifts across different time periods in the Russian language.

RuSemShift was created through an extensive manual annotation process using the crowdsourcing platform Yandex.Toloka.² The annotation followed the DUREl framework (see Section 2.3.3), which quantifies semantic change based on the average relatedness between pairs. Native Russian speakers, all university-educated and over 30 years old, participated as annotators. They assessed the relatedness of meanings for a target word in two sentences—both containing the word—on a four-point scale. For each target word, three groups of sentence pairs were formed: EARLIER (pairs from the earliest time period), LATER (pairs from the most recent period), and COMPARE (pairs where one sentence is from the earlier period and the other from the later one) [25]. Each group consisted of 20 sentence pairs.

The primary source of textual data for RuSemShift was the Russian National Corpus (RNC)³, a balanced corpus that includes Russian texts from various genres produced from the mid-18th century to the early 21st century. RuSemShift encompasses three distinct periods in the history of the Russian language, following previous studies:

- **Pre-Soviet:** Texts produced between 1682 and 1916 (before the 1917 Revolution).
- **Soviet:** Texts produced between 1918 and 1990 (during the Soviet Union).
- **Post-Soviet:** Texts produced between 1991 and 2017 (after the fall of the Soviet Union).

² <https://toloka.yandex.ru/>

³ <http://ruscorpora.ru>

Two datasets were constructed to analyze semantic change across historical periods: RuSemShift1, covering the transition from pre-Soviet to Soviet times, and RuSemShift2, spanning the shift from the Soviet to post-Soviet era.

For RuSemShift1, the initial list of 43 target words—presumed to have undergone semantic change—was selected based on prior research. To ensure a balanced dataset, “filler” words, assumed to be semantically stable, were included, bringing the total to 71 words for annotation. For RuSemShift2, a set of 42 target words (35 nouns and 7 adjectives) was compiled, primarily drawing from the dictionary New Words and Meanings [66]. After incorporating filler words, the final annotation set comprised 69 words. In total, 140 words were annotated across both datasets, providing a robust foundation for analyzing lexical semantic change over time.

The test set was constructed from the annotated words. For each word in the pre-constructed lists, all unique sentence contexts were extracted from the corresponding two corpora (pre-Soviet/Soviet or Soviet/post-Soviet). Subsequently, 60 sentence pairs per word were randomly sampled (20 for each group: EARLIER, LATER, and COMPARE). These pairs were presented to the annotators for evaluating the semantic relationship. After annotation, two measures of the degree of semantic change were calculated for each word: Δ LATER (the difference between the average relationship of the LATER and EARLIER groups) and COMPARE (the average relationship within the COMPARE group). The final dataset, which is publicly available, contains these annotations and can be utilized to evaluate semantic change detection systems.

This dataset was presented as part of the RuShiftEval shared task [18], which focused on semantic change detection in Russian. The competition results marked a significant shift in the field: For the first time in shared tasks on semantic change detection, architectures based on contextualized embeddings (token-based models) clearly outperformed their static embedding (type-based) counterparts. This contrasts with earlier competitions, such as SemEval 2020 Task 1 and DIACR-Ita, where static embeddings consistently demonstrated superior performance. The results of RuShiftEval indicate that the research community has made significant progress in effectively leveraging contextualized embeddings for semantic change detection, marking a new direction for future methods in the field.

2.4.2. WUG

The first annotated test sets for LSCD were introduced in 2020 as part of SemEval-2020 Task 1. These datasets covered four languages: English, German, Swedish, and Latin, and were constructed using the following historical corpora [34]:

- **English:** Clean Corpus of Historical American English (CCOHA) [67], (periods: 1810 to 1860 and 1960 to 2010).
- **German:** DTA corpus [68], along with a combination of the BZ and ND corpora (periods: 1800-1899 and 1946-1990).
- **Swedish:** Kubhist corpus (periods: 1790-1830 and 1895-1903) [69].
- **Latin:** LatinISE corpus (periods: -200 to 0 and 0 to 2000) [70].

The authors built the datasets by selecting 100–200 target words suspected of undergoing semantic change, drawing on historical and etymological dictionaries. A second annotator

refined this list by examining 50 usage samples from each corpus (C_1 and C_2) for each word to identify observable meaning differences. To create a balanced dataset, the researchers also selected stable control words—one for each changing word—that matched in part of the speech and exhibited similar frequency trends between the two periods. They then manually verified the diachronic stability of these control words.

Schlechtweg et al. [4] report that for each target word, they randomly selected 100 instances from both C_1 and C_2 for annotation, resulting in a total of 200 instances per target word. If a target word had fewer than 100 instances, all available examples were annotated. The usage samples for each target word were then merged into a single set, U , and annotated using an extended version of the DUREl framework [25, 30].

Annotators rated pairs of usages from different time periods on a 4-point semantic relatedness scale (see Table 2.3). Unlike the original DUREl framework, which relied on random sampling of usage pairs to calculate average relatedness over time, this extended version introduced a strategic sampling method. The researchers aimed to construct a sparsely connected graph of usages—(WUG)—in which nodes represent individual usages and edges reflect the median relatedness scores assigned by annotators to each usage pair.

In this graph, nodes represent individual usages, and edges correspond to the median judgments provided by annotators. As noted by [34], the main challenge of this annotation method was to minimize the number of annotated edges while retaining sufficient information for meaningful clustering. Since fully annotating the usage graph is infeasible, this strategy provides an efficient yet informative representation of the underlying semantic relationships.

As described in Section 2.3.3, usages were clustered in the WUG to produce per-period sense-frequency distributions; here it was therefore applied that established pipeline (clustering the unified WUG, inducing period-specific subgraphs, and deriving normalized SFDs P1 and P2) specifically to the current dataset to obtain its SFDs used in subsequent analyses.

The frequency of a sense within a corpus is calculated by counting the number of usages assigned to the cluster representing that sense. To detect LSC, the resulting sense frequency distributions from C_1 and C_2 are then compared. This comparison supports both binary and graded formulations of change: by identifying gained or lost senses (binary change) or by computing the JSD between the normalized distributions (graded change).

These initial datasets not only established a shared evaluation framework for LSCD but also catalyzed the development of diverse modeling approaches. Early state-of-the-art systems included both type-based models, which relied on distributional representations aggregated over time periods, and token-based models, which used contextualized embeddings and sense clustering techniques. Notably, in the SemEval 2020 Task 1 results, type-based methods outperformed token-based approaches, particularly in the binary classification task. These benchmarks set the foundation for subsequent LSCD research, including the creation of multilingual datasets and more refined annotation methodologies.

Table 2.5 provides an overview of the original WUG-based datasets released for SemEval-2020 Task 1, which later served as the foundation for the construction of additional resources in other languages.

Following this line of research, additional datasets were constructed using the same WUG-based annotation framework for multiple languages. DWUG ES [50] extends the methodology to Spanish, preserving pairwise semantic relatedness annotations and WUG-derived gold-standard data, while incorporating adaptations to support additional analyses (see Chapter 3). Similarly, datasets for Norwegian [51] and Chinese [71] adopt the same annotation framework but differ in their sampling and pairing procedures. In particular, both datasets

Table 2.5: Overview of datasets. LGS = languages, Id = identifier of the dataset, n = number of target words, C1 = old period of time, C2 = modern period of time.

LGS	Id	n	C1	C2
English	DWUG EN	37	1810-1860	1960-2010
German	DWUG DE	48	1800-1899	1946-1990
Latin	DWUG LA	40	-200-0	0-2000
Swedish	DWUG SV	31	1790-1830	1895-1903

select 40 usages per target word (20 from each time period) and employ alternative strategies for constructing usage pairs.

2.4.3. Sense-based style

The DWUG DE Sense dataset extends the existing DWUG DE dataset [34] with sense annotations. The original DWUG DE dataset was created for lexical semantic change detection and contains German word usages sampled from two historical periods (1800–1899 and 1946–1990) together with semantic relatedness judgments between usage pairs.

For each target word, 50 usages were randomly sampled from the original DWUG DE dataset (25 per time period, from a maximum of 100 usages available in the original resource) and annotated with the sense definition that best described the meaning of the target word in context.⁴ Annotators were instructed to assign the label *andere* (‘other’) when none of the available sense definitions adequately described the usage. They were also allowed to skip instances that were considered ambiguous or unclear.

The annotation was performed by three annotators with linguistic expertise: a professional computational linguist, a linguistics graduate, and a university student majoring in German studies. The annotators had no prior access to the dataset before the annotation process.

The annotation procedure was carried out in two rounds. In the first round, only the computational linguist annotated the data. During this phase, additional sense definitions were introduced for four target words (*abdecken*, *Fuß*, *Manschette*, and *Schmiere*) after the original inventory was judged insufficient to account for some usages. These new definitions were subsequently integrated into the sense inventory and presented together with the original definitions to the remaining annotators during the second annotation round.⁵

The following example illustrates the annotation format used in DWUG DE Sense for the target word *Ohrwurm*.

use: *Im **Ohrwurm** ist der obere Magenmund inwendig mit einigen Zähnen in zwey Reihen besetzt.*, ‘In the **earwig** the upper stomach mouth is occupied inside with some teeth in two rows.’

sense1: *Insekt, (von) dem der Volksglaube annimmt, daß es gerne Schläfern ins Ohr kriecht*, ‘insect, which is popularly believed to like to crawl into the ear of sleepers’

sense2: *eingängige Melodie*, ‘catchy melody’

⁴ Some target words contain repeated usages, e.g. *Ohrwurm*. Similarly, some usages contain multiple occurrences of the target word, each of which was treated as a separate instance. Consequently, the same sentence may occur multiple times with different target-word positions, as in the case of *Abgesang*. These instances were retained in order to preserve the dataset’s original characteristics.

⁵ The annotated dataset and all derived labels are available at <https://doi.org/10.5281/zenodo.8197552>.

Table 2.6: Statistics for the latest version (2.3.0) of DWUG DE and the new DWUG DE Sense dataset. Both datasets contain German word usages from two historical corpora covering 1800–1899 and 1946–1990 respectively. n = no. of target words, N/V/A = no. of nouns/verbs/adjectives, $|U|$ = no. usages per word ($t_1 + t_2$), AN = no. of annotators, $|J|$ = avg. no. judgments per annotation instance, KRI = Krippendorff’s α , STYLE = annotation style.

Data	n	N/V/A	$ U $	AN	$ J $	KRI	STYLE
DWUG DE	50	34/14/2	$\leq 100 + \leq 100$	8	1.7	.67	use-use
DWUG DE Sense	24	16/7/1	25+25	3	2.9	.87	use-sense

sense3: *Zuträger, Schmeichler*, ‘informer, sweet-talker’

sense4: *andere*, ‘other’

In this example, all three annotators assigned the label *sense1*. The following example for *abgebrüht* illustrates a case of disagreement among annotators:

use: [...] *das war ein finsterer Herr mit dem harten Blick eines **abgebrühten** Schellfisches.*, ‘[...] that was a sinister gentleman with the hard look of a **blanched/hard-nosed** haddock.’

sense1: *mit kochendem Wasser übergossen*, ‘doused with boiling water’

sense2: *gefühllos, frech, dickfellig, abgehärtet gegen sittl. Eindrücke etc.*, ‘callous, insolent, thick-skinned, hardened to moral impressions, etc.’

sense3: *andere*, ‘other’

In the example of *abgebrüht*, two annotators assigned ‘sense1’, while one annotator assigned ‘sense2’, illustrating that some usages allow multiple plausible interpretations and may result in annotator disagreement.

Table 2.6 provides an overview of DWUG DE Sense and compares it to the original DWUG DE dataset. DWUG DE Sense contains approximately half the number of target words available in DWUG DE and between $\frac{1}{3}$ and $\frac{1}{4}$ of the annotated usages per word. In contrast, it includes a higher average number of judgments per annotation instance (2.9 vs. 1.7), despite relying on fewer annotators (3 vs. 8).⁶ This difference reflects the lower annotation effort required by the use-sense annotation paradigm (see Section 2.2), which makes it possible to obtain multiple annotations for the same usage. The resulting annotations were subsequently used to derive gold sense labels and construct cleaned versions of the dataset for evaluation purposes.

Agreement

Table 2.7 reports pairwise agreement between the three annotators as well as the overall agreement (full), excluding the 134 judgments assigned the label ‘other’. Krippendorff’s α reaches 0.87 across all annotators, while pairwise agreement ranges between 0.84 and 0.89, indicating a high level of annotation consistency. Percentage agreement and pairwise Cohen’s Kappa [72] yield comparable results, with average values of 0.88 and 0.87, respectively.

⁶ The average number of judgements per instance is slightly below 3 because a small number of usages were skipped by some annotators during the annotation.

Table 2.7: Agreement (Krippendorff’s α) between annotators on sense definition annotation. Left: Pairwise agreement between annotators on full data. Right: Overall agreement between annotators per time period.

	A	B	C	full	old	new
A		.84	.89			
B			.89			
full				.87	.83	.90

Table 2.8: Statistics of cleaned datasets. $|U|$ = average number of usages per time period, $|S|$ = average number of senses (in all cases the median number of senses is 2).

Data	$ U $	$ S $ full	$ S $ old	$ S $ new
<i>maj</i> ₂	23+23	2.9	2.3	2.5
<i>maj</i> ₃	16+18	2.8	1.8	2.4

According to Erk et al. [30], sense annotation studies based on the WordNet sense inventory typically report percentage agreement values between 0.67 and 0.78. The agreement obtained for DWUG DE Sense is therefore higher than that reported in many previous synchronic sense annotation studies. It should be noted, however, that agreement values are influenced by several factors, including the granularity of the sense inventory.⁷

For diachronic use-use annotation datasets, reported agreement values range from 0.52 [49] to 0.67 [34].⁸ Overall, the reported agreement suggests that DWUG DE Sense constitutes a reliable benchmark for evaluating computational models of sense induction and semantic change.

Table 2.7 also presents agreement scores separately for the two time periods. Agreement is lower for the historical data than for the modern data (0.83 versus 0.90), indicating that assigning sense labels is more challenging in historical contexts. This observation is consistent with the expectation that annotators, despite their linguistic expertise, are generally more familiar with contemporary language usage than with historical language varieties.

The difference in agreement across time periods further suggests that computational models may also face greater difficulty when processing historical data. However, the data-cleaning procedures described in Section 2.4.3 substantially reduce annotation uncertainty and are expected to mitigate part of this discrepancy.

Cleaning and label derivation

Following the annotation process, two cleaning conditions were defined. In the first condition, all instances for which the three annotators assigned different labels were removed. Usages with at least two missing annotations were also excluded, resulting in 1117 usages (from 1200). For each remaining use, the sense definition selected by the majority of annotators (at least two) was assigned as the gold label. This version of the dataset is referred to as *maj*₂ (majority label with agreement of at least two annotators).

⁷ See Section 3.4 for factors that may influence annotation agreement.

⁸ These datasets employ a four-point relatedness scale, making direct comparison difficult.

In the second condition, all instances for which the three annotators did not assign the same label were removed. Usages with missing annotations were likewise excluded, leaving 826 usages. Each use was then assigned the unanimously selected sense label. This version is referred to as *maj₃* (majority label with agreement of all three annotators).

The subsets corresponding to the historical and modern time periods are denoted as “DWUG DE Sense Old” and “DWUG DE Sense New”, respectively. Sense clusters for the WSI task are derived from the aggregated gold sense labels. From these clusters, sense frequency distributions are obtained and binary as well as graded semantic change labels are inferred following the methodology of Schlechtweg et al. [4]. The binary change score indicates whether a sense was gained, lost, or remained stable across time.⁹ The graded change score is computed as the Jensen–Shannon distance between sense probability distributions.

As an illustration, Figure 2.3 presents the sense frequency distribution of *Ohrwurm* under the *maj₃* cleaning condition. The word exhibits two senses with relatively balanced frequencies in the full dataset (19 and 16 usages, respectively). However, only one of these senses occurs in each time period, indicating both sense loss and sense gain. Since no sense is shared across the two periods, the word undergoes complete semantic replacement, resulting in a Jensen–Shannon distance of 1.0 (see also Figure 2.6).

A different pattern can be observed for *Manschette* in Figure 2.4. The full dataset contains three senses with a strongly dominant sense. In the historical portion, only this dominant sense is attested, whereas the two less frequent senses emerge in the modern period. Consequently, these senses can be interpreted as gains over time. The corresponding Jensen–Shannon distance is 0.46, reflecting a moderate-to-high degree of semantic change.

For each of the two cleaning conditions (*maj₂* and *maj₃*), binary WiC or semantic proximity labels are additionally derived for all usage pairs of a target word using a procedure similar to that of Pilehvar and Camacho-Collados [9]. Pairs of usages assigned the same majority sense label receive a label of ‘1’, whereas pairs assigned different sense labels receive a label of ‘0’. As a result, the dataset also provides WiC annotations that can be used for evaluating semantic proximity models.

Table 2.8 summarizes statistics for the cleaned versions of the dataset. The cleaning procedure results in a slight reduction in the average number of usages per time period under the *maj₂* condition and a larger reduction under the *maj₃* condition. The average number of senses in the full dataset is 2.9 and 2.8 for *maj₂* and *maj₃*, respectively. Across all data partitions (full, old, and new), the average number of senses is lower under the stricter *maj₃* condition. This difference is particularly pronounced for the historical portion of the data (2.3 vs. 1.8 senses on average), reflecting the higher level of annotator disagreement for older language usages. Figure 2.5 presents the number of senses per target word under the *maj₃* condition. Most words exhibit two senses across the two time periods, although several exceptions can be observed, including words with one, six, and seven senses (*artikulieren*, *Schmiere*, and *Fuß*, respectively). Figure 2.6 shows the Jensen–Shannon distances between sense probability distributions for each target word under the *maj₃* condition, corresponding to the graded semantic change values derived from the annotated and cleaned data. The values span nearly the entire range between 0 and 1, indicating substantial variation in the degree of semantic change across target words. Some words exhibit complete semantic change, such as *Ohrwurm* and *Seminar*, whereas others show virtually no semantic change,

⁹ Only senses exceeding a predefined occurrence threshold are considered. The code used for aggregation, cleaning, and label derivation is available at <https://github.com/Garrafao/WUGs>.

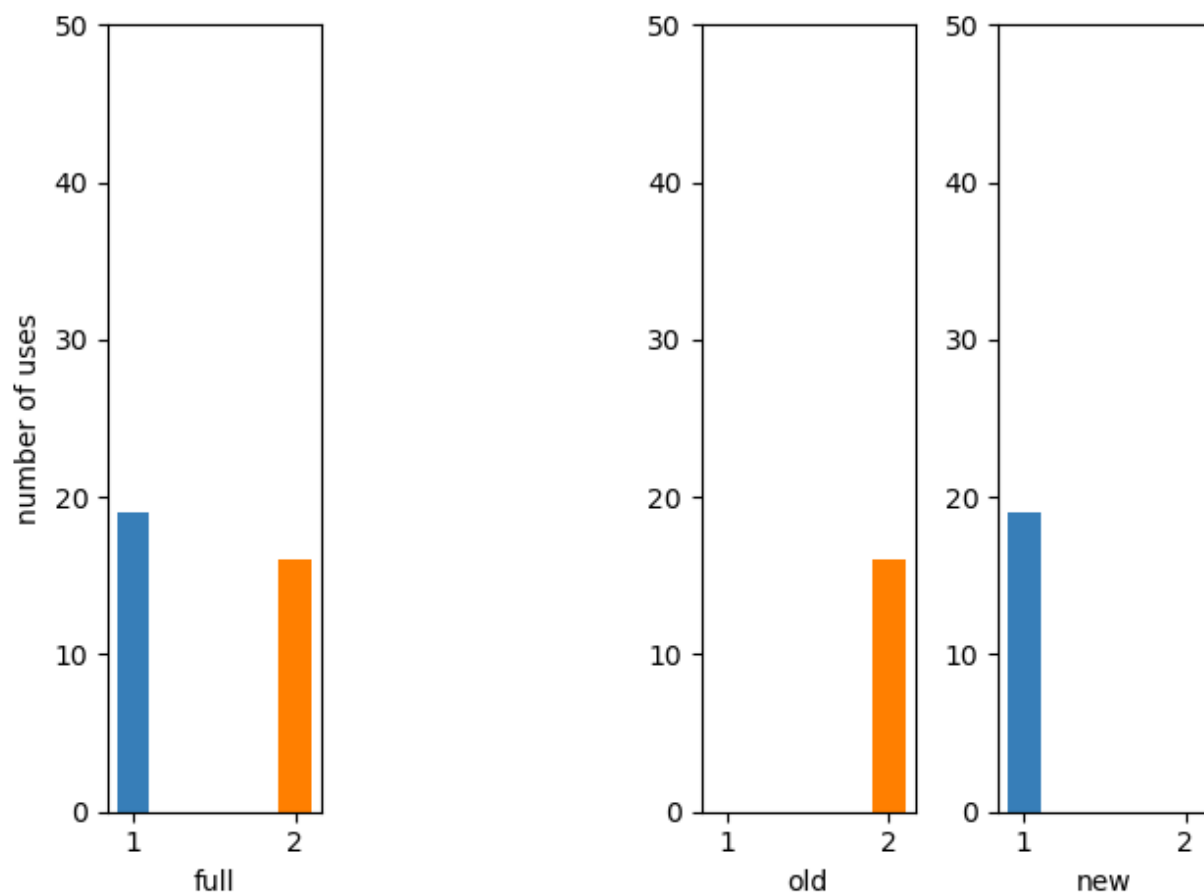


Figure 2.3: Sense frequency distribution for *Ohrwurm* from cleaned dataset (maj₃).

such as *artikulieren*.¹⁰

2.5. Computational approaches

The computational models designed to detect semantic change are grounded in the distributional hypothesis [73], which posits that words appearing in similar contexts tend to share similar meanings. This hypothesis also serves as the foundation for Vector Space Models [74], which represent words as vectors whose elements are derived from their occurrences in various contexts.

Before SemEval-2020, most approaches were type-based: they aggregated statistics over all occurrences of a word to produce a single vector representation per period and compared these period-wise vectors to detect change, an approach that—while effective to some degree—relies on simplifying assumptions and struggles to capture fine-grained variation across individual usages. The release of SemEval-2020 Task 1 provided multilingual datasets and standardized targets that enabled systematic comparison of methods; although top systems in the shared task remained largely type-based (using averaged or condensed vector

¹⁰ These values are obtained after data cleaning. Different cleaning criteria may lead to different semantic change scores.

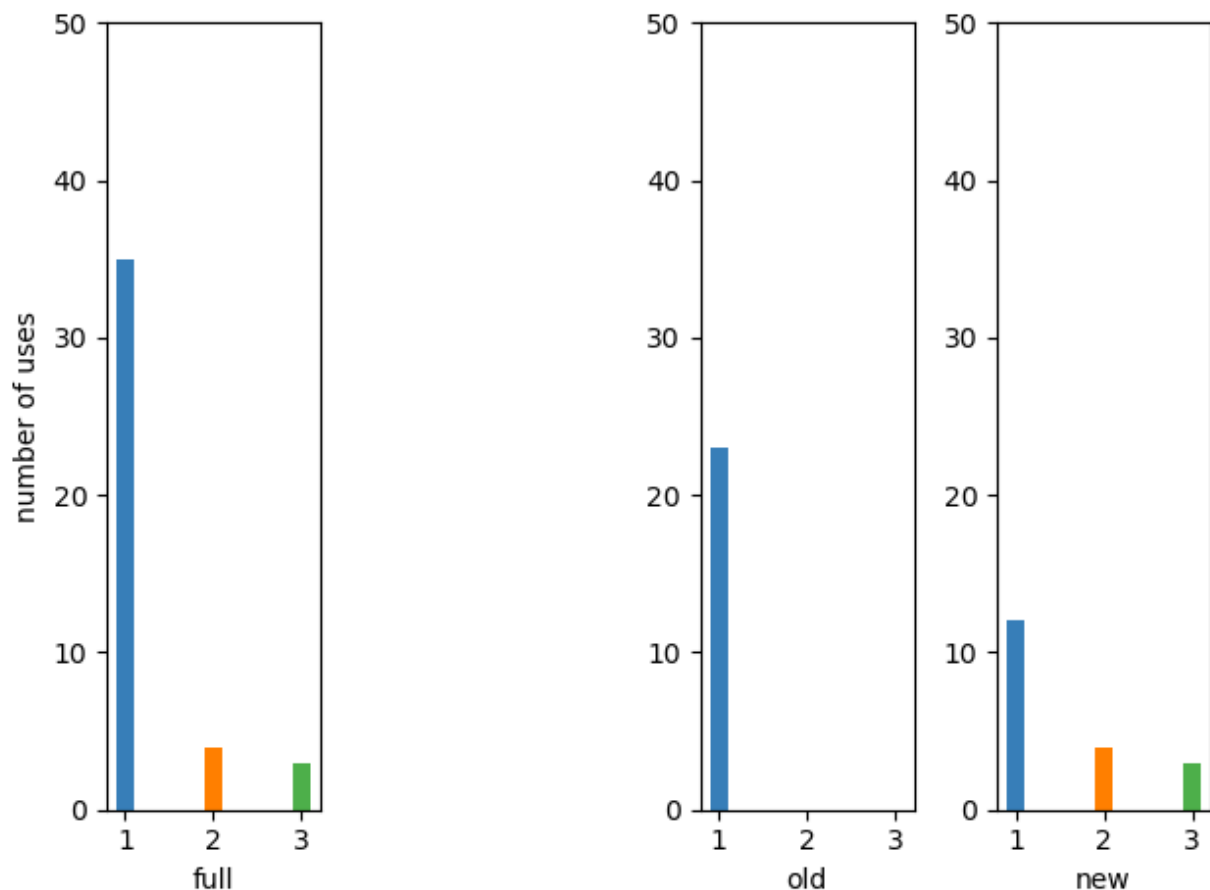


Figure 2.4: Sense frequency distribution for *Manschette* from cleaned dataset (maj₃).

representations), the competition highlighted the limitations of such representations and motivated the development of more usage-aware approaches.

Subsequently, the availability of larger, high-quality LSCD resources—including the Russian DUREL-style RuSemShift [49]—coincided with and helped accelerate a broader shift toward token-based approaches. These models represent each individual use of a word with its own contextualized vector, enabling a more nuanced modeling of different senses. Many of these systems incorporated clustering techniques to infer sense groupings from usage vectors, thereby estimating both the number and distribution of meanings across time. This allowed researchers not only to quantify semantic change more precisely but also to approximate sense-level dynamics directly from corpora.

Despite these advances, early token-based approaches still did not explicitly incorporate the human annotation process, which plays a central role in LSCD evaluation. The most recent wave of models has therefore moved toward WiC-based approaches [19, 20], which aim to simulate human relatedness judgments by predicting whether two usages express the same meaning. These predictions can then be combined with clustering methods to approximate the creation of gold data, mirroring the annotation pipeline more closely. Such models currently represent the state of the art across multiple LSCD datasets in different languages.

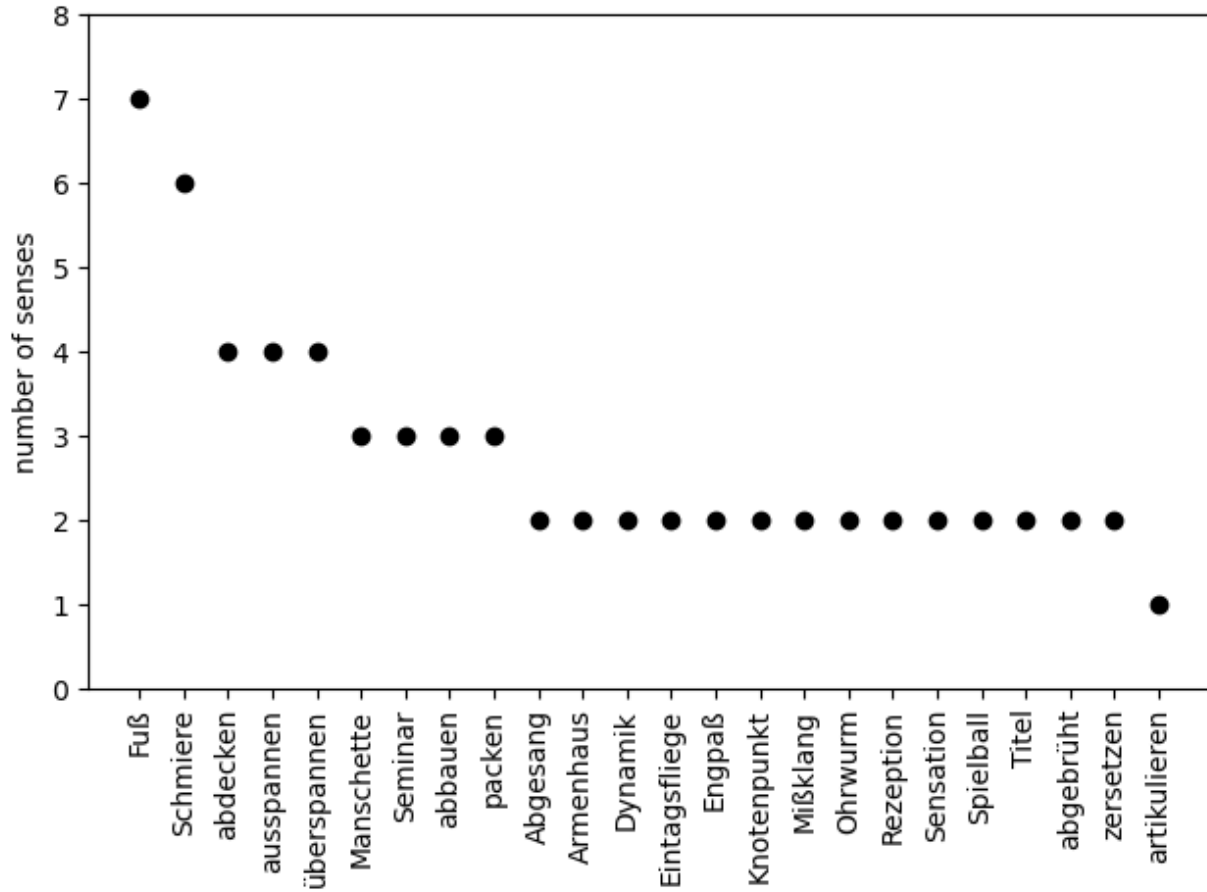


Figure 2.5: Number of senses per word for cleaned dataset (maj₃).

2.5.1. Type-based LSCD models

Type-based models are designed to encapsulate the various usages of a word into a single, condensed vector representation. These models are characterized by the following features:

- A semantic representation mapping each word to a vector,
- An alignment method; and
- A change measure to quantify variations in word meanings over time.

Although type-based models assign a single vector to each word per time period, they still capture the predominant usage (sense and topical context) within each period. However, since these vectors are learned in separate spaces, they are not directly comparable. To address this, an alignment step is required to map the representations into a common space. Once aligned, a metric is applied to quantify the degree of semantic change over time.

2.5.1.1. Semantic representation

Type-based semantic representations aggregate information from all occurrences of a word within a given time period to produce a single vector or distribution per word per period. Common instantiations include count-based co-occurrence matrices (optionally reweighted,

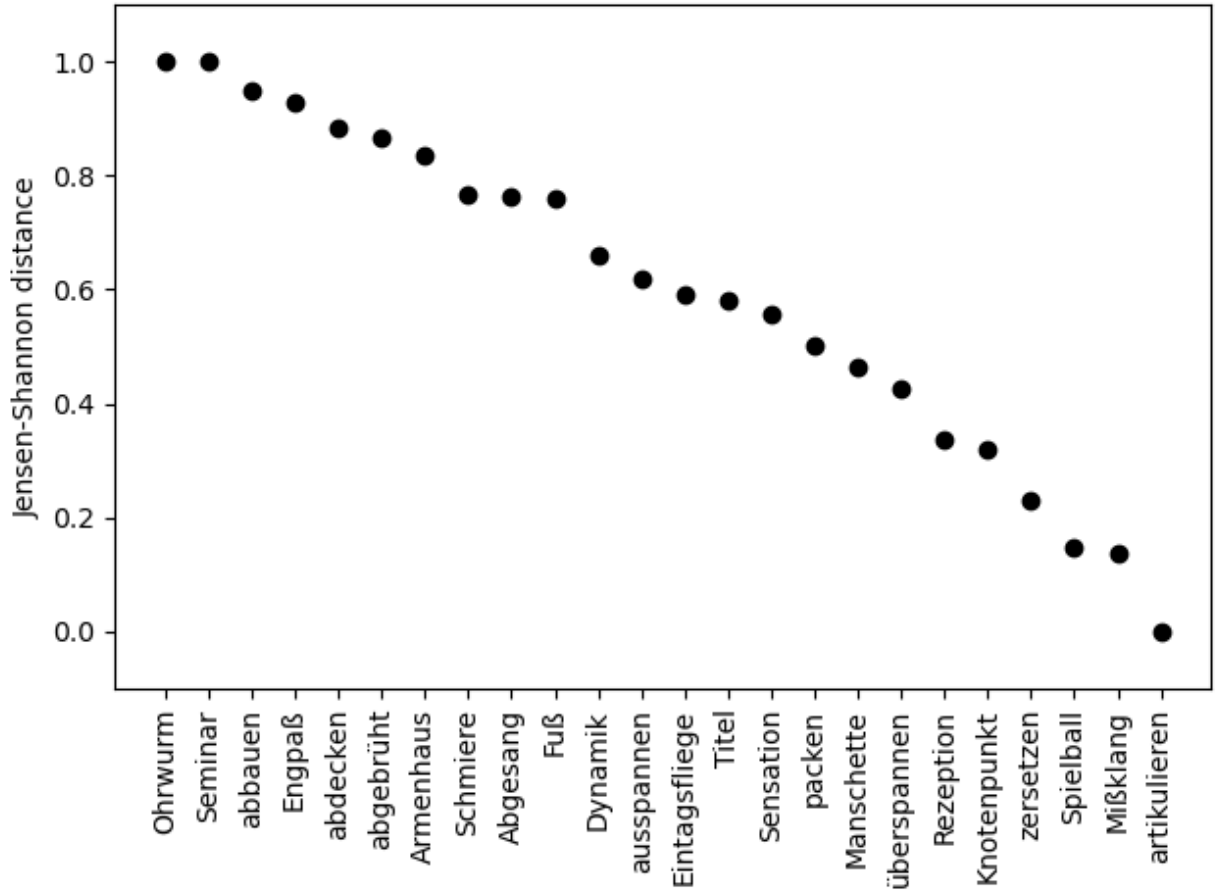


Figure 2.6: Jensen-Shannon distance between sense probability distributions for cleaned dataset (maj₃).

for example with Positive Pointwise Mutual Information), dimensionality-reduced representations obtained via matrix factorization or Random Indexing, and prediction-based static embeddings such as Skip-gram with Negative Sampling and Continuous Bag of Words. Because these representations are learned independently per period, an alignment step (for example, Orthogonal Procrustes, Temporal Referencing, or Column Intersection) is typically required before applying a change metric (for example, cosine or Euclidean distance) to quantify diachronic shift. Type-based methods provide a compact, computationally efficient baseline but are limited in capturing fine-grained, usage-level sense variation; token-based, contextualized approaches address this limitation by representing individual usages. For formal definitions, see Appendix C.

2.5.2. Token-based LSCD models

Token-based models, particularly contextualized language models such as BERT [16], leverage the transformer architecture to generate distinct vector representations for different usages of a word. The transformer architecture, introduced by Vaswani et al. [75], employs self-attention mechanisms to capture contextual relationships within a sequence of text. This allows the model to produce dynamic embeddings that reflect the surrounding context of each word occurrence. As a result, contextualized models are capable of effectively distinguishing

between different meanings and usages of a word depending on its context.

The key components of contextualized models for detecting LSC include [3, p. 70]:

- A semantic representation that maps each word usage to a distinct vector;
- An optional clustering step to combine information across multiple contexts; and
- A change measure to quantify variations in word meanings over time.

2.5.2.1. Semantic representation

Semantic representations in LSCD typically rely on contextualized embeddings generated by language models such as ELMo [76], BERT [16], and others. These embeddings are derived by encoding each individual usage of a word in its specific context, resulting in a high-dimensional vector that reflects semantic and syntactic properties for each individual usage of a specific word. Unlike static embeddings—which assign a single vector to a word regardless of context—contextualized models are capable of distinguishing between different senses and usages of the same word.

BERT and many of its derivative models are based on the masked language modeling (MLM) objective, in which a subset of input tokens is randomly masked, and the model is trained to predict these masked tokens using their surrounding bidirectional context. This training strategy enables the model to capture rich contextual dependencies and subtle meaning variations. In addition to MLM, BERT introduces a second pretraining objective, next sentence prediction (NSP), which requires the model to determine whether a given pair of sentences appear consecutively in the original text. The inclusion of NSP facilitates the learning of sentence-level semantics and discourse-level coherence, further enhancing the quality of the resulting contextualized embeddings.

These training strategies enable BERT and similar models to produce contextualized embeddings that are sensitive to usage-specific meaning. In practical applications, embeddings are typically extracted from the final hidden layer by selecting the vector corresponding to the target token, or by averaging over subword tokens when applicable.

2.5.2.2. Clustering methods

Unlike type-based models, where embeddings are learned independently for each time period—leading to representations that are often not directly comparable due to the stochastic nature of neural network training [12, 45]—token-based models generate embeddings within a shared vector space [48, 77]. Consequently, the alignment step commonly required in type-based approaches becomes unnecessary in token-based settings. Nevertheless, clustering methods can be applied to contextualized embeddings to induce sense clusters, where each cluster corresponds to a distinct sense or usage pattern of the target word.

K-means is a clustering method that partitions a set of observations into K distinct clusters based on their features. Given a set of n data points $\{x_1, x_2, \dots, x_n\}$, each represented as a vector in $\mathbb{R}^d$, the goal of K-means is to minimize the within-cluster sum of squared distances between points and their assigned cluster centroids [78].

Formally, K-means aims to find a set of K cluster centroids $\{\mu_1, \mu_2, \dots, \mu_n\}$ that minimize the objective function:

$$J = \sum_{k=1}^K \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$$

where C_k denotes the set of points assigned to cluster k , and $\|\cdot\|$ is the Euclidean norm.

Correlation clustering is a graph clustering technique which minimizes the sum of cluster disagreements, i.e., the sum of negative edge weights within a cluster and the positive edge weights across clusters [4]. Correlation Clustering has hyperparameters for splitting edge weights into positive and negative (`threshold_cc`), for the maximum number of senses, and for maximum attempts and maximum iterations for simulated annealing.

Agglomerative clustering refers to a class of hierarchical clustering algorithms that build a nested sequence of clusters by iteratively merging the two closest clusters until a stopping criterion is met, such as reaching a predetermined number of clusters. Starting with each data point as a singleton cluster, the algorithm merges clusters based on a linkage criterion that measures the distance between clusters. Common linkage methods include single linkage, complete linkage, and average linkage [79].

Given two clusters, C_i and C_j , the distance between them can be defined, for example, by average linkage as:

$$d(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{\mathbf{x}_p \in C_i} \sum_{\mathbf{x}_q \in C_j} \|\mathbf{x}_p - \mathbf{x}_q\|$$

where $|C_i|$ and $|C_j|$ denote the number of points in clusters C_i and C_j , respectively, and $\|\cdot\|$ is a chosen norm, often the Euclidean distance.

Agglomerative clustering produces a dendrogram representing the hierarchical structure of clusters, enabling the exploration of cluster granularity at different levels.

Weighted Stochastic Block Model (WSBM) is a probabilistic generative model used to detect community or block structures in weighted networks. It extends the classical stochastic block model by modeling the weights of edges between nodes as random variables drawn from parameterized distributions, depending on the block memberships of the nodes [80].

Given a Graph $G = (V, E, W)$ with nodes V , edges E , and edge weights $W = \{w_{ij}\}$ the WSBM assumes that each node i belongs to a latent block $z_i \in \{1, \dots, K\}$, where K is the number of blocks. The weight w_{ij} of the edge between nodes i and j is then generated according to a distribution $P(\cdot \mid \theta_{z_i z_j})$, parameterized by $\theta_{z_i z_j}$ specific to the block pair (z_i, z_j) .

The likelihood of the observed weighted adjacency matrix W given the block assignments $Z = \{z_i\}$ and parameters $\Theta = \{\theta_{kl}\}$ is:

$$P(W \mid Z, \Theta) = \prod_{i < j} P(w_{ij} \mid \theta_{z_i z_j})$$

Inference typically involves estimating the block assignments Z and parameters Θ that maximize this likelihood, often via expectation-maximization [81] or Bayesian methods.

Spectral clustering (SC) is a class of algorithms that apply clustering to a low-dimension projection of the affinity matrix of the graph [82]. Let $W \in \mathbb{R}^{n \times n}$ be the similarity matrix, where W_{ij} measures the similarity between points i and j . The degree matrix D is defined as a diagonal matrix with entries:

$$D_{ii} = \sum_{j=1}^n W_{ij}$$

The (unnormalized) graph Laplacian is then given by:

$$L = D - W$$

Spectral clustering proceeds by computing the first k eigenvectors of L (or a normalized variant), forming a matrix $U \in \mathbb{R}^{n \times k}$. Each row of U is treated as a representation of the corresponding data point, and a clustering algorithm such as K -means is applied to partition the points into k clusters.

Chinese Whispers (CW) is an efficient, randomized clustering algorithm with a time complexity linear with respect to the number of edges [83]. The algorithm first assigns all nodes to different clusters. Then the nodes are processed in randomized order for a small number of iterations (we set this hyperparameter to 20) and are assigned to the strongest cluster in the local neighborhood, i.e., the cluster whose sum of edge weights to the current node is maximal. The calculation of edge weights is controlled by the **weighting** hyperparameter, which takes three different values:

1. **lin**: This calculates the weight of an edge between two nodes in a graph using linear weighting, which is the edge weight divided by the degree of the destination node.
2. **log**: This computes the weight of an edge between two nodes in a graph using logarithm weighting, which is the edge weight divided by the logarithm of the degree of the destination node.
3. **top**: This keeps edge weight as is.

2.5.2.3. Change measure

Once contextualized embeddings or sense clusters have been obtained for target words across different time periods, quantifying lexical semantic change requires appropriate change measures. These measures aim to capture the degree and nature of semantic shifts by comparing representations from different temporal contexts.

Average Pairwise Distance (APD) is a metric for quantifying semantic change, often applied over contextualized word embeddings. For a given word, the APD computes the average of the pairwise distances—typically cosine distances—between all combinations of its contextualized embeddings observed across two time slices. Let $U = \{u_1, u_2, \dots, u_n\}$ and $V = \{v_1, v_2, \dots, v_m\}$ be the sets of contextualized embeddings of a word in two different corpora or time periods. Then, the APD is defined as:

$$\text{APD}(U, V) = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n d(\mathbf{u}_i, \mathbf{v}_j)$$

where $d(\cdot, \cdot)$ is a distance function, such as cosine distance. A higher APD value indicates greater semantic divergence, suggesting that the word has undergone more substantial change in meaning between the two periods. This method provides a way to compute the COMPARE metric (see Section 2.3.3), yielding a fine-grained and context-sensitive quantification of semantic change and enabling the detection of gradual or subtle shifts in usage [84].

The **JSD** metric is a commonly used metric for quantifying the divergence between two probability distributions and is particularly suitable for comparing sense distributions of a

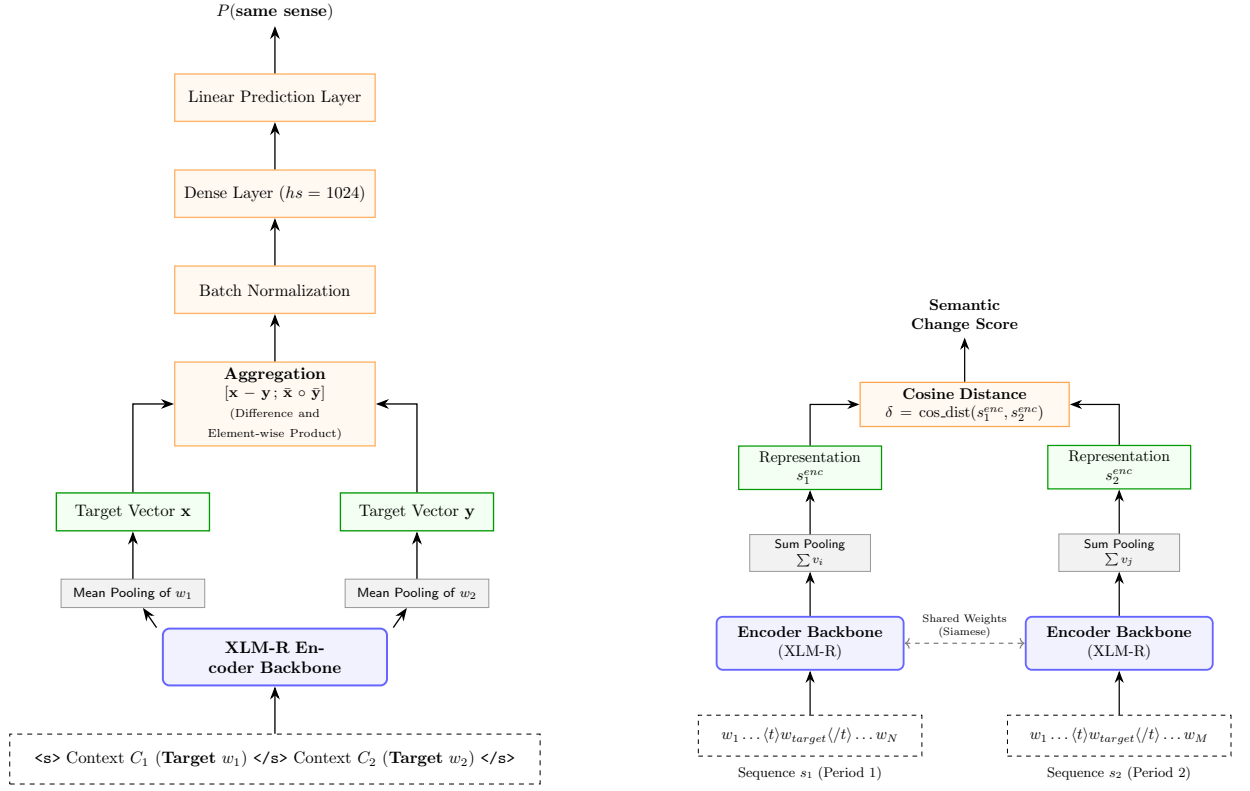


Figure 2.7: Comparison of the DeepMistake (left) and XL-LEXEME (right) architectures, illustrating their respective mechanisms for generating graded change predictions based on contextualized embeddings.

word across different time periods. It is used to quantify graded semantic change at the level of sense distributions (see Section 2.3.3.1).

2.5.3. WiC-based LSCD models

The core principle underlying WiC-based models is to assess whether two contextualized instances of a target word express the same sense. To this end, these models operate over contextual embeddings and produce a continuous score—either a probability or a distance-based measure—that quantifies the degree of semantic change between two usages.

These systems emulate the annotation procedure carried out by human annotators, with the key difference that they do not rely on a discrete rating scale. Instead, they generate continuous outputs that reflect the likelihood that two usages convey the same sense. Despite this difference in representation, the objective remains the same: to determine whether a pair of usages expresses the same sense.

WiC-based models are typically trained or fine-tuned on annotated WiC datasets to approximate this decision process. Two representative examples are DeepMistake [19] and XL-LEXEME [20]. The former models semantic proximity through probability estimates, whereas the latter relies on cosine distance; both ultimately aim to decide whether two contextualized usages express the same meaning.

From an architectural perspective, WiC-based models implement pairwise comparison mechanisms over contextualized representations, but differ in how this comparison is realized.

DeepMistake follows a bi-encoder design, in which each sentence is independently encoded using a shared multilingual transformer, and a classification layer predicts the probability that both instances express the same sense. In contrast, XL-LEXEME adopts a Siamese-style architecture, where contextualized embeddings of the target word are directly compared using a distance metric such as cosine similarity. In both cases, the encoder is typically based on XLM-RoBERTa (XLM-R) [17], enabling cross-lingual generalization (see Figure 2.7).

Through fine-tuning, these models learn a task-specific proximity function that structures the representation space so that usages expressing the same sense are mapped closer together, while those corresponding to different senses are positioned further apart.

This pairwise semantic proximity signal provides the foundation for extending WiC-based models to related sense-oriented tasks. In particular, it enables clustering-based approaches to WSI, where multiple contextualized usages of a word are grouped according to their pairwise proximity relations in order to identify distinct senses.

Building on this signal, approaches to LSCD differ in how they aggregate or structure these pairwise proximity scores. Two main methodologies are commonly employed:

- **APD**, which aggregates the model’s pairwise outputs across all usage pairs of a target word and interprets this aggregation as a measure of semantic change.
- **WUG-based approach**, where the outputs of the WiC model are used to construct a graph in which nodes represent individual usages and edges encode pairwise semantic proximity relations. Clustering methods are then applied to this graph to identify groups of usages corresponding to distinct senses of the target word, after which divergence-based measures are used to quantify how the distribution of these senses changes across time periods.

Importantly, the WUG-based approach closely mirrors the human annotation workflow used to construct gold-standard datasets: annotators first assess pairwise semantic proximity between usages, and sense distinctions emerge from the structure induced by these judgments.

Chapter 3

Building a High-Quality Spanish LSCD Dataset and Evaluating State-of-the-Art Approaches

3.1. Introduction

In recent years, several LSCD shared tasks have been organized [4, 18, 54]. These tasks have contributed to a better understanding of LSCD, but have also had their shortcomings: (i) they have used mainly small pre-selected sets of target words creating an unrealistic evaluation scenario for the application of computational models in historical semantics and lexicography where researchers typically aim to cover the full vocabulary of a language [85], (ii) different formalizations of the LSCD task have been proposed including binary classification and ranking tasks [3, 25, 86] and these have been employed inconsistently, and (iii) none of them have focused on Spanish, despite the fact that there are more than 450 million native speakers of this language.

We tackle these shortcomings by organizing a shared task on Spanish diachronic data with a more realistic evaluation scenario requiring participants to provide LSC predictions for the full corpus vocabulary (Discovery). Additionally, we cover previous scenarios by asking participants to predict LSC only in the limited sample of annotated target words (Detection). By offering a range of additional optional tasks (defined on the same annotated data) participants are able to evaluate and compare models on various formalizations of the LSCD task. In order to derive gold LSC labels for target words, we annotate and publish the largest existing data set of semantic proximity judgments covering 100 words with approximately 62k judgments from 12 human native speakers.¹¹

3.2. Background

The detection of lexical semantic changes is of great interest in research areas such as historical semantics, lexicography, linguistics and NLP. For a comprehensive review of the literature on the area we refer the reader to the recent surveys [45, 87, 88]. In previous years several shared tasks have been organized: SemEval-2020 Task 1 [4] for English, German, Latin, and

¹¹ The data set is available at <https://zenodo.org/record/6300104>.

Swedish, DIACR-Ita for Italian [54], and RuShiftEval for Russian [18].¹² All shared tasks applied an evaluation setup where LSC was measured between pairs of time periods.

SemEval used a total of 156 target words for all languages with no development/test split. Approximately half of these were drawn from etymological dictionaries or research literature, while the other half was drawn from the corpus vocabularies by selecting lemmas with similar POS and frequency as the first half of target words. Target word occurrences in sentences (usages) were combined into pairs and these were annotated for their semantic proximity [34]. Target words were excluded if they had a high number of undecidable use pairs or were annotated too sparsely. Sense clusters were inferred from the annotation. From the clusters a binary (sense loss/gain vs. none) and a graded (JSD between cluster distributions) change score were derived and used to evaluate participants on a corresponding binary classification and ranking task.

DIACR-Ita used a total of 18 target words with no development/test split. All of these were drawn from an etymological dictionary. Target word usages were annotated with word sense definitions. Words with a high number of OCR errors and annotator disagreements were excluded. From the annotation Binary Change scores similar to SemEval were derived and used to evaluate participants on a binary classification task.

RuShiftEval used a total of 111 target words (all nouns) split into 12 for development and 99 for testing. These were selected in a similar procedure to SemEval: approximately half of these were drawn from etymological dictionaries, research literature or “invented” by the authors, while the other half was drawn from the corpus vocabularies by selecting lemmas with similar POS and frequency as the first half of target words. Target word usages from different time periods were combined into usage pairs and annotated for semantic proximity. From these the DUREL COMPARE score (see Section 3.3.3 for more details) [25] was derived, which can be seen as an approximation of SemEval’s Graded Change score [3]. Participants were evaluated in a ranking task on the COMPARE scores.

As we can see, target words in previous shared tasks have been strongly preselected and systems have been evaluated on different tasks. They have also yielded (seemingly) contradictory results: while type-based model architectures have dominated in SemEval and DIACR-Ita, token-based architectures have dominated in RuShiftEval. In all tasks clustering-based models have shown rather low performance.

3.3. Task description

Our task was designed in two phases:

1. Graded Change Discovery, and
2. Binary Change Detection.

Note that *discovery* introduces additional difficulties for models as compared to the more simple semantic change *detection*, e.g. because a large number of predictions is required and the target words are not preselected, balanced or cleaned [cf. 85]. Yet, discovery is an important task, with applications such as lexicography where dictionary makers aim to cover the full vocabulary of a language.

¹² There was also a student shared task on German data [89].

3.3.1. Phase 1: Graded Change Discovery

Similar to Kurtyigit et al. (2021) [85], we define the task of **Graded Change Discovery** as follows:

Given a diachronic corpus pair C_1 and C_2 , rank the intersection of their (content-word) vocabularies according to their degree of change between C_1 and C_2 .

The participants were asked to rank the set of content words in the lemma vocabulary intersection of C_1 and C_2 according to their degree of semantic change between C_1 and C_2 where a higher rank means stronger change. The true degree of semantic change of a target word w was given by the Jensen-Shannon distance [63, 64] between w 's word sense frequency distributions in C_1 and C_2 [cf. 4]. The two word sense frequency distributions were estimated via human annotation of word usage samples for w from C_1 and C_2 (see Section 3.4.4). Participants' predictions were *not* evaluated on the full set of target words, as this would be unfeasible to annotate, but on an (unpublished) random sample of words from the full set of target words. The predictions were scored against the ground truth via Spearman's rank-order correlation coefficient [26].

3.3.2. Phase 2: Binary Change Detection

Similar to Schlechtweg et al. (2020) [4], we define the task of **Binary Change Detection** as follows:

Given a target word w and two sets of its usages U_1 and U_2 , decide whether w lost or gained senses from U_1 to U_2 , or not.

The participants were asked to classify a pre-selected set of content words into two classes, 0 for no change and 1 for change. The true binary labels of word w were inferred from w 's word sense frequency distributions in C_1 and C_2 (see Subsection 3.3.1). Participants' predictions were scored against the ground truth with the following metrics: F1 (main metric), Precision, and Recall. A crucial difference compared to Graded Change Discovery was that the public target words corresponded exactly to the hidden words on which we evaluated. Also, we published the usages sampled for annotation. Hence, participants could work with the exact annotated data, which was not possible in the first phase where participants could only work with the full corpora (from which the usages for annotation were sampled).

3.3.3. Optional tasks

Participants could submit predictions for several optional tasks:

Graded Change Detection was defined similar to Graded Discovery. The only difference was that the public target words corresponded exactly to the hidden words on which we evaluated. Participants were scored with Spearman correlation.

Sense Gain Detection was similar to Binary Change Detection. However, only words which gained (not lost) senses receive label 1. Participants were scored with F1, Precision and Recall.

Sense Loss Detection was similar to Binary Change Detection. However, only words which lost (not gained) senses received label 1. Participants were scored with F1, Precision and Recall.

COMPARE asked participants to predict the negated DUREL COMPARE metric [25]. This metric is defined as the average of human semantic proximity judgments of usage pairs

Table 3.1: Sizes of both corpora.

Corpus	Time period	Tokens
Old corpus (C_1)	1810–1906	$\sim 13M$
Modern corpus (C_2)	1994–2020	$\sim 22M$

for w between C_1 and C_2 .¹³ It can be seen as an approximation of JSD (Graded Change) [3]. Participants were scored with Spearman correlation.

Participants’ submission files only needed to include predictions corresponding to the obligatory tasks in order to get a valid submission. They did not see the leaderboard while the evaluation phases were running. Furthermore, participants only had three valid submissions for each evaluation phase.¹⁴

3.4. Data

In this section, we describe the corpora, the selection process of target words, the sampling of usages and their annotation. Moreover, we explain how the target words were presented to the participants considering the two phases of the shared task.

3.4.1. Corpora

We created two corpora covering disjoint time periods: 1810 to 1906 (old corpus, C_1) and 1994 to 2020 (modern corpus, C_2) (see Table 3.1). The former was created using different sources freely available from Project Gutenberg¹⁵ and the latter using different sources available from the OPUS project¹⁶ [90]. For the old corpus, all the sources collected were concatenated. As for the modern corpus, four datasets were used: Spanish portion of TED2013, Spanish portion of News-Commentary v16, Spanish portion of MultiUN and Spanish version of Europarl corpus. TED2013 was used in its entirety, while 50 snippets with 5000 lines each were extracted from the other datasets by cutting the corpora into snippets of the mentioned size and randomly choosing 50 of them.

Both corpora were parsed using spaCy [91].¹⁷ Each corpus contains four versions of the original dataset (raw, tokenized, lemmatized and POS-tagged).

3.4.2. Target words

3.4.2.1. Phase 1 (Graded Discovery)

Public target words was a list of 4385 words created in the following way: we first took the corpus vocabulary intersection from the lemmatized versions of both corpora. Then we

¹³ Contrary to the original metric we first take the median of all annotator judgments for each usage pair and then average these values. For details see: <https://github.com/Garrafao/WUGs>.

¹⁴ We decided not to include the binary subtasks in phase 1, as the usage samples were not published which meant that participants needed to work with the full corpora instead of the samples on which the gold scores were inferred. We assumed that the sampling error between usages in the full corpora and our samples is much larger for Binary Change than for Graded Change [cf. 3].

¹⁵ <https://www.gutenberg.org/browse/languages/es>

¹⁶ <https://opus.nlpl.eu/>

¹⁷ Find detailed issues in Appendix A.

removed words below a minimum frequency threshold of 40 for the old corpus and 73 for the modern corpus.¹⁸ Then we removed all non-content words, i.e., we left only nouns, verbs, adjectives and adverbs. The final list of target words was published and participants were required to submit results for all 4385 words in the development and evaluation phase 1.

Hidden target words The large number of public target words was crucial to our task. However, it was not feasible to annotate all of them. Hence, we only annotated a subset of the public target words for semantic change. Participants’ predictions for development and evaluation phase 1 were evaluated only on this subset of target words, which remained hidden from the participants. We selected the hidden target words in the following way: Initially, a list of 15 changing words was selected by scanning etymological dictionaries and consulting with a linguistic specialist to obtain words for changes from *C1* to *C2*. Likewise, it was verified that these words were in both corpora. Additionally, a list of 85 words were randomly sampled from the public target words. The $85 + 15 = 100$ words were annotated as described in Section 3.4.4. Then, 20 words were excluded based on inter-annotator agreement.¹⁹ The remaining set of 80 target words were split randomly into two groups, 20 words for the development set and 60 for the evaluation set (see Table 2.3). Uploaded submissions were scored against these 20/60 annotated words during development/evaluation phases.

3.4.2.2. Phase 2 (Binary Detection)

The target words corresponded to the 20/60 hidden words from Phase 1 for development/evaluation. There it was no distinction here between public and hidden target words. Participants also got access to the annotated usages (20+20 from each corpus). Uploaded submissions were scored against the 20/60 public annotated words.

3.4.3. Word usages

All occurrences of the target words per corpus were extracted according to the lemma. Then, 20 usages were randomly sampled per target word from each corpus.

3.4.4. Annotation

We applied the SemEval procedure to annotate target word usages, as described in Schlechtweg et al. (2020, 2021) [4, 34]. Annotators were asked to judge the semantic relatedness of pairs of word usages, such as the two usages of *servidor* in (3.1) and (3.2), on the scale in Table 2.3.

- (3.1) Todo esto lo hago con mi iPhone; se va derecho al **servidor**, allí se hace el trabajo de archivo, clasificación y ensamble.
*‘I do all this with my iPhone; it goes straight to the **server**, there the work of archiving, sorting and assembling is done.’*
- (3.2) Llamó a grandes voces a sus **servidores**, y únicamente le contestó el eco en aquellas inmensas soledades, y se arrancó los cabellos y se mesó las barbas, presa de la más espantosa desesperación.

¹⁸ 40 was chosen by us for the old corpus and then we calculated 73 for the new corpus to reflect the same proportion of the frequency threshold to corpus size.

¹⁹ We removed target words with agreements of less than 0.3 Krippendorff’s α and less than 0.3 on a version of Krippendorff’s α where expected disagreements were calculated from the full annotated data (instead of for each word separately). The latter measure is less sensitive to skewed judgment distributions for individual words.

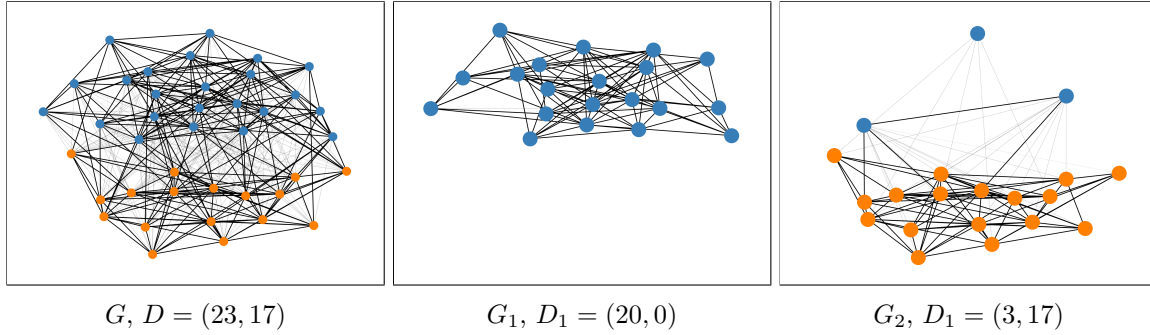


Figure 3.1: Word Usage Graph *servidor* (left), subgraphs for old corpus G_1 (middle) and for modern corpus G_2 (right). The colors correspond to the clusters. **black**/gray lines indicate **high**/low edge weights.

*‘He called out to his **servants**, and only the echo in those immense solitudes answered him, and he pulled out his hair and ruffled his beard, prey to the most frightening desperation.’*

The annotated data of a word was represented in a WUG, where vertices represented word usages, and weights on edges represented the (median) semantic relatedness judgment of a pair of usages such as (3.1) and (3.2). The final WUGs were clustered with correlation clustering [4, 34, 92] (see Figure 3.1, left) and split into two subgraphs G_1 and G_2 representing nodes from subcorpora C_1 and C_2 respectively (middle and right). Clusters were then interpreted as word senses and changes in clusters over time as lexical semantic change.²⁰

In contrast to Schlechtweg et al. (2020) [4], we used the openly available DUREl interface for annotation and visualization.²¹ This also implied a change in sampling procedure, as the system implemented only random sampling of usage pairs (without SemEval-style optimization, i.e., sampling in rounds with connection of clusters). For each target word we sampled $|U_1| = |U_2| = 20$ usages (sentences) per subcorpus (C_1, C_2) and uploaded these to the DUREl system, which presented usage pairs to annotators in randomized order. We recruited twelve Spanish native speakers (4 Chileans, 4 Colombians, 2 Cubans, 1 Spaniard and 1 Venezuelan). All had university level education, while seven had a background in linguistics of which two had one in historical linguistics. We monitored agreement between annotators during the annotation process and discussed some strong annotation disagreements with certain annotators. This led to the exclusion of one annotator early in the process who often completely inverted the annotation scale (e.g. judged 1 while agreeing that the two usages have identical meanings).

Similar to Schlechtweg et al. (2020) [4], we ensured the robustness of the obtained clusterings by continuing the annotation of a target word until all clusters in its WUG were connected by at least one judgment.²² For 16 words the annotation had to be stopped before this condition was met. We manually inspected the unconnected clusters of some words and concluded that missing connections did not lead to clustering errors.

²⁰ We used Schlechtweg et al. (2020, 2021)’s [4, 34] code provided at <https://www.ims.uni-stuttgart.de/data/wugs>.

²¹ <https://www.ims.uni-stuttgart.de/data/durel-tool>.

²² Note that this condition was more strict than Schlechtweg et al. (2021) [4]’s where only connection of multi-clusters (clusters with more than one usage) was guaranteed. Their condition was always met in our data.

Table 3.2: Overview target words. n = no. of target words, N/V/A = no. of nouns/verbs/adjectives+adverbs, $|U|$ = avg. no. of usages per word, AN = no. of annotators, JUD = total no. of judged usage pairs, AV = avg. no. of judgments per usage pair, KRI = Krippendorff’s α , SPR = weighted mean of pairwise Spearman, UNC = avg. no. of uncomparing multi-cluster combinations, LOSS = avg. of normalized clustering loss * 10, $LSC_{B/G}$ = mean binary/Graded Change score.

Data set	n	N/V/A	U	AN	JUD	AV	KRI	SPR	UNC	LOSS	LSC_B	LSC_G
development	20	13/4/3	40	10	12k	40	.53	.59	0	.53	.55	.39
evaluation	60	30/14/16	40	12	38k	40	.58	.60	0	.45	.47	.37
discarded	20	8/6/6	40	12	12k	40	.27	.33	0	.52	.30	.18
full	100	51/24/25	40	12	62k	40	.53	.57	0	.48	.45	.34

We finally labeled a target word as Binary Change if it gained or lost a cluster over time. For instance, *servidor* in Figure 3.1 was labeled as change as it gained the orange cluster from C_1 to C_2 . Consequently, *servidor* was also labeled as gaining a sense; but not as losing a sense, since the blue cluster persists. Graded Change was defined as the Jensen-Shannon distance between the normalized cluster frequency distributions D_1 and D_2 yielding a high value of 0.82 (ranges between 0.0 and 1.0) for *servidor*, as sense probabilities changed drastically. The negated COMPARE score was derived by averaging over all graph edges with nodes from different time periods and negating this value, yielding a high score of -1.97 (ranges between -4.0 and -1.0) for *servidor*.²³ Following Schlechtweg et al. (2020) [4] we used k and n as lower frequency thresholds for the binary notions to avoid that small random fluctuations in sense frequencies caused by sampling variability or annotation error were misclassified as change. As proposed in Schlechtweg [3, p. 48] for comparability across sample sizes we set $k = 1 \leq 0.01 * |U_i| \leq 3$ and $n = 3 \leq 0.1 * |U_i| \leq 5$, where $|U_i|$ was the number of usages from the respective time period.²⁴ This resulted in $k = 1$ and $n = 3$ for all target words.

Find an overview over the final set of WUGs in Table 3.2. We reached an inter-annotator agreement of Krippendorff’s $\alpha = .53$ and Spearman’s $\rho = .57$ which was comparable to previous studies [e.g. 25, 49, 85, 93].²⁵

3.5. Analysis of Inter-Annotator Disagreement

Although the dataset comprises 60 target words, performing a detailed qualitative analysis for every item would be impractical and would considerably increase the length of the discussion. Instead, we conduct a focused case-study analysis on the word *planta*, which exhibited substantial annotation disagreement across annotators from different regional backgrounds. This word is particularly suitable for analysis because it is highly polysemous and appears in both modern and historical contexts with multiple semantically distinct usages, including botanical, architectural, and ornamental senses. The objective of this analysis is not

²³ Find a more detailed discussion of different change scores in Schlechtweg et al. (2020) [4] and Schlechtweg (2023) [3].

²⁴ That is, k was always between 1 and 3. There are three possible cases: $k = 1$ if $0.01 * |U_i| \leq 1$, $k = 0.01 * |U_i|$ if $1 < 0.01 * |U_i| < 3$, $k = 3$ if $0.01 * |U_i| \geq 3$. Similarly for n .

²⁵ We provide WUGs as Python NetworkX graphs, descriptive statistics, inferred clusterings, change values and interactive visualizations for all target words and the respective code at <https://www.ims.uni-stuttgart.de/data/wugs> (DWUG ES).

Table 3.3: Annotation disagreement patterns for the word *planta*.

Example	A (CO)	B (ES)	C (CU)	D (CL)
Example 3.3 – 3.4	1	4	4	3
Example 3.5 – 3.6	1	1	4	–

A = Colombian annotator, B = Spaniard annotator, C = Cuban annotator, D = Chilean annotator.

to generalize from a single word to the entire dataset, but rather to illustrate the types of linguistic and contextual factors that contributed to lower inter-annotator agreement in the Spanish annotations, including semantic ambiguity, differences in sense granularity, historical language variation, and potential effects of regional or cultural interpretation.

- (3.3) ... La investigación ha demostrado que las **plantas** de colza, maíz y remolacha azucarera dañan la flora y la fauna más que los cultivos convencionales ...
*‘Research has shown that canola, corn, and sugar beet **plants** harm wildlife and flora more than conventional crops’*
- (3.4) Al dedicar la mezquita al ejercicio de nuestras sagradas ceremonias religiosas, es de presumir que, salvo las modificaciones necesarias á las exigencias del nuevo culto, en nada se tocaria á la **planta** y ornamentacion de la fábrica árabe, pues no solo este género estaba muy en uso y se conservaban templos cristianos pertenecientes á esta arquitectura...
*‘By dedicating the mosque to the performance of our sacred religious ceremonies, it may be assumed that, except for the modifications necessary to the requirements of the new worship, nothing would be changed in the building’s **planta** and ornamentation of the Arab structure, since not only was this style widely in use but Christian temples belonging to this architecture were preserved ...’*

Table 3.3 summarizes the annotation patterns associated with the selected examples. The table highlights substantial variability in proximity judgments across annotators from different regional backgrounds, including several cases of maximal disagreement (1 vs 4). These disagreements provide qualitative evidence that semantic interpretation may be influenced not only by lexical polysemy and contextual ambiguity, but also by differences in linguistic familiarity and the degree of semantic granularity applied during annotation.

In the examples shown in Examples 3.3 and 3.4, the modern usage refers to biological organisms, whereas the historical usage refers to the architectural layout of a building. The disagreement may stem from the coexistence of semantically distant senses of *planta*, as well as from the interpretive difficulty introduced by the historical and architectural context of the second usage. Additionally, the historical sentence belongs to an older period of Spanish and contains syntactic and stylistic constructions that may sound unfamiliar or unusual to contemporary speakers, potentially increasing annotation difficulty. The variability in scores across annotators from different regional backgrounds further suggests that semantic interpretation may be influenced by differences in linguistic familiarity and the level of sense granularity applied during annotation.

- (3.5) Además, por tercer año consecutivo, China produjo más electricidad eólica que

energía nuclear. Es por ello que el argumento de que China dependerá de **plantas** nucleares para producir energía ...

*‘Additionally, for the third consecutive year, China generated more wind power electricity than nuclear power. That is why the argument that China will rely on nuclear **plants** to produce energy ...’*

- (3.6) Y a las plantas a veces les resulta bastante difícil moverse porque puede haber ciudades u otras cosas de por medio. Así que si toda la vida humana depende de las **plantas**, ¿acaso no tendría sentido que intentáramos salvarlas?...

*‘And sometimes **plants** find it quite difficult to move because there may be cities or other things in the way. So if all human life depends on the plants, wouldn’t it make sense for us to try to save them?...’*

The pair of usages in Examples 3.5 and 3.6 illustrates another case of disagreement between annotators. The first usage refers to industrial facilities (*plantas nucleares*), whereas the second refers to biological organisms. The disagreement may stem from differences in how annotators interpreted the semantic relationship between these usages. Some annotators treated the industrial meaning as semantically distinct from the biological sense, assigning low proximity scores, while others may have perceived a broader semantic connection through the shared notion of an organized or functional structure. This example further illustrates how polysemy and differences in sense granularity can contribute to annotation variability in LSCD tasks.

3.6. Systems

We now summarize the baseline systems as well as the systems and resources used by the participating teams.

3.6.1. Baselines

For both phases we use five baselines:

baseline1 *Skip-Gram with Negative Sampling + Orthogonal Procrustes + Cosine Distance (SGNS+OP+CD)* This approach learned vector representations for each word (type-based) in two input corpora with a shallow neural language model [94, 95].²⁶ These were then aligned using Orthogonal Procrustes [12]. For phase 1, the method computed Graded Change as the cosine distance between old and modern vectors for all words in the vocabulary. This same value was used in the COMPARE subtask. In phase 2, binary predictions were computed by setting a threshold to the cosine distances, which was calculated as the sum between the mean and the standard deviation (std) of all these distances [97]. All words with values above the threshold were classified as *change*, and values below were classified as *no change*. This approach has shown high performance in several previous studies and shared tasks [97–100].

baseline2 *Normalized Log-Transformed Frequency Difference (FD)* For phase 1, this method calculated the frequency of each target word in each of the two corpora, normalized it by the logarithm of the total corpus frequency and then calculated absolute differences between these values as a measure of change. We submitted these values for the change

²⁶ As parameters we chose dim=100, window size=10, epochs=5, number of negative samples=5, subsampling threshold=0.001 [cf. 96].

graded and COMPARE subtasks. For phase 2, the method applied the same thresholding approach used in baseline1. For the sense loss subtask, it first verified that the target word presents change using the value of the change binary subtask. Then, if the differences were negative, the words were classified as loss = 1 and as loss = 0 otherwise. For sense gain the labeling is reversed.

baseline3 *Grammatical profiles* were generated from tagged and parsed corpora [101]. These profiles were essentially frequency vectors of various morphological and syntactic features (for example, *case = Nominative*, or *syntax role = subject*) for a given word in a given historical corpus. The cosine distance between the profile vectors of the same word for the two periods was used as an estimate of graded semantic change and COMPARE. Binary predictions were generated from ordered lists of graded scores for all target words by applying an offline change-point detection algorithm based on dynamic programming. The algorithm finds a point (a word) in an ordered list of scores, where the scores become significantly higher. This word and all words with score values above it were assigned the class “changed”. This baseline did not produce predictions for the sense loss and sense gain subtasks.²⁷

baseline4 *Minority class* This baseline produced predictions by labeling each word with the minority class label of the respective Binary Change score (change binary, loss, gain). This is label 1 (change) in all cases. It only applied to phase 2.

baseline5 *Random baseline* This baseline produced random predictions for all subtasks in both phases. For phase 1, we generated random values between 0 and 1 from a uniform distribution for all hidden target words and computed Spearman correlation with the gold scores. This process was repeated 100 times and we reported the average performance over all repetitions. For phase 2, we used a parallel procedure generating random labels $\in \{0, 1\}$ from a uniform distribution.²⁸

3.6.2. Participating systems

Below we present a summary of the methods developed by the participants:²⁹

HSE [102] This team participated with two different methods. The first consisted of fine-tuning BERT [16] on the lemmatized versions of the corpora in order to extract embeddings of the target words separately for each period, which are then clustered using K-means. Graded Change was estimated as the average cosine distance between all pairs of cluster centroids in the first and second periods. In order to estimate Binary Change, the Graded Change scores were thresholded by clustering them into two clusters.

The second method was based on grammatical profiles [101]. The frequency of morphological and syntactic categories for each target word in both corpora (parsed with UdPipe [103]) were counted and used as features in two time-specific vectors. Graded Change was measured by the cosine distance between these vectors, while Binary Change was measured by thresholding the graded scores.

GlossReader [104] This system fine-tuned the XLM-R multilingual language model [17] as part of a gloss-based Word Sense Disambiguation (WSD) system on a large English WSD dataset. It employed zero-shot cross-lingual transferability to build contextualized embed-

²⁷ All results for baseline3 were computed and submitted by Andrey Kutuzov using the code at <https://github.com/glnmario/semchange-profiling>.

²⁸ Baseline3, baseline4 and baseline5 were added after the shared task finished.

²⁹ The descriptions are based on the system description papers submitted by the participating teams, with the exception of Rombek who did not provide a paper but gave us a brief description by e-mail.

dings for Spanish data. The Graded Change score for each word was calculated as the Average Pairwise (Manhattan) Distance (APD) between the embeddings for (non-preprocessed) word usages in the old and new corpus. Binary changes were estimated by thresholding these scores. For the sense gain and sense loss subtasks the same predictions were reused.

UAlberta [105] This team applied different methods to the two subtasks. For Graded Change Discovery, they followed the design of CIRCE [99] and computed distances based on both static (type-based) and contextual (token-based) embeddings, with their relative weights tuned on the development set. For static embeddings, they used SGNS+OP+Euclidean Distance on the lemmatized versions of the corpora. For contextual embeddings, the XLM-R model was trained on the combined corpus (tokenized) to predict masked instances of the target words and Graded Change was measured using Euclidean APD. For Binary Change Detection, they framed the task as a WSD problem, creating sense frequency distributions for target words in the old and modern corpus with an end-to-end WSD system [106]. It was assumed that the word semantics has changed if: (1) a sense is observed in the modern corpus but not in the old corpus (or vice versa), or (2) the relative change for any sense exceeds a tuned threshold.

CoToHiLi [107] This team proposed a type-based embedding model combined with hand-crafted linguistic features. The system computed several features for every target word based on embedding distances between time periods and linguistic hand-crafted features, which were then weighted into an ensemble model to predict the final score. First, the system obtained word embeddings separately on the two corpora (tokenized) with the Continuous Bag-of-Words (CBOW) model [94, 95], which were then aligned to obtain a common embedding space. The alignment algorithms used were: supervised alignment using a seed word dictionary and a linear mapping method, a semi-supervised algorithm and unsupervised alignment based on adversarial training [108–111]. Finally, cosine distance between embeddings of the same word in different corpora was used as an indicator of graded semantic change. For the binary task, the system used thresholding the graded scores.

DeepMistake [112] This team employed a WiC model, i.e., a model designed to determine if a particular word has the same meaning in two given contexts. In essence, they attempted to directly apply a model trained on a related task to our problem. The WiC model was initially trained by fine-tuning the XLM-R language model on the Multilingual and Cross-lingual Word-in-Context (MLC-WiC) dataset [113]. Subsequently, it was further fine-tuned on the provided annotations for the development set in this shared task and on the Spanish portion of the multi-language XL-WSD dataset [114]. Graded Change was measured similarly to APD by averaging same-sense probabilities between embeddings for usages (no preprocessing) from different time periods. For the change binary subtask, the authors applied thresholding to the Graded Change scores, for the sense gain and sense loss subtasks the same predictions were reused.

They also experimented with clustering by representing word usages and their same-sense probabilities in a weighted undirected graph, which was then clustered with Correlation Clustering. Graded Change was measured with JSD, while Binary Change was measured with the Binary Change score definition from Section 3.4.4.

BOS [115] The system described by this team was based on generating lexical substitutes that describe old and new senses of a given word. These were generated using the XLM-R masked language model. For polysemous words, lexical substitutes depended on the meaning expressed in a particular context. For each target word, usages were sampled from both corpora, lemmatized and used to generate lexical substitutes. Next, two sets of vectors were

built for old and new usages where each usage is represented by a vector of the probabilities of its substitutes. For Graded Change the Cosine APD between old and new vectors was computed, while for Binary Change a threshold was applied to this score. The authors also proposed three different approaches based on pairwise distances for the sense gain and loss subtasks.

Rombek This system adapted ideas from the WSI task. Lexical substitutes were generated in the same way as with the BOS system (see above) and arranged in a matrix. Agglomerative clustering was then applied to each target word to obtain clusters with candidate senses. JSD was applied between clusters to obtain Graded Change estimates. Thresholding was applied to produce binary predictions.³⁰

3.6.3. Summary

Most systems were based on three main components: (i) a semantic representation of words or word usages as vectors, (ii) an aggregation method over vectors, and (iii) a change measure. Type-based systems usually employed an additional alignment step over semantic representations. Also, the preprocessing of data was crucial for the performance of contextualized embeddings [14].

Preprocessing Some teams only used the tokenized version of the shared task dataset (CoToHiLi, UAlberta), while other teams only used the lemmatized version (UAlberta, BOS, HSE). One team varied the preprocessings with systems (UAlberta): lemmatization for type-based embeddings and tokenization, lemmatization and POS-tagging for the WSD system. Two teams did not use any sort of preprocessing (GlossReader, DeepMistake), while two teams used substitution with dynamic patterns (e.g. *<mask> (y [target]), [target] (por ejemplo <mask>))* for their lexical substitution models (BOS, Rombek).

Semantic representations Most systems used token-based contextualized embeddings such as BERT (HSE) and XLM-R (DeepMistake, GlossReader, Rombek, UAlberta, BOS). Some teams further fine-tuned these embeddings on Language Modeling, WSD or WSI/WiC tasks. One team (DeepMistake) fine-tuned on the semantic proximity judgments from the published development data. Only three teams used type-based semantic representations including SGNS (UAlberta), CBOW (CoToHiLi) and Grammatical Profiling (HSE).

Vector aggregation Participating teams used different approaches to aggregate vectors into more abstract semantic representations. A common strategy was to model the COMPARE score by computing APD between vectors from different time periods (DeepMistake, GlossReader, UAlberta, BOS). This strategy has shown to perform well in various previous studies and shared tasks [14, 19, 84, 85]. Another strategy was to cluster the vectors (HSE, Rombek, DeepMistake). Clustering algorithms used are: Agglomerative Clustering (Rombek), K-means (HSE) and Correlation Clustering (DeepMistake). One system used a WSD system to assign cluster labels (UAlberta).

Change Measure For Graded Change most teams using contextualized embeddings directly relied on APD scores as described above. They used different distance measures such as: Cosine (BOS), Euclidean (UAlberta) and Manhattan (GlossReader) distances. One team averaged same-sense probabilities (DeepMistake). The teams relying on clustering mostly used the JSD to measure Graded Change (Rombek, DeepMistake). One team instead used cosine distance between cluster centroids (HSE). The teams relying on type-based representations used either Cosine (CoToHiLi, HSE) or Euclidean distance (UAlberta). For Binary

³⁰ This team did not submit a paper to the shared task.

Change most teams relied on thresholding the graded predictions (DeepMistake, GlossReader, Rombek, HSE, CoToHiLi, BOS). This strategy has shown high performance in several previous studies and shared tasks [4, 85, 97]. Two teams using a clustering approach measured Binary Change by applying exactly the definition from the annotation process (DeepMistake) or a similar definition (UAlberta).

3.7. Results

The results shown in Tables 3.4, 3.5 and 3.6 correspond to the best submissions per subtask.³¹

Graded Change Discovery As shown in Table 3.4, **GlossReader** and **DeepMistake** obtained first and second place in the main task of evaluation phase 1, while **HSE** came third.³² These were the only teams that managed to outperform baseline1 (SGNS+OP+CD) and baseline3 (Grammatical Profiles). The three winning systems were based on fine-tuned versions of contextualized embeddings with average vector aggregation (GlossReader, DeepMistake) or clustering (HSE). Interestingly, the top two systems did not model the JSD between cluster distributions (as done on the annotation to derive gold scores), but instead model the COMPARE score (with APD). We discuss this observation further in Subsection 3.7.1.

Table 3.4: Summary of system performance in phase 1. Teams are ranked according to SPR score for the Graded Change subtask in decreasing order. The values corresponding to the three best systems are highlighted in bold type.

Task		Change graded	COMPARE
#	Team name	SPR	SPR
1	GlossReader	0.735 (1)	0.842 (1)
2	DeepMistake	0.702 (2)	0.829 (2)
3	HSE	0.553 (3)	0.558 (4)
4	baseline1	0.543 (4)	0.561 (3)
5	baseline3	0.508 (5)	0.459 (5)
6	Rombek	0.497 (6)	0.456 (6)
7	CoToHiLi	0.282 (7)	–
8	baseline2	0.092 (8)	0.088 (7)
9	baseline5	0.064 (9)	-0.072 (8)
10	BOS	-0.125 (10)	-0.129 (9)

COMPARE Discovery GlossReader and DeepMistake also reached the first and second place on the COMPARE task in evaluation phase 1. This is not surprising, because they actually modeled the COMPARE score with APD. Consequently, also the correlation was considerably higher than with Graded Change (e.g. $\rho = 0.842$ vs. 0.735). Baseline1 took the third place.

Binary Change Detection For Phase 2 (Tables 3.5 and 3.6), again **GlossReader** performed best, this time followed by **UAlberta** and **Rombek**. Interestingly, with the

³¹In the case of HSE who used two different systems, the displayed results correspond to the token-based system.

³²Since not all users reported a team name on Codalab, some leaderboard entries are filled with usernames.

Table 3.5: Summary of the results of Phase 2 for subtasks Graded Change, COMPARE and Binary Change. Teams are ranked according to F1 score for subtask Change binary in decreasing order. The values corresponding to the three best systems are highlighted in bold type.

Task		Change binary			Change graded	COMPARE
#	Team name	F1	P	R	SPR	SPR
1	GlossReader	0.716 (1)	0.615 (3)	0.857 (3)	0.735 (1)	0.842 (1)
2	UAlberta	0.709 (2)	0.549 (7)	1.000 (1)	–	–
3	Rombek	0.687 (3)	0.590 (4)	0.821 (4)	0.535 (5)	0.546 (5)
4	BOS	0.658 (4)	0.510 (8)	0.929 (2)	0.209 (8)	0.163 (7)
5	DeepMistake	0.655 (5)	0.633 (2)	0.679 (6)	0.676 (2)	0.821 (2)
6	CoToHiLi	0.636 (6)	0.553 (6)	0.750 (5)	0.282 (7)	–
7	baseline4	0.636 (6)	0.467 (11)	1.0 (1)	–	–
8	HSE	0.586 (7)	0.567 (5)	0.607 (7)	0.553 (3)	0.558 (4)
9	baseline3	0.548 (8)	0.500 (9)	0.607 (7)	0.373 (6)	0.423 (6)
10	baseline1	0.537 (9)	0.846 (1)	0.393 (9)	0.543 (4)	0.561 (3)
11	baseline5	0.508 (10)	0.484 (10)	0.536 (8)	0.064 (10)	-0.072 (9)
12	baseline2	0.222 (11)	0.500 (9)	0.143 (10)	0.092 (9)	0.088 (8)

exception of GlossReader the systems used in Phase 1 did not obtain a good performance in Phase 2. However, participants managed to outperform all baselines with the exception of HSE not outperforming baseline4 (minority class). Two out of the winning systems used thresholding (GlossReader, Rombek), i.e., they modeled the COMPARE score or the JSD and then thresholded these scores to obtain Binary Change predictions. From these teams only UAlberta inferred sense clusters. Hence, here we saw again what we saw for phase 1: the top-performing teams were often not modeling the annotation procedure.

Sense Gain/Loss Detection The top performance for sense gain ($F1 = 0.591$) was clearly lower than for Binary Change, while for loss the top performance ($F1 = 0.688$) approaches the one for Binary Change. The best results for sense gain were obtained by **DeepMistake**, followed by **BOS** and **GlossReader**. In the sense loss subtask, **GlossReader** obtained the best performance, followed by **Rombek** and **BOS**. GlossReader and DeepMistake submitted the same results to both subtasks as for Binary Change Detection implicitly assuming that gain and loss always occur together. In this way, they mostly outperformed Rombek and BOS who tried a more principled approach.

Graded Change/COMPARE Detection The top performance for these tasks was the same in evaluation phase 1 and 2 ($\rho = 0.735$ and 0.842). Some teams had the same results in both phases (GlossReader, HSE, CoToHiLi) and thus likely submitted the same predictions. Two teams improved their results (Rombek, BOS), while one team had lower results (DeepMistake). We are unsure about the impact of the published target words and their usages on these results, as teams did not consistently report whether they used this information in phase 2.

3.7.1. Discussion

The Graded Change Discovery subtask was solved with a rather high performance by the winning team ($\rho = 0.735$). This is comparable to the top performance in SemEval ($\rho = 0.725$ for DE) obtained with type-based embeddings. The COMPARE Discovery subtask was

Table 3.6: Summary of the results of Phase 2 for subtasks Sense loss and Sense gain. The values corresponding to the three best systems are highlighted in bold type.

Task		Sense gain			Sense loss		
#	Team name	F1	P	R	F1	P	R
1	GlossReader	0.511 (3)	0.333 (5)	0.929 (2)	0.688 (1)	0.564 (2)	0.880 (2)
2	DeepMistake	0.591 (1)	0.433 (1)	0.929 (2)	0.582 (5)	0.533 (3)	0.640 (4)
3	HSE	0.250 (8)	0.192 (9)	0.357 (5)	0.364 (7)	0.421 (5)	0.320 (5)
4	baseline1	–	–	–	–	–	–
5	Rombek	0.50 (4)	0.409 (2)	0.643 (4)	0.681 (2)	0.727 (1)	0.640 (4)
6	baseline3	–	–	–	–	–	–
7	BOS	0.520 (2)	0.361 (4)	0.929 (2)	0.610 (3)	0.529 (4)	0.720 (3)
8	baseline2	0.211 (9)	0.400 (3)	0.143 (6)	0 (8)	0 (8)	0 (7)
9	UAlberta	0 (10)	0 (10)	0 (7)	0 (8)	0 (8)	0 (7)
10	CoToHiLi	0.462 (5)	0.316 (6)	0.857 (3)	0 (8)	0 (8)	0 (7)
11	baseline4	0.378 (6)	0.23 (8)	1.0 (1)	0.588 (4)	0.416 (6)	1.0 (1)
12	baseline5	0.333 (7)	0.313 (7)	0.357 (5)	0.367 (6)	0.375 (7)	0.36 (6)

solved with even higher performance ($\rho = 0.842$). This is comparable to the top performance in RuShiftEval ($\rho = 0.822$). However, the results in our shared task were obtained under harder conditions, i.e., for a large number of uncleaned target words (Discovery).³³ This suggests that, as far as Graded Change is concerned, LSCD systems are applicable to solve real-world problems and may be useful in historical semantics or lexicography. However, the more relevant task for these fields is Binary Change Detection/Discovery [86]. The results for Binary Change Detection were lower ($F1 = 0.716$), but still clearly higher than the best baseline (0.636). Results in SemEval were mixed, but mostly not higher than $F1 = 0.7$ (DE), while results in DIACR-Ita were high with an accuracy of 0.94, which was, however, obtained with a different metric and on a very small and strongly preselected set of target words. A future challenge will thus be to improve performance on the binary task.

Our shared task was clearly dominated by token-based systems. Out of seven participants only two used a (standalone) type-based system which also performed much worse than the winning teams (CoToHiLi, HSE).³⁴ Also, our type-based baseline1 was clearly outperformed by a number of token-based systems (three in phase 1 and six in phase 2). This confirms the tendency observed in RuShiftEval where token-based systems outperformed type-based ones on LSCD. Before that, in SemEval and DIACR-Ita the type-based systems had dominated. Potential reasons for this switch are the understanding of biases in contextualized embeddings [14], their optimization through fine-tuning [19, 116] and the optimization of vector aggregation methods [14, 19, 84].

In our task, we saw clustering methods amongst the best-performing systems (HSE, UAlberta) for the first time. This is an important development, because the current top-performing system (GlossReader), as well as many other systems not relying on clustering, did not model the target word annotation procedure (cf. Subsection 3.4.4). Instead, it ex-

³³ We assume that the performance of participating systems obtained on the hidden target words generalizes roughly to the full set of public target words as the sample was taken largely random.

³⁴ The result reported by the HSE team in the leaderboard corresponds to the first method described in Section 3.6.2.

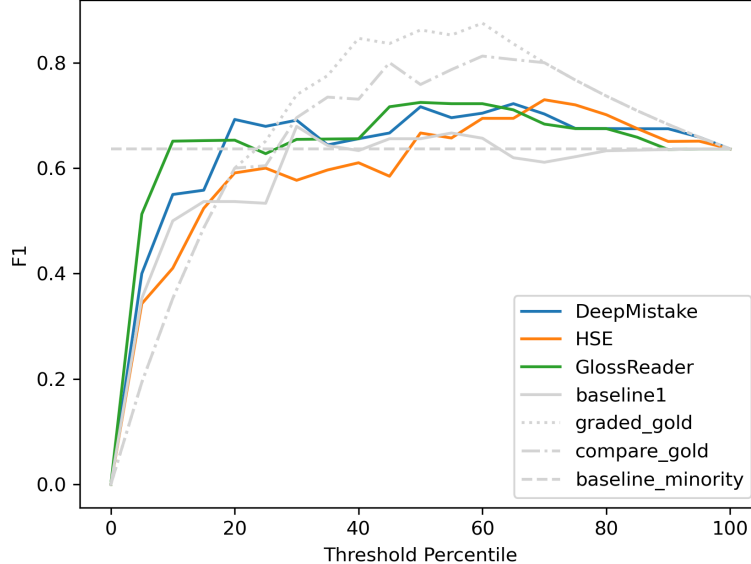


Figure 3.2: F1 scores over binarization thresholds based on percentiles on submitted Graded Change predictions for top four teams in evaluation phase 1.

exploited correlations between the COMPARE score and JSD as well as Binary Change. These scores are known to correlate strongly in current LSCD datasets [3], including ours. The correlation between gold (negated) COMPARE and JSD scores in our dataset is 0.92, while it is 0.69 for gold (negated) COMPARE and Binary Change. This means that modeling the COMPARE score is a good predictor for Graded as well as Binary Change. However, this also means that, the current best-performing systems have a clear upper bound on their potential to solve LSCD tasks (where this upper bound is higher for Graded than for Binary Change). Hence, if we want to break through this upper bound in the future, we need to develop or improve other system types possibly relying on clustering to model the annotation procedure.³⁵

In order to see how far the current approach of thresholding COMPARE/JSD/graded scores carries, we compared performance of the top three systems in evaluation phase 1 across binarization thresholds in Figure 3.2. As we can see, the three systems had a similar maximum performance of roughly $F1 = 0.72$ around a binarization threshold of 50 – 70 %.³⁶ At 100 % they all converged to the minority class baseline (all target words labeled as 1). The upper bound on this approach was given by the maximum performance of the gold JSD (graded_gold) and the gold COMPARE score (compare_gold). These upper bounds were 0.88 and 0.81 respectively. This means that perfectly modeling the COMPARE or even the JSD score can reach high but never perfect performance on Binary Change.

³⁵ Homskiy and Arefyev [112] had promising results with applying the clustering framework used in the annotated data and semantic proximity graphs derived from fine-tuned contextualized embeddings.

³⁶ Interestingly, HSE here obtained maximum performance amongst all systems (0.73), much higher than their submission in evaluation phase 2. A similar observation holds for our baseline1. This shows how crucial threshold selection is in this approach.

3.8. Conclusion

We conducted the first shared task on semantic change discovery and detection in Spanish. We manually annotated 100 Spanish words for semantic change between two corpora, an old one covering the period between 1810 and 1906, and a modern one covering the years between 1994 and 2020. The discovery part of our task imposed several computational challenges for participants, as it required calculating semantic change scores for all words in the vocabulary.

We received predictions from six teams in phase 1 and seven teams in phase 2. Participants applied systems using static and contextualized word embeddings in combination with various fine-tuning procedures, vector aggregation methods and change measures. Graded Change Discovery was solved with high performance while Binary Change Detection still remains far from being solved. The most successful method winning both main tasks is a system fine-tuning contextualized multilingual XML-R embeddings on WSD data, aggregating vectors into cross-corpus pairs and measuring change as the average of their distances, or a binarization of these values. However, we showed that this approach has a clear upper bound which will not allow to solve the tasks completely reliably in the future. Another interesting result from our task was that clustering approaches are amongst the winning teams for the first time.

We hope that this shared task will help pave the way for future research in the discovery and detection of semantic lexical changes for the Spanish language, and that our data can be used in the future for the proposal of novel ideas and techniques.

Chapter 4

Optimizing WSI for LSCD

This chapter builds on the conceptual and methodological foundations introduced in Chapter 2, particularly the relationship between WiC, WSI, and LSCD, as well as token-based and WiC-based modeling approaches (see Section 2.5.3). In Chapter 3, the analysis of the Spanish LSCD shared task revealed that several of the strongest-performing systems relied on clustering-based and sense-oriented representations. These observations suggest that approaches which more closely model the sense-level structure underlying the LSCD annotation process may provide advantages for semantic change detection. Furthermore, the results of Chapter 3 indicated that aggregation-based approaches may be approaching a performance ceiling, motivating the exploration of clustering architectures as an alternative means of improving LSCD performance. Motivated by these findings, the present chapter investigates whether improving the WSI component of LSCD pipelines leads to improved semantic change detection performance. More specifically, we examine whether optimization performed at the level of WSI transfers to LSCD and whether a single configuration can effectively support both tasks.

4.1. Introduction

To investigate whether optimization at the level of WSI transfers to LSCD, we adopt the LSCD pipeline proposed by Davletov et al. [117], which combines WiC-based semantic proximity predictions with graph clustering techniques to generate sense inventories and semantic change scores. Semantic proximity judgments produced by WiC models are used to construct weighted graphs over word usages, which are subsequently clustered to induce word senses. The resulting clusters are evaluated both as solutions to the WSI task and as the basis for LSCD predictions. We evaluate this approach on German, Spanish, and English datasets, allowing us to analyze the relationship between WSI and LSCD across different languages and annotation settings.

The German DWUG DE Sense v1.0.0³⁷ dataset (see Section 2.4) provides both gold-standard sense annotations and graded semantic change labels, making it possible to evaluate the complete pipeline at the WSI and LSCD levels. In addition, experiments are conducted on the English DWUG EN v3.0.0³⁸ dataset and the Spanish DWUG ES v4.0.2³⁹ dataset, introduced in Chapter 2 and Chapter 3, respectively. While these datasets were originally

³⁷ <https://zenodo.org/records/8197553>

³⁸ <https://zenodo.org/records/14028531>

³⁹ <https://zenodo.org/records/14891659>

created for semantic change detection, they provide reference sense clusters derived from manual annotation, which are used for WSI evaluation, as well as graded semantic change scores for LSCD evaluation. Consequently, all three datasets allow us to examine how clustering quality relates to semantic change detection performance under a common experimental framework.

The experiments focus on graded semantic change detection, as this has become the predominant evaluation setting in recent LSCD research. Consequently, most state-of-the-art systems and benchmark results are reported for graded semantic change, making it the most appropriate setting for comparing the proposed approach with existing methods. In addition, graded semantic change provides a more fine-grained characterization of lexical change than binary classification and has been shown to be less sensitive to sampling variability [3, p. 66], making it a more robust evaluation setting.

To provide a competitive point of comparison, the experiments also include the APD method as a reference baseline. The results presented in Chapter 3 showed that APD consistently achieves strong performance on graded semantic change detection, establishing a highly competitive benchmark for this task. At the same time, the findings from Chapter 3 suggest that aggregation-based approaches have an upper bound in their ability to model graded semantic change. Since these methods estimate semantic change directly from contextualized representations without explicitly modeling word senses, further performance improvements may be increasingly difficult to obtain within this paradigm.

In contrast, clustering-based approaches induce sense inventories prior to computing semantic change scores, thereby modeling semantic change at the sense level and more closely reflecting the annotation process underlying semantic change datasets. This characteristic suggests additional opportunities for improvement and provides a principled framework for investigating the relationship between WSI and LSCD, which constitutes the primary objective of this chapter. Consequently, APD is included as a strong reference baseline, while the main focus of the analysis is on understanding the potential of sense-aware clustering methods for semantic change detection.

4.2. Background

Chapter 2 reviewed the main datasets, tasks, and computational approaches used in LSCD research. This chapter focuses specifically on approaches that model semantic change through sense induction and clustering.

Several LSCD systems rely on clustering word usages into sense groups and estimating semantic change from shifts in the resulting sense-frequency distributions. Examples include the approaches proposed by Giulianelli et al. [48], Laicher et al. [14], and Davletov et al. [117]. In these systems, WSI effectively functions as an intermediate step for LSCD, since the induced sense inventories are used to derive semantic change scores. However, despite this close relationship, previous work has not systematically examined whether improvements in WSI performance translate into improved LSCD performance.

4.3. Datasets

The experiments in this chapter are conducted on three datasets: DWUG DE Sense, the Spanish LSCD dataset introduced in Chapter 3, and the English LSCD dataset described in Chapter 2. All three datasets provide reference sense clusters derived from manual annotation

together with graded semantic change scores, enabling evaluation at both the WSI and LSCD levels.

DWUG DE Sense contains 24 German target words sampled from two historical periods (1800–1899 and 1946–1990) and provides manually assigned sense labels for individual usages. Following the methodology described in Section 2.4, two cleaned versions of the dataset are considered. The maj_2 version retains usages for which at least two annotators agreed on the assigned sense label, whereas the maj_3 version retains only usages with unanimous agreement. The resulting sense labels are treated as gold-standard clusters and are used to derive graded semantic change scores following Schlechtweg et al. [4]. Detailed statistics of both cleaned versions are reported in Table 2.8.

For the Spanish and English datasets, reference sense clusters derived from manual annotation are used for WSI evaluation, while the corresponding graded semantic change scores serve as gold-standard LSCD labels. Since these datasets do not require the cleaning procedure applied to DWUG DE Sense, the original cluster annotations are used throughout the experiments.

4.4. Tasks and Evaluation

The objective of this chapter is to investigate the relationship between WSI and LSCD within a unified clustering-based framework. As discussed in Section 4.1, the framework first induces sense inventories from semantic proximity predictions and subsequently derives semantic change scores from the resulting sense distributions. This naturally gives rise to two complementary evaluation tasks: WSI and LSCD.

The experiments are conducted on the German, Spanish, and English datasets described in previous chapters. All three datasets provide reference sense clusters derived from manual annotation, which allows the evaluation of induced sense inventories through WSI. In addition, all datasets provide graded semantic change scores, enabling evaluation at the LSCD level. This setup makes it possible to analyze how improvements in clustering quality affect semantic change detection performance across different languages and annotation settings.

For the WSI task, clustering quality is evaluated using the ARI [118], which measures the agreement between predicted and reference cluster assignments. The reference clusters correspond to the manually annotated sense inventories available for each dataset.

For the LSCD task, graded semantic change scores are derived from the induced sense frequency distributions across time periods. System predictions are then evaluated using Spearman’s rank correlation coefficient [119] between predicted and gold semantic change rankings. This evaluation directly measures the ability of a system to capture differences in semantic change across target words.

By evaluating both WSI and LSCD within the same experimental framework, we can assess the extent to which improvements in sense induction transfer to semantic change detection, which constitutes the central research question of this chapter.

4.5. Models

This section provides a detailed description of the model architectures that we employ in our experiments to solve the tasks defined above. We present each component part of the architecture and provide a comprehensive explanation of its role in the overall system. By doing so, we aim to provide a clear understanding of how the system operates and how it

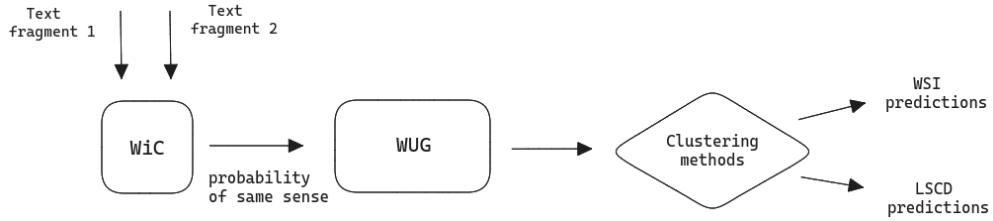


Figure 4.1: Overview of the architecture used to generate predictions for the WSI and LSCD tasks.

was designed to address the specific research questions of interest.

Our architecture is based on state-of-the-art models applying components used in one of the recent LSCD shared tasks [4, 49, 50]: First, we use a WiC model [112], as described in Section 4.5.1, to generate predictions for pairs of usages that assess the relatedness of a target word based on its contextual information. Second, the WiC model predictions are used to construct a WUG [34], where nodes represent usages and edges represent the probability that two usages are similar. Finally, clustering algorithms are applied to the WUG, producing clusters that correspond to the different senses of the target word.⁴⁰ Overall our pipeline is based on three main steps (see Figure 4.1):

1. apply WiC model to score all pairs of word usages of each target word separately,
2. build the WUG, i.e., a graph with nodes corresponding to different usages of the same target word and edges weighted with WiC scores for the corresponding pairs of usages,
3. apply clustering algorithms to word usages to potentially establish meaningful clusters representing word senses.

From the full cluster graph, we can construct two time-specific subgraphs with two time-specific frequency cluster distributions, from which we calculate graded change predictions with Jensen Shannon distance (see Section 3.4). Find an example of a clustered WUG in Figure 3.1. It shows the full graph G on the left and the time-specific graphs G_1 (old) and G_2 (new) with their respective cluster frequency distribution.

We describe the three parts of our pipeline with more detail in the subsections below.⁴¹

4.5.1. Word-in-Context

The WiC task (see Section 2.3.1) constitutes the first step of our pipeline. To solve this task we employ several *DeepMistake* models [19, 112], which had previously shown SOTA or

⁴⁰ We also tried experimenting with substitution-based state-of-the-art models, find more details in Appendix D.

⁴¹ The architecture of our pipeline implies that there are considerable differences between the derivation of the gold clusters (Section 2.4.3) and model clusters: Gold clusters are derived from use-sense judgments not requiring clustering as we filter out all ambiguous judgments. Model clusters are derived from (pairwise) use-use similarity predictions from a WiC model with an additional clustering algorithm. LSCD is then indeed measured in the same way from gold and model clusters. Hence, even if the WiC model were the ideal approximator of human (WiC/use-use) judgements, then model WSI and LSCD results are not bound to be perfect because the ground truth was not constructed with a clustering of use-use judgments, but from use-sense judgments. Of course, there is a certain correspondence between human use-use and use-sense judgments, but this correspondence can be ambiguous.

near-SOTA results in two shared tasks [18, 50]. These models were trained as probabilistic binary classifiers predicting the probability that two given occurrences of some target word have the same sense. We use this probability as a measure of relatedness between word usages, with higher probability corresponding to more similar usages. All models we employ have the same architecture, which was shown to perform best in the previous studies. The difference between the models lies in the data they were trained on. Since none of the DeepMistake models was trained on any labeled WiC data in German, we take the models that had previously shown the best results in the LSCD shared tasks in Russian and Spanish, which were trained on a mixture of diachronic and synchronic WiC data in these languages. For comparison, we additionally employ the models trained on a synchronic WiC dataset in 5 languages, and also on subsets in English or in Russian only. In total, 6 WiC models were involved in the process of hyperparameter selection.

4.5.1.1. The architecture of DeepMistake

Given a pair of usages of some target word, these usages are concatenated and encoded by the XLM-R large backbone [17]. Specifically, the backbone gets the input in the following format: `<s>usage1</s>usage2</s>`.⁴² For each of two occurrences of the target word the outputs from the last Transformer layer at the positions of subwords of the target word are averaged (i.e. mean pooling over subwords of the target word is done). Thus, we get two contextualized embeddings for two occurrences of the target word.

Then two contextualized embeddings are combined and passed to the classification head. Following [112], we concatenate the L1-distance and the dot product between the normalized embeddings: $(\|\bar{x} - \bar{y}\|_1, \langle \bar{x}, \bar{y} \rangle)$. The classification head consists of a batch normalization and a linear layer followed by the softmax over two classes. Models were trained on several WiC datasets using the CE loss. Both the XLM-R backbone and the classification head were fine-tuned.

Taking into account the symmetric nature of the same-sense relation, i.e. whether two occurrences of some target word have the same sense or different senses should not depend on the order in which these occurrences appear, DeepMistake employs both training time and test time augmentation. During training, for each `<s>usage1</s>usage2</s>` a symmetric example `<s>usage2</s>usage1</s>` was automatically generated, thus increasing the number of training examples. During inference for each edge in a WUG two (ordered) pairs of usages are generated and scored by DeepMistake, then the scores are averaged to get the final edge weight.

4.5.1.2. WiC training and model variants

In this work we experimented with several DeepMistake models differing in how they were trained and which data they were trained on. The training examples were taken from several source datasets which are summarized in table 4.1.

1. **MCL-WiC** is a synchronic Multilingual and Cross-Lingual Word-in-Context dataset [113]. For training the authors of DeepMistake employed the original training set in English, 70%⁴³ of each development set in French, Russian, Arabic, and Chinese, and also all the examples from the trial sets including few cross-lingual examples.

⁴² When an input is longer than $L = 500$ subwords, we shorten it to be exactly this length. Specifically, we shorten left and right contexts of each usage that are longer than $L/4$ subwords proportionally to their extra length and leave those that are shorter than $L/4$ intact.

⁴³ The rest 30% were used as validation data for early stopping.

Table 4.1: Training data of the employed DeepMistake models. Each example is a pair of usages of the same word in the same language, except for en-nen* which are cross-lingual examples with one usage in English and another in French, Russian, Arabic or Chinese.

Data source	Languages	size	#targets	Avg. len.
MCL-WiC	en-en	8008	3728	48
	ru-ru	708	352	41
	fr-fr	708	352	46
	ar-ar	708	354	45
	zh-zh	708	342	-
	en-nen*	32	16	51
RuSemShift	ru-ru	3898	70	51
Spanish DWUG	es-es	5443	15	167
Spanish XL-WSD	es-es	8260	310	98

2. **RuSemShift** is a diachronic dataset proposed in [49] and consisting of pairs of usages in Russian from pre-Soviet, Soviet and post-Soviet epochs. The original labels are values from 1 to 4. For training DeepMistake all pairs with labels 3 or larger were treated as positive examples and the rest as negative ones. Compared to the Russian part of MCL-WiC, RuSemShift contains 5x more examples but 5x less different target words.
3. **Spanish DWUG** is a diachronic dataset in Spanish from [50] containing pairwise human annotations from 1 to 4 similarly to RuSemShift. For training DeepMistake only pairs with the gold scores of 1 and 4 were employed as negative and positive examples correspondingly, all other pairs were filtered out. This kind of filtering was shown to be beneficial for the model performance [112].
4. **Spanish XL-WSD** is the Spanish part of the synchronic XL-WSD Word Sense Disambiguation dataset [114], which was converted to the WiC format. Specifically, the Spanish development and test subsets of this WSD dataset were taken, and all pairs of examples sharing the same target word lemma were generated. Those pairs having identical sense labels for both usages of the target word were labeled as positive, and the rest as negative. Compared to the Spanish DWUG, this dataset suggests much higher diversity of target words.

We experimented with the following DeepMistake models.

1. **MCL→RU** (or “mean+dist_l1ndotn-hs0 on $MCL_{CE}^{nen-acc} \rightarrow RSS_{CE}^{dev2-sentSpear}$ ” in the notation of [19]) is the DeepMistake model that has shown the best results on the Russian RuShiftEval-2021 dataset⁴⁴. It was fine-tuned in two stages, first on the multilingual MCL-WiC dataset and then on the diachronic RuSemShift dataset in Russian.
2. **MCL→ES** (a.k.a. “MCL→DWUG_es^{bin2}+XL-WSD” in [112]) is the DeepMistake model that has achieved the best Spearman’s correlation with the gold COMPARE

⁴⁴ This is an improved version of the best model from [19], it has achieved the average Spearman’s correlation with the gold COMPARE scores across all pairs of time periods equal to 0.85, thus, outperforming the winner of the RuShiftEval-2021 competition. See <https://github.com/Daniil153/DeepMistake/> for all metrics of this model.

Table 4.2: Hyperparameters to build a WUG.

Parameter	Values
binarize	[True, False]
quantile	[1,...,10]
WiC model	[enMCL, ruMCL, MCL→RU MCL, MCL→ES, ALL]
fill_diagonal	[True, False]
use_disconnected_edges	[True, False]

scores among all participants of the Spanish LSCDiscovery-2022 shared task. The first stage of fine-tuning is identical to the previous model, but the second stage was performed on both synchronic and diachronic data from Spanish DWUG and Spanish XL-WSD.

3. **ALL** (a.k.a. “MCL+RSS+DWUG_es^{bin2}_{ALL}+XL-WSD” in [112]) is the model that has achieved the best Spearman’s correlation with the gold JSD scores among all Deep-Mistake models in LSCDiscovery-2022. It was fine-tuned on all training data listed in table 4.1 in a single stage.
4. **MCL** (a.k.a. “MCL-WiC(CE)” in [19]) is fine-tuned in one stage on all the MCL-WiC data in 5 languages.
5. **enMCL** (a.k.a. “MCL-WiC_train-en-en(CE)” in [19]) is fine-tuned in one stage on the English part of MCL-WiC only.
6. **ruMCL** (a.k.a. “MCL-WiC_ru-ru(CE)” in [19]) is fine-tuned in one stage on the examples from MCL-WiC in Russian only. Additional 1000 examples from the test set in Russian were added, resulting in 1708 examples in total.

4.5.2. Building the Word Usage Graph

With the predictions of the WiC model, a complete graph is constructed in which the nodes represent usages and the weighted edges denote the semantic proximity between pairs of usages. The resulting graph serves as a structural representation of the dataset that allows clustering algorithms to identify different senses of a target word based on the weighted edges. We explore different mechanisms for constructing the graph, which can be expressed through the following hyperparameters:

- **threshold**: All edges with weights less than this hyperparameter are dropped. During grid search, it acquires the values of the 10%, 20%, ..., and 90% percentile of all edge weights, along with a default value of 0.5.
- **binarize**: When set to true, all edges are assigned a weight of 1 (subsequent to dropping the edges with weight less than **threshold**).

Table 4.3: Clustering method hyperparameters.

Clustering method	Parameter	Values
Chinese Whispers	weighting	[lin, top, log]
Correlation Clustering	threshold_cc	0.5
WSBM	distribution	[binomial, poisson, geometric, normal, exponential]
Spectral Clustering	n_c	[Silhouette, Calinski-Harabasz, Eigengap]

- **fill_diagonal**: This parameter controls if the main diagonal of the adjacency matrix is filled with 1.0 or 0.0 as value.
- **use_disconnected_edges**: When set to true, edges with a weight less than the cutoff threshold will be kept as part of the network with a weight of 0.0. For Chinese Whispers and Spectral Clustering (see Section 4.5.3), this value will default to True, for WSBM and Correlation Clustering we will only keep them in case of binarization.⁴⁵

We additionally scale the weights of each graph with a MinMaxScaler fitted on the full set of semantic proximity predictions of the respective WiC model because scores are tightly center around 0.5. Table 4.2 shows a summary of the hyperparameters needed to build a WUG.

4.5.3. Graph Clustering

In the final step of our pipeline, we apply graph clustering methods to our WUG to obtain word senses in an unsupervised fashion. Below, we describe all the graph clustering methods that we consider in this study. Additionally, Table 4.3 provides a comprehensive overview of the clustering methods along with the parameter ranges applicable to each model.

4.5.3.1. Chinese Whispers

Chinese Whispers (CW) is an efficient, randomized clustering algorithm with a time complexity linear with respect to the number of edges [83]. The algorithm first assigns all nodes to different clusters. Then the nodes are processed in randomized order for a small number of iterations (we set this hyperparameter to 20) and are assigned to the strongest cluster in the local neighborhood, i.e., the cluster whose sum of edge weights to the current node is maximal. The calculation of edge weights is controlled by the **weighting** hyperparameter, which takes three different values:

1. **lin**: This calculates the weight of an edge between two nodes in a graph using linear weighting, which is the edge weight divided by the degree of the destination node.
2. **log**: This computes the weight of an edge between two nodes in a graph using logarithm weighting, which is the edge weight divided by the logarithm of the degree of the destination node.

⁴⁵ 0-weighted edges are removed for WSBM and Correlation Clustering (without binarization) because otherwise wrongly influence the clustering algorithm.

3. top: This keeps edge weight as is.

We use the implementation provided by Ustalov et al. [120].

4.5.3.2. Correlation Clustering

We use a variation of Correlation Clustering (CC) [92] in our experiments (see Section 2.5.2.2). This method has, to our knowledge, not previously been employed for the WSI task. However, it has been used extensively in the LSCD context to cluster human annotations [34, 50–52, 85, 93, 121], but also to cluster model predictions [112]. We set the hyperparameter `threshold_cc`, which controls the separation of edge weights into positive and negative, to 0.5. The maximum number of senses is set to 10, while the maximum number of simulated annealing attempts and iterations are set to 2000 and 50000, respectively. Note that `threshold_cc` is normalized using the same scaler applied to all edge weights (see above).

4.5.3.3. Weighted Stochastic Block Model

We use a Bayesian formulation of the Weighted Stochastic Block Model (WSBM) (see Section 2.5.2.2), a generative model for random graphs popular in biology, physics and social sciences [122, 123]. The model has been first applied on WUGs by Schlechtweg et al. [43] and subsequently by Kotchourko [124] and Tunc [125]. Fitting the model is equivalent to determining the optimal latent block structure providing a clustering of word usages.⁴⁶ We use the WSBM version described in Schlechtweg et al. [43] marginalizing out edge probabilities.

The WSBM model supports several distributions from which edge weights are drawn through the `distribution` parameter which can take one of the following values: exponential, normal, poisson, binomial, and geometric.⁴⁷ Exponential and normal distribution are only applicable to real-valued edge weights while poisson, binomial, geometric distribution are only applicable to discrete edge weights. Hence, we apply the former two distributions only to non-binarized graphs while we apply the latter three distributions only to binarized graphs.

4.5.3.4. Spectral Clustering

We employ Spectral Clustering as described in Section 2.5.2.2, using the scikit-learn⁴⁸ implementation with default hyperparameters. In our setup, we use K-means to identify the clusters. As this method requires the number of clusters as an input parameter, we explore several strategies for selecting an appropriate value of k .

Silhouette Score

This is defined as the mean silhouette coefficient of all nodes. The coefficient is the difference of the mean nearest-cluster distance and the mean intra-cluster distance, divided by the maximum of the two [126].

Calinski-Harabasz Score

The score is defined as ratio of the sums of between-cluster dispersion and of within-cluster dispersion [127].

Eigengap heuristic

This technique selects the number of clusters k as the value which maximizes $\lambda_{k+1} - \lambda_k$, where $\lambda_1, \lambda_2, \dots, \lambda_k$ are the eigenvalues of the affinity matrix Laplacian [82].

⁴⁶ https://graph-tool.skewed.de/static/doc/autosummary/graph_tool.inference.BlockState.html

⁴⁷ <https://graph-tool.skewed.de/static/doc/demos/inference/inference.html>

⁴⁸ <https://scikit-learn.org>

In the case of Silhouette and Calinski-Harabasz Score, the clustering with the highest score was selected, as these metrics are designed to favor more representative clusterings. Additionally, in cases where multiple clusterings had the same score, we selected the one with the fewest clusters.⁴⁹ We refer to Spectral Clustering with the respective method to choose the number of clusters by “.../s” for Silhouette, “.../c” for Calinski-Harabasz, and “.../e” for Eigengap.

4.5.4. Methodological Scope and Alternative Approaches

The experimental framework proposed in this thesis focuses on graph-based approaches for modeling semantic proximity relations between word usages. More specifically, the methods explored throughout this work operate over WUGs, where nodes represent individual usages and weighted edges encode semantic proximity scores estimated by WiC-based models or LLM-generated annotations. This design choice is motivated by the sense-oriented perspective adopted in this thesis, where LSCD is treated as a downstream consequence of how effectively semantic structures and sense distributions are recovered from contextualized usages.

During the design of the experimental framework, alternative approaches such as Dynamic Topic Models (DTMs) [128] and Deep Modularity Networks (DMoN) [129] were considered. However, these approaches were ultimately excluded due to methodological considerations related to the objectives of this thesis and to the modeling assumptions underlying current sense-oriented LSCD approaches.

Dynamic Topic Models were not incorporated because they operate under a substantially different representation paradigm from the one adopted in this work. DTMs are designed to model the temporal evolution of latent topics at the corpus level by capturing changes in word distributions across time slices. While this objective is related to diachronic language analysis, the methods developed in this thesis focus instead on usage-level semantic proximity modeling, where semantic change is inferred from pairwise relations between contextualized word usages. In our framework, the central objective is not to model document-level topical evolution, but rather to recover sense-level semantic structures through clustering over semantic proximity graphs.

This distinction is particularly important because topic distributions do not necessarily correspond to lexical senses. Changes in topic prevalence may reflect shifts in discourse, genre, or thematic focus without implying genuine semantic change at the lexical level. In contrast, the approaches investigated in this thesis explicitly operate over contextualized usages and aim to recover semantic groupings that approximate sense distinctions. Furthermore, recent LSCD research has increasingly shifted toward contextualized and usage-based representations, particularly through WiC-based models and graph clustering approaches, which are more directly aligned with the objectives of sense induction and graded semantic change detection explored in this work.

Deep Modularity Networks (DMoN) [129] were also considered but ultimately excluded for two main technical reasons. First, DMoN is primarily designed for settings in which node features and graph structure jointly contribute to the clustering process. In our case, however, the clustering signal is already fully encoded in the graph edges, whose weights

⁴⁹ We use an adjacency matrix representing sentence similarity, rather than a feature matrix, to calculate the Silhouette Score and Calinski-Harabasz Score.

correspond to semantic proximity scores produced by the WiC-based models or by LLM-generated annotations, representing the estimated likelihood that two usages share the same sense. Methods such as Chinese Whispers and the WSBM directly exploit this weighted edge structure to infer cluster membership. By contrast, DMoN incorporates graph information indirectly through graph neural network message passing and learned node representations, introducing additional representational transformations that are not clearly advantageous in a setting where the semantic information is already explicitly encoded in the edge weights.

Second, DMoN optimizes a modularity-based objective, which is known to suffer from the resolution limit problem [130]. This limitation is particularly relevant in dense semantic proximity graphs such as those used in this thesis, where modularity optimization tends to merge small but semantically meaningful communities into larger clusters. In the context of WSI and LSCD, this behavior is undesirable because it may obscure fine-grained distinctions between word senses that the clustering process is intended to capture.

Overall, the methods selected in this thesis were chosen because they operate directly over semantic proximity graphs and are more closely aligned with the sense-oriented objectives of the proposed LSCD framework. Nevertheless, alternative approaches such as Dynamic Topic Models and graph neural clustering architectures constitute interesting directions for future research, particularly for investigating whether topic-level temporal representations or learned graph embeddings can complement semantic proximity-based models of lexical semantic change.

4.6. Experiments

In this section, we describe the experiments conducted to evaluate the clustering-based framework presented in Section 4.5. Using the German, Spanish, and English datasets introduced in previous chapters, we investigate whether optimization at the WSI level transfers to LSCD performance. Specifically, we compare different WiC models, clustering algorithms, and hyperparameter configurations, evaluating their impact on both induced sense inventories and semantic change predictions. To ensure a robust assessment, all experiments are conducted within a cross-validation framework with hyperparameter optimization [131].

For each dataset, the target words are partitioned into five folds, yielding dataset-specific cross-validation splits. Within each cross-validation iteration, an exhaustive hyperparameter grid search is performed over the WiC model, WUG configuration, and the parameters specific to the clustering method under consideration. The optimal configuration identified on the training folds is subsequently evaluated on the corresponding test fold. This procedure produces five performance scores for each clustering method, which are summarized using their mean and standard deviation. WSI performance is measured using ARI, while LSCD performance is measured using Spearman correlation. Results are reported separately when optimization is guided by WSI and LSCD performance. This experimental design enables the analysis of the relationship between the two tasks and allows us to assess whether clustering configurations that perform well for WSI also generalize to semantic change detection across different datasets and languages.

The complete set of hyperparameter configurations explored during cross-validation, including WiC models, WUG settings, and clustering-specific parameters for each dataset and method, is reported in Appendix E, which provides full per-fold details of the model selection process.

Table 4.4: Results on the German dataset. The table reports the rank (in parentheses) obtained in the cross-validation experiments together with the corresponding performance metrics (ARI for WSI and Spearman’s ρ for LSCD) for each clustering method.

Tasks	WSI		LSCD	
	ARI	SPR	ARI	SPR
Chinese Whispers	(6) .622 $\pm$.130	(6) .768 $\pm$.241	(5) .602 $\pm$.078	(4) .768 $\pm$.241
Correlation Clustering	(5) .622 $\pm$.132	(2) .873 $\pm$.110	(3) .666 $\pm$.096	(5) .715 $\pm$.207
WSBM	(3) .710 $\pm$.068	(1) .911 $\pm$.156	(4) .613 $\pm$.158	(2) .828 $\pm$.169
Spectral Clustering/s	(1) .771 $\pm$.090	(2) .873 $\pm$.141	(1) .715 $\pm$.067	(1) .835 $\pm$.130
Spectral Clustering/c	(2) .755 $\pm$.087	(4) .834 $\pm$.077	(2) .702 $\pm$.087	(3) .774 $\pm$.111
Spectral Clustering/e	(4) .633 $\pm$.144	(5) .807 $\pm$.202	(6) .426 $\pm$.208	(6) .596 $\pm$.213

4.6.1. Analysis of the German Dataset

Table 4.4 provides a summary of the results obtained from our 5-fold cross-validation experiments. It displays the average test score across folds along with the standard deviation. The WiC models used as hyperparameters may vary between folds. Tables E.1 and E.2 detail which WiC model was selected for each fold. Additionally, the reported standard deviation quantifies how the metrics vary from one fold to another, depending on both the hyperparameters selected in each fold and the test subset corresponding to that fold. For Spectral Clustering, we present the results for each method to choose the number of clusters separately. The table presents two distinct approaches to the hyperparameter selection: optimization for WSI (left) and for LSCD (right). For both approaches, we report also the performance on the other task which we do not directly optimize for in order to understand how well the performance translates to the other task. Among the methods evaluated, Spectral Clustering stands out as the most effective, particularly when utilizing the Silhouette score to select the number of clusters. This configuration consistently obtains the highest ARI score considering the WSI task and the highest SPR in the LSCD task.

As we can see in Table 4.4, WSBM is the most effective model when translating parameters from WSI to LSCD. Conversely, when applying the optimized parameters from the LSCD task to address the WSI task, the Spectral Clustering method, specifically using the Silhouette validation method, demonstrates superior performance, yielding the most favorable results. An additional noteworthy observation is that, with the exception of Chinese Whispers, all methods consistently yield superior results when utilizing parameters optimized for the WSI task to address the LSCD task, compared to using parameters optimized for the LSCD task. We conjecture that this is because the model selection criterion for LSCD overfits more easily because of the notoriously small number of annotated examples (words) compared to the WSI task (usages). This problem affects all existing LSCD datasets [e.g. 34, 50, 52, 132]. Hence, we consider this finding relevant for the LSCD community in general. It suggests that optimization for WSI should be preferred over optimization for LSCD in small data scenarios, even if LSCD is the task of interest.

Table 4.5: Results on the Spanish dataset. The table reports the rank (in parentheses) obtained in the cross-validation experiments together with the corresponding performance metrics (ARI for WSI and Spearman’s ρ for LSCD) for each clustering method.

Tasks	WSI		LSCD	
	ARI	SPR	ARI	SPR
Chinese Whispers	(1) .475 $\pm$.049	(3) .590 $\pm$.260	(2) .453 $\pm$.066	(4) .498 $\pm$.254
Correlation Clustering	(3) .412 $\pm$.113	(1) .661 $\pm$.106	(3) .411 $\pm$.115	(1) .651 $\pm$.121
WSBM	(2) .444 $\pm$.022	(5) .553 $\pm$.186	(1) .480 $\pm$.078	(2) .602 $\pm$.243
Spectral Clustering/s	(5) .351 $\pm$.108	(4) .578 $\pm$.173	(4) .371 $\pm$.112	(5) .420 $\pm$.187
Spectral Clustering/c	(4) .363 $\pm$.097	(6) .420 $\pm$.239	(5) .316 $\pm$.113	(6) .301 $\pm$.271
Spectral Clustering/e	(6) .310 $\pm$.084	(2) .595 $\pm$.152	(6) .301 $\pm$.094	(3) .567 $\pm$.189

4.6.2. Analysis of the Spanish Dataset

Table 4.5 summarizes the results obtained on the Spanish dataset using the same 5-fold cross-validation framework described above. As in the German experiments, the table reports the average performance across folds together with the corresponding standard deviation. The left-hand side presents results obtained when hyperparameters are selected according to WSI performance, while the right-hand side reports results obtained when optimization is guided by LSCD performance. For each setting, we additionally report performance on the complementary task in order to analyze the relationship between WSI and LSCD.

The results reveal a different pattern from that observed on the German dataset. Chinese Whispers achieves the highest ARI score when optimizing for WSI, indicating that it is the most effective clustering method for reproducing the reference sense inventories in the Spanish data. However, the strongest LSCD results are obtained by Correlation Clustering, which reaches the highest Spearman correlation under both optimization strategies. The interpretation of this result requires considering how the Spanish dataset was constructed. Unlike the German DWUG DE Sense dataset, whose sense inventories were manually annotated, the reference sense inventories for the Spanish dataset were derived using Correlation Clustering during dataset creation. Consequently, the strong performance of Correlation Clustering is not entirely unexpected, since the evaluation is naturally aligned with the clustering procedure used to construct the reference annotations. More generally, this implies that part of the observed ranking of clustering methods is influenced by the construction procedure of the reference sense inventories and should therefore not be interpreted as a fully independent comparison of clustering algorithms. Nevertheless, despite this expected methodological bias, the results still indicate that the clustering configuration that best captures the reference sense inventories is not necessarily the one that best models semantic change, as Chinese Whispers achieves the highest WSI performance while Correlation Clustering produces the strongest LSCD results.

Despite this methodological characteristic of the benchmark, the results still provide evidence of a relationship between WSI and LSCD performance. For several clustering methods, including Correlation Clustering and Spectral Clustering/e, the hyperparameters selected through WSI optimization yield LSCD results that are comparable to or better than those obtained when directly optimizing for LSCD. In particular, Correlation Clustering achieves its highest LSCD performance when the model selection criterion is based on WSI rather

Table 4.6: Results on the English dataset. The table reports the rank (in parentheses) obtained in the cross-validation experiments together with the corresponding performance metrics (ARI for WSI and Spearman’s ρ for LSCD) for each clustering method.

Tasks	WSI		LSCD	
	ARI	SPR	ARI	SPR
Chinese Whispers	(1) .243 $\pm$.098	(1) .700 $\pm$.209	(1) .256 $\pm$.129	(1) .598 $\pm$.198
Correlation Clustering	(4) .192 $\pm$.104	(5) .307 $\pm$.194	(3) .080 $\pm$.067	(4) .293 $\pm$.189
WSBM	(2) .226 $\pm$.144	(3) .562 $\pm$.297	(6) .045 $\pm$.025	(2) .555 $\pm$.214
Spectral Clustering/s	(5) .279 $\pm$.095	(2) .600 $\pm$.205	(2) .088 $\pm$.034	(3) .505 $\pm$.233
Spectral Clustering/c	(3) .106 $\pm$.027	(6) .157 $\pm$.425	(4) .059 $\pm$.042	(5) .288 $\pm$.215
Spectral Clustering/e	(6) .064 $\pm$.038	(4) .395 $\pm$.207	(5) .054 $\pm$.050	(6) .126 $\pm$.232

than LSCD. Similarly, Spectral Clustering/e reaches a higher Spearman correlation when using WSI-selected hyperparameters.

Unlike the German dataset, no single clustering method dominates both tasks. The Spanish results therefore suggest that the relationship between WSI and LSCD is weaker and more method-dependent. Nevertheless, the overall pattern remains consistent with the central hypothesis of this chapter: optimization at the WSI level often yields competitive LSCD performance, even when semantic change detection is the final task of interest. This finding further supports the use of WSI as a reliable model-selection criterion in scenarios where LSCD annotations are limited and prone to overfitting.

4.6.3. Analysis of the English Dataset

Table 4.6 summarizes the results obtained on the English dataset. As in the previous experiments, we report the average performance across the cross-validation folds together with the corresponding standard deviations. The table presents the two optimization settings considered throughout this chapter: hyperparameter selection based on WSI performance (left) and hyperparameter selection based on LSCD performance (right). For each setting, we also report the performance obtained on the complementary task in order to assess the extent to which improvements transfer between WSI and LSCD.

In contrast to the German and Spanish datasets, the English dataset exhibits substantially lower ARI scores across all clustering methods, indicating weaker agreement between the induced clusters and the reference sense inventories. As in the Spanish dataset, the English reference sense inventories were obtained by applying Correlation Clustering during dataset construction. Therefore, one might expect Correlation Clustering to reproduce the reference inventories more accurately than the remaining clustering algorithms. However, this expected advantage is not observed. Despite being used to derive the gold-standard inventories, Correlation Clustering does not achieve the strongest WSI or LSCD performance. This observation is particularly noteworthy because, unlike the Spanish dataset, the English benchmark was created using an iterative edge-sampling strategy designed to progressively construct well-connected WUGs before clustering, rather than relying solely on randomly sampled usage pairs. Consequently, the clustering algorithm used during dataset creation alone cannot explain the observed performance differences.

Several factors may contribute to the comparatively lower WSI performance observed on

the English dataset. Besides differences in the quality of the semantic proximity judgments and the structural properties of the resulting usage graphs, the English benchmark includes a *noise cluster*, which is absent from the German dataset and differs in its distribution from the Spanish benchmark. Assignments to this cluster may vary substantially across clustering methods, thereby reducing agreement with the reference inventories as measured by ARI. Consequently, lower WSI scores should not necessarily be interpreted as evidence of poorer sense induction, but may instead reflect characteristics of the benchmark itself. Explicitly modeling the noise cluster represents an interesting direction for future work.

Chinese Whispers obtains the strongest overall results on this dataset. It achieves the highest LSCD performance when hyperparameters are optimized for WSI ($\rho = .700$), while also obtaining the best LSCD result under LSCD optimization ($\rho = .598$). In addition, it produces the strongest WSI performance under LSCD optimization (ARI = .256). These results indicate that Chinese Whispers provides the most stable behavior across both tasks on the English dataset.

A different pattern emerges for Spectral Clustering with Silhouette validation. Although it achieves the highest ARI score when optimized for WSI (ARI = .279), its LSCD performance remains below that of Chinese Whispers. This contrasts with the German dataset, where Spectral Clustering consistently produced the strongest results, suggesting that the relationship between clustering quality and semantic change detection is more dataset-dependent than previously observed.

WSBM also exhibits interesting behavior. While its clustering performance is only moderate, it achieves the second-best LSCD performance in both optimization settings ($\rho = .562$ and $\rho = .555$ respectively). Similar observations were made for the Spanish dataset, where WSBM also produced competitive semantic change scores despite not being the strongest clustering method according to ARI. This further supports the observation that improvements in WSI performance do not always translate directly into improvements in LSCD performance.

As in the German dataset, hyperparameters selected through WSI optimization generally transfer well to LSCD. Five of the six clustering methods achieve higher LSCD performance when optimized for WSI rather than directly for LSCD, with Spectral Clustering/c constituting the only exception. However, the magnitude of these improvements is smaller than that observed for the German dataset, indicating that the benefits of WSI-based model selection are present but less pronounced.

Overall, the English experiments reveal a weaker relationship between WSI and LSCD performance than that observed for the German dataset. While clustering quality remains an important factor, the results suggest that capturing shifts in sense distributions across time may be more relevant for semantic change detection than maximizing agreement with reference sense inventories. This finding reinforces the importance of jointly evaluating WSI and LSCD when studying the transferability of clustering configurations across tasks.

4.6.4. Comparison with APD

The comparison between clustering-based approaches and the APD baseline is presented in Table 4.7. For the clustering framework, we report the LSCD performance obtained by the configuration selected through WSI optimization, which corresponds to the main objective of this chapter: evaluating whether optimization at the WSI level transfers to semantic change detection.

The results reveal different patterns across datasets. For the Spanish dataset, APD

Table 4.7: Comparison between the best clustering-based LSCD configuration and the APD baseline across datasets. Results are reported in terms of Spearman’s rank correlation coefficient (ρ).

Dataset	Best WSI -> LSCD	APD
German	.873 $\pm$.141	.760 $\pm$.121
Spanish	.590 $\pm$.260	.653 $\pm$.250
English	.600 $\pm$.205	.638 $\pm$.292

achieves a Spearman correlation of 0.653, outperforming the best clustering-based configuration selected through WSI optimization, which reaches 0.590. A similar trend can be observed for the English dataset, where APD obtains a correlation of 0.638 compared to 0.600 for the clustering-based approach. These findings are consistent with the results reported in Chapter 3, where aggregation-based methods demonstrated strong performance for graded semantic change detection.

For the German dataset, the best clustering-based configuration achieves a Spearman correlation of 0.873, outperforming APD, which obtains a correlation of 0.760. Although APD remains highly competitive, the clustering-based framework achieves the strongest overall performance. This result is particularly noteworthy because the German dataset provides high-quality sense annotations specifically designed for evaluating both WSI and LSCD. The performance improvement suggests that explicitly modeling sense structure can be beneficial when reliable sense inventories are available and semantic change is evaluated through shifts in sense distributions.

Taken together, these results indicate that the relative strengths of aggregation-based and clustering-based approaches depend on the characteristics of the dataset. While APD achieves the strongest performance on the Spanish and English datasets, clustering-based methods outperform APD on the German dataset and remain competitive on the other two datasets. Moreover, clustering-based approaches provide explicit sense inventories and interpretable representations of semantic change, whereas APD estimates semantic change directly from contextualized representations without modeling sense structure.

It is also important to note that WSI performance should be interpreted with caution. Lower agreement with the reference sense inventories may partly reflect characteristics of the gold-standard annotations rather than deficiencies of the clustering methods themselves. In particular, the presence of noise clusters—grouping usages that cannot be reliably assigned to a well-defined sense—and the sparsity of the annotated graph can reduce ARI even when the induced clusters capture the underlying semantic structure relevant for LSCD. Incorporating explicit mechanisms to model such noise and annotation sparsity constitutes a promising direction for future sense-aware clustering approaches.

Overall, the results support the view that aggregation-based methods currently constitute strong baselines for graded semantic change detection. At the same time, the strong performance achieved by clustering-based approaches, particularly on the German dataset, suggests that sense-aware methods remain a promising direction for future research. Furthermore, the fact that competitive LSCD performance can be obtained from configurations selected exclusively through WSI optimization provides additional evidence that the two tasks are closely related.

Detailed results for each cross-validation fold, including the hyperparameter configurations selected during WSI and LSCD optimization, are provided in Appendix E. These tables

report the complete model selection process for all clustering methods and datasets and allow inspection of the variability of the selected configurations across folds.

4.6.5. Summary of Findings

The central objective of this chapter was to investigate the relationship between WSI and LSCD and, in particular, to determine whether optimization at the level of WSI transfers to semantic change detection. To this end, we evaluated a common clustering-based framework across three datasets covering German, Spanish, and English, and analyzed the impact of different clustering configurations on both tasks.

The results show that the relationship between WSI and LSCD is consistently observable across datasets. In all experimental settings, clustering quality, measured through ARI, is associated with the quality of the induced semantic change predictions. However, the strength of this relationship varies across clustering methods and datasets. No single clustering algorithm emerges as universally optimal, suggesting that the effectiveness of a clustering strategy depends on the characteristics of the dataset and the semantic proximity signals produced by the underlying WiC models.

The German dataset provides the most direct evidence for studying this relationship, as it contains gold-standard sense annotations specifically designed for evaluating both WSI and LSCD. The results indicate that configurations selected using WSI performance frequently yield competitive LSCD performance, supporting the hypothesis that WSI can serve as a useful proxy task for model selection in semantic change detection. Similar tendencies are observed for the Spanish and English datasets, where clustering-based approaches selected through WSI optimization achieve competitive semantic change detection results.

The comparison with APD further contextualizes these findings. On the Spanish and English datasets, APD achieves the strongest LSCD performance, although the gap with the clustering-based framework remains relatively small. In contrast, the best clustering-based configuration outperforms APD on the German dataset. These results suggest that the relative strengths of aggregation-based and clustering-based approaches depend on the characteristics of the dataset and the availability of reliable sense information.

Overall, the experiments demonstrate that clustering-based methods constitute a competitive alternative to aggregation-based approaches for graded semantic change detection. While APD remains a strong and robust baseline, clustering-based methods provide explicit sense inventories and interpretable representations of semantic change, thereby more closely reflecting the annotation process underlying semantic change datasets. Most importantly, the observed transfer between WSI and LSCD across all three datasets indicates that improvements in sense induction can translate into improvements in semantic change detection, supporting the use of WSI as a proxy task for the development and selection of LSCD systems.

4.7. Conclusions

This chapter investigated the relationship between WSI and LSCD within a unified clustering-based framework. Building on semantic proximity predictions produced by WiC models, we evaluated several clustering algorithms and analyzed the extent to which optimization for WSI transfers to semantic change detection.

The experiments were conducted on three datasets covering German, Spanish, and English. Together, these datasets enabled the evaluation of clustering quality through gold-

standard sense inventories and the assessment of semantic change through graded LSCD annotations. Across datasets, the results demonstrate a consistent relationship between WSI and LSCD performance, indicating that improvements in sense induction can translate into improvements in semantic change detection.

A central finding of this chapter is that configurations selected through WSI optimization frequently yield competitive LSCD performance. In particular, optimization based on WSI proved to be a reliable strategy for model selection, while optimization directly on LSCD did not consistently provide superior results. These findings support the use of WSI as a proxy task for LSCD, especially in scenarios where semantic change annotations are scarce and model selection based solely on LSCD performance may be prone to overfitting.

The experiments also revealed that no single clustering algorithm dominates across all datasets. While Spectral Clustering achieved strong and stable performance on the German dataset, different clustering methods emerged as the most effective solutions for Spanish and English. This suggests that the optimal clustering strategy depends on the characteristics of the dataset and the semantic proximity signals generated by the underlying WiC models.

Finally, we compared clustering-based approaches with the APD aggregation method. Consistent with the findings presented in Chapter 3, APD remains a highly competitive baseline for graded semantic change detection. However, clustering-based methods achieved comparable performance while additionally producing explicit sense inventories and interpretable representations of semantic change. These results suggest that, although aggregation-based approaches currently remain strong solutions for LSCD, sense-aware clustering methods provide a promising and linguistically motivated alternative with potential for future improvements.

Chapter 5

Enhancing LSCD predictions through LLMs

Chapter 3 introduced a Spanish LSCD dataset and established strong aggregation-based baselines for graded semantic change detection. Chapter 4 subsequently investigated clustering-based approaches on German, Spanish, and English datasets, analyzing the relationship between WSI and LSCD. Building on these findings, the present chapter explores an alternative paradigm based on LLMs. Focusing on the Spanish and English datasets, we investigate whether semantic proximity judgments generated by LLMs can provide an effective basis for semantic change detection and how their performance compares to the aggregation-based approaches presented in Chapter 3 and the clustering-based approaches examined in Chapter 4.

5.1. Introduction

Over time, LSCD has benefited from the development of progressively more advanced methods to capture meaning variation. Early approaches relied on counting-based techniques and static word embeddings [45, 87, 133], but these were soon surpassed by contextualized models [134], which enabled more nuanced representations of word usage in context. Among these, models trained on the WiC task have proven particularly effective for LSCD, since both tasks involve determining whether a pair of word usages share the same meaning. The key difference lies in LSCD’s diachronic nature, where usage comparisons span across different time periods.

Notably, DeepMistake [19] and XL-LEXEME [20] (see Section 2.5.3) have demonstrated state-of-the-art performance on several LSCD datasets. DeepMistake achieved leading results in the Russian [18] and Spanish [50] shared tasks, while XL-LEXEME outperformed previous methods on English, Swedish, and German datasets. These models leverage the close relationship between the LSCD and WiC tasks to learn effective contextualized representations for semantic change.

The findings of Chapter 4 further support this perspective. There, we showed that optimizing the WSI component of a clustering-based LSCD pipeline leads to improved semantic change detection performance, suggesting that careful modeling of the underlying sense-related subproblems can benefit the overall LSCD task. Since these pipelines also depend critically on WiC-based semantic proximity judgments, a natural next step is to investigate whether the WiC component itself can be improved. Recent advances in LLMs provide an opportunity to explore this question, as LLMs may generate semantic proximity judgments

directly through natural-language prompting without requiring task-specific fine-tuning.

More recently, LLMs such as GPT-3 [22] have emerged as general-purpose tools capable of strong performance across a wide variety of NLP tasks [135, 136]. These models are trained on massive text corpora and can generate high-quality responses from natural language prompts alone, without task-specific fine-tuning. Despite their success, LLMs have not yet been extensively studied for LSCD, particularly for graded semantic change detection.

Despite the widespread success of LLMs across various NLP tasks, their performance has been shown to be highly sensitive to the choice of prompt [137, 138]. This sensitivity highlights a key limitation: crafting effective prompts often requires substantial human expertise and manual effort. Previous studies have shown that optimized prompts can significantly enhance model performance compared to unrefined or manually designed prompts, yielding improvements in tasks such as classification, question answering, and others [139–141].

In this chapter, we investigate whether LLMs can replace the WiC component traditionally used in LSCD pipelines and whether such substitution can improve semantic change detection performance. This comparison is motivated by the practicality of LLMs: (i) they can be applied without fine-tuning on downstream tasks, and (ii) given the promising outcomes reported across several researches [135, 136]. In addition to the model comparison, we also explore whether automatic prompt selection can enhance the semantic proximity judgments produced by LLMs and check if those models can outperform much smaller and faster specialized models.

This chapter addresses the thesis research questions concerning the use of LLMs for LSCD. In particular, we evaluate whether automatic prompt optimization improves performance, whether LLMs can compete with state-of-the-art WiC-based systems on graded semantic change detection, and whether their semantic proximity judgments align with human annotations at the usage-pair level.

5.2. Background

This section provides an overview of recent studies on LSCD that employ LLMs, leveraging their contextual and generative capabilities to model semantic change. Building upon this literature, an important question remains open: existing work has not systematically evaluated whether LLM-based semantic proximity judgments can replace WiC-based models within LSCD pipelines, nor how prompt optimization affects their reliability and performance across languages.

LLMs + LSCD

Wang and Choi [142] show that the use of LLMs, specifically GPT-4 [143], yields superior results compared to BERT [16] and traditional methods, including statistical approaches and static embedding techniques [95], for detecting semantic change. The authors employ an experimental setup based on the TempoWiC dataset [144], which involves two tasks: instance-level meaning shift detection and corpus-level LSC detection. In the first task, they utilize a binary classification approach to determine whether a target word has changed its meaning within the entire corpus. The second task involves classifying whether a target word retains similar meanings across sentence pairs from two different time periods. Predictions from the LLMs are generated using various prompts. Ultimately, the authors demonstrate that LLMs outperform the other methods in both tasks.

LLMs + WiC

In a similar vein, Periti et al. [145] present a series of experiments comparing the performance of the BERT model [16] with that of ChatGPT, utilizing both the web interface and the foundation model via the API. Rather than employing LSCD datasets, the authors evaluate ChatGPT’s performance on the WiC task [9], given its relevance to LSCD [19, 20].

To achieve this, they utilize two datasets: TempoWiC [144] and DWUG EN [4].⁵⁰ TempoWiC is employed to evaluate changes in meaning over a short time span, while DWUG EN assesses the model’s ability to detect semantic change in a long time span. Predictions from the LLMs are generated using various prompts. The results indicate that BERT outperforms the LLMs in this evaluation. Ultimately, the authors conclude that ChatGPT faces challenges in detecting semantic change, both in diachronic contexts and over short time periods. These results are contradictory to the studies of Wang and Choi [142]; however, the experimental setups in both cases differ significantly.

Yadav et al. [146] find that automating the annotation process by reusing human-tailored instructions is a major challenge. To substantiate their claims, they use the DWUG EN dataset [34] comprising 46,000 pairs of word usages annotated following a 4-point semantic relatedness scale: 1 (unrelated meanings), 2 (distantly related meanings), 3 (closely related meanings), and 4 (identical meanings) [25]. By utilizing ChatGPT through its API, the authors design a series of prompts to guide the annotation of LSCD samples. They divide the dataset into three subsets: training, development, and testing, to optimize and evaluate the prompts for achieving accurate predictions from ChatGPT. The results indicate that ChatGPT performs poorly in annotating the samples when prompted with the full annotation guidelines presented in a human-like style particularly when assessed using the Krippendorff’s alpha score [147]. In contrast, ChatGPT yields better results when custom prompts are employed.

Periti and Tahmasebi [148] present a systematic comparison of contextualized models for the LSCD task. In one of their experiments, they evaluate several models used as computational annotators, including GPT-4 only for English. Their findings show that GPT-4 achieves the highest Spearman correlation with human annotations for the English dataset, outperforming all other models, including XL-LEXEME. In addition, GPT-4 also shows superior performance in the Word Sense Induction (WSI) task. However, its performance in the WiC task falls short compared to that of XL-LEXEME.

However, their work does not address the challenges or potential of prompt engineering when applying LLMs to LSCD. In particular, they do not explore how prompt formulation can impact model performance—a key consideration when working with general-purpose LLMs.

WiC model + LSCD

The state-of-the-art models for the LSCD task consist of two WiC models. The DeepMistake model, introduced by Arefyev et al. [19], is built on the XLM-R encoder. This model consists of two components: an XLM-R-based encoder jointly encoding two word usages and a classification head. All weights are trained on the MCL-WIC dataset [149], and then undergo language-specific fine-tuning. For Spanish they are fine-tuned on the Spanish version of the synchronic XL-WSD [114], and the development version of the DWUG ES [50] datasets. For Russian the RuSemShift [49] dataset is employed. DeepMistake is the state-of-the-art for LSCD in Spanish [19] and Russian [112].

⁵⁰ Periti et al. [145] renamed this dataset as HistoWiC.

The XL-LEXEME model [20] is a pre-trained bi-encoder built on the XLM-R multilingual model. This model has been trained and fine-tuned on the MCL-WiC [149] and XL-WiC [150] datasets. XL-LEXEME currently holds the state-of-the-art position for English, Swedish, and German datasets [4].

Our work provides a systematic comparison between general-purpose, open-weight LLMs and specialized LSCD models for the LSCD task. Motivated by the practical advantages of LLMs—such as avoiding the need for task-specific fine-tuning—we assess their performance across multiple evaluation methodologies and datasets. Unlike previous studies, our work introduces the use of automatic prompt optimization as a key strategy to enhance LLM performance for LSCD, aiming to adapt these models more effectively to the task. In doing so, we prioritize a comprehensive evaluation of their capabilities relative to dedicated LSCD models.

5.3. Data

In this study, we employ the DWUG EN dataset described in Section 2.4.2 and the DWUG ES dataset described in Chapter 4. Table 5.1 summarizes their main characteristics, including the number of target words and the corresponding time periods.

Table 5.1: Overview of datasets. LGS = languages, Id = identifier of the dataset, n = number of target words, C1 = old period of time, C2 = modern period of time.

LGS	Id	n	C1	C2
Spanish	DWUG ES	60	1810-1906	1994-2020
English	DWUG EN	37	1810-1860	1960-2010

Additionally, we employ the DWUG EN resampled dataset⁵¹, along with the development version of the DWUG ES dataset (the test set is reported in Table 5.1). As will be detailed later, these two datasets are utilized to optimize the prompts designed for the LLMs to determine whether a pair of word usages have the same meaning. For this optimization, we partition the resampled DWUG EN and the development version of the DWUG ES datasets into training, development, and test sets, allocating 60% of the data for training and 20% each for development and testing.

Table 5.2 presents the frequency distribution of annotation labels in both datasets, which are used to optimize two prompts: one in English and the other in Spanish. As observed, the classes are imbalanced. To address this issue, we apply stratified random sampling to ensure equal-sized samples for each class across the training, development, and testing sets.

Table 5.2: Class distribution across DWUG EN resampled and development version of DWUG ES datasets.

Dataset	Class 1	Class 2	Class 3	Class 4
DWUG EN (resampled)	641	1,658	1,472	2,075
DWUG ES (dev)	1,406	1,522	2,343	3,433

After implementing a stratified sampling strategy, we obtain balanced samples for each

⁵¹ <https://zenodo.org/records/14025941>

class across all partitions in both datasets, as shown in Table 5.3. This balanced allocation enhances the robustness of training, evaluation, and testing for both languages, effectively addressing the imbalances present in the original datasets.

Table 5.3: Statistics for each partition of the DWUG EN (resampled) and the development version of the DWUG ES datasets used to optimize the prompts for annotating semantic proximity examples.

Datasets	Class 1	Class 2	Class 3	Class 4	Target words
DWUG EN (resampled) training	300	300	300	300	15
DWUG EN (resampled) dev	100	100	100	100	15
DWUG EN (resampled) test	100	100	100	100	15
DWUG ES (dev) training	500	500	500	500	20
DWUG ES (dev) dev	200	200	200	200	20
DWUG ES (dev) test	225	225	225	225	20

Unlike Chapter 4, we do not include the German DWUG-DE Sense dataset in this chapter. This decision is primarily driven by computational and budgetary constraints, as the experiments in this chapter require repeated LLM inference for prompt optimization and multilingual evaluation, which significantly increases the overall cost of experimentation. Instead, we focus on the English and Spanish datasets, which are widely used benchmarks in the LSCD literature and enable direct comparison with prior work. In particular, the English dataset serves as a standard evaluation benchmark in existing LSCD studies, while the Spanish dataset provides continuity with both Chapter 3 and Chapter 4, where it was introduced and extensively analyzed. This selection allows us to balance computational feasibility with methodological comparability across datasets and experimental settings.

5.4. Experimental setup

In this section, we present the proposed methodologies designed to demonstrate the feasibility of using large language models to tackle the Graded Change LSCD task.

As outlined in Section 5.3, our experiments utilize two datasets: DWUG EN and DWUG ES. We generate LSC predictions using a WSI baseline approach, as described by Schlechtweg et al. [151]. Building upon the prompting strategies introduced in Section 5.5, we include results from both initial and optimized prompt configurations for the LLMs. These prompts are designed to generate semantic proximity judgments consistent with the 4-point scale used in the manual annotations [25].

In this study, we employ three open-source language models: Llama 3.1⁵², Llama 3.3, and Mixtral 8x7B⁵³ (Mixtral:8x7B) [152]. We specifically use the 8 billion-parameter versions of Llama 3.1 (Llama3.1:8B) and the 70 billion-parameter version of Llama 3.3 (Llama3.3:70B). We selected these models to systematically assess the impact of model scale on LSCD performance. This range enables us to examine how increasing model capacity interacts with prompt optimization and whether larger models consistently offer advantages across different evaluation approaches and languages. Given the computational and financial cost associated with repeated inference during prompt optimization and multilingual evaluation, we conduct

⁵² <https://ai.meta.com/blog/meta-llama-3-1/> (for more information) and available for download at <https://ollama.com/library/llama3.1>.

⁵³ <https://ollama.com/library/mixtral>

our experiments using quantized versions of the selected models [153].

In the following, we describe the pipeline used to compute predictions from both LLMs and WiC models:

- **WiC** We annotate gold pairs of usages in the English and Spanish datasets. The annotations produced by XL-LEXEME [20] are the cosine distances indicating whether two word usages share the same meaning. Additionally, we employ three versions of DeepMistake: MCL->es, enMCL, and MCL, as this model has been fine-tuned on various WiC datasets across multiple languages [cf. 19, 112]. The scores produced by DeepMistake are the probabilities that two word usages have the same meaning. Finally, we prompt Llama 3.1, Mixtral, and Llama 3.3 to annotate the same usage pairs by generating semantic proximity scores aligned with the 4-point scale employed in the manual annotation of the employed datasets [25].
- **WUG** See Section 2.3.3.
- **Clustering methods** We apply one of the clustering methods to the constructed WUGs to infer word senses. Next, to quantify the semantic change for each word, we calculate the JSD between the SFD of the inferred senses for the old and new usages.

In addition, we generate LSC scores based on the APD approach [84]:

- The average pairwise distance, derived from the scores generated by the WiC models or LLMs’ annotations, is returned as a measure of semantic change for a word.

5.4.1. WUG

We select several hyperparameters for the WUG construction procedure: `normalize`, `fill_diagonal`, and `threshold` (see Section 4.5.2).

5.4.2. Clustering methods

We employ in our experiments three clustering methods on WUGs: Spectral clustering⁵⁴ [154], Agglomerative clustering⁵⁵ [79], and WSBM⁵⁶ [80]. Spectral clustering and Agglomerative clustering methods need to know in advance the number of clusters. We test values from 2 to 5 [151] and identify the optimal value using the Silhouette score [155] (see Sections 2.5.2.2 and 4.5.3).

Following the methodological considerations discussed in Section 4.5.3, we restrict our analysis to clustering methods that directly exploit probabilistic edge structure.

5.5. Prompts and their optimizations

To address the first research question concerning prompt optimization for LSCD, we design an experiment comparing the performance of LLMs using manually crafted prompts with that of LLMs using optimized prompts informed by training data.

Building on the work of Yadav et al. [146], we utilize the manually optimized prompt identified by the authors as the most effective for classifying semantic proximity samples.

⁵⁴ <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.SpectralClustering.html>

⁵⁵ <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.AgglomerativeClustering.html>

⁵⁶ <https://graph-tool.skewed.de/static/doc/autosummary/graph-tool.inference.BlockState.html>

Originally crafted in English, this prompt was translated into Spanish to create an additional version tailored for semantic proximity samples in Spanish. Additionally, we extend the non-optimized prompts with semantic proximity examples provided to the annotators in the web of the DUREL framework [25]. Based on these examples, we create three prompts:

- **Zero-shot** We provide instructions to annotate the samples using the 4-point scale, along with an explanation for each point on the scale.
- **Few-shot** We provide instructions to the model to annotate the samples using the 4-point scale, and include an example of a pair of sentences for each possible annotation.
- **Chain of Thought** The same as the Few-shot approach with the key difference that we ask the model to reason step by step and we give them explanations also why the selection of the score for each example.

As mentioned in Section 5.1, hand-optimizing prompts is a challenging task, as even slight adjustments in the instructions can result in significantly different interpretations or responses from the model. To tackle this issue, we further refine both prompts using the DSPy framework [156], leveraging the data presented in Table 5.3.

We develop a DSPy program that employs the MIPROv2 (Multi-prompt Instruction PROposal Optimizer) optimizer [157] in conjunction with the accuracy metric to optimize the prompts, utilizing the default parameters for this optimizer.⁵⁷ Additionally, we conduct several experiments varying the number of samples per class provided to the optimizer to explore how this parameter influences the quality of the generated prompts. We observe a consistent pattern across all settings: increasing the number of samples per class generally leads to the generation of better prompts, resulting in higher accuracy on the test set.

MIPROv2 optimizes DSPy prompts by learning how instructions and demonstrations affect task performance. It generates demonstrations via bootstrapping and proposes instructions using grounding, where a language model receives contextual information about the dataset, program, and past prompts. New configurations—combinations of instructions and demonstrations—are proposed using Bayesian optimization and evaluated on mini-batches. The best-performing configurations are then selected based on full training set performance [156].

We then annotate the test sets using both English and Spanish prompts for the English data, and both English and Spanish prompts for the Spanish data. Subsequently, we select the best prompt based on accuracy to annotate the DWUG EN and DWUG ES datasets.

Table 5.4: Accuracy of non-optimized prompts (NOP) and optimized prompts (OP) for DWUG EN and DWUG ES datasets, using prompts in English (PrE) and Spanish (PrS). Results are reported for Mixtral, Llama 3.1, and Llama 3.3 as the underlying LLMs.

Dataset-Prompts	Llama 3.1		Mixtral		Llama 3.3	
	NOP	OP	NOP	OP	NOP	OP
DWUG ES - PrS	26.8	29.5	32.6	31.22	32.33	39.77
DWUG ES - PrE	26.5	35.5	33.7	34.80	40.88	46.33
DWUG EN - PrS	25.75	31.0	32.8	40.75	37.25	45.5
DWUG EN - PrE	28.75	37.25	33.25	38.75	35.75	49.25

⁵⁷ <https://dspy.ai/learn/optimization/optimizers/>

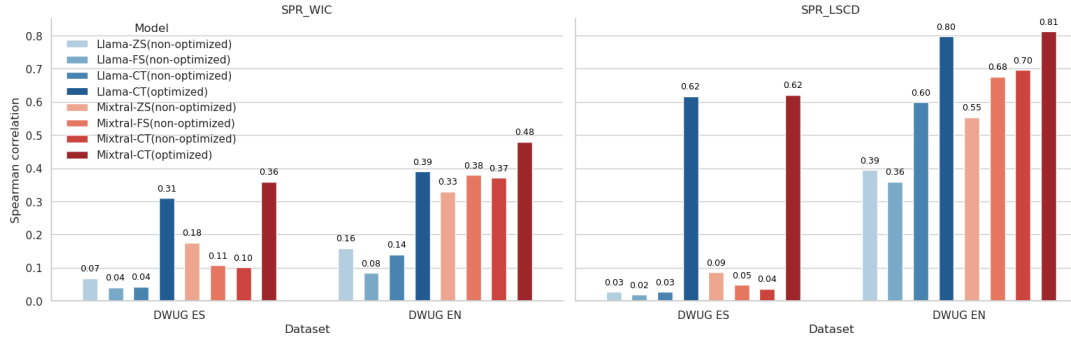


Figure 5.1: Spearman correlation between human and model annotations for word usage pairs (SPR_WiC), and between gold graded change scores and model predictions using the APD-based approach (SPR_LSCD). “ZS” denotes Zero-Shot, “FS” stands for Few-Shot, and “CT” refers to Chain-of-Thought prompting.

Table 5.4 presents the accuracy results for non-optimized and optimized prompts in classifying pairs of usages according to their labels on the 4-point scale. Bolded results indicate the best-performing prompt type for each dataset. Notably, the best prompt for a given dataset is not always in the same language as the dataset itself. Finally, we select the best-performing prompt for each dataset to annotate the LSCD samples.

Figure 5.1 presents a comparison between manually engineered prompts and automatically optimized prompts, evaluated using two metrics: the Spearman correlation between human and LLM-generated annotations, and the Spearman correlation between gold graded change scores and predictions derived from the APD-based approach using LLM annotations. The results show that optimized prompts substantially outperform manually crafted ones across both languages and experimental setups.

These results show that automatic prompt optimization consistently improves the performance of medium-sized models (i.e., Llama3.1:8B and Mixtral:8x7B) on the LSCD task. Next, we extend our investigation to a larger and more recent model (specifically, Llama3.3:70B). All subsequent experiments will be conducted using LLM annotations generated with the best performing optimized prompts obtained using the methodology presented here.

5.6. Experiments for graded change detection

In this section, we describe the methodology employed to evaluate whether LLMs can achieve strong performance on the graded LSCD task and compete with state-of-the-art WiC-based models.

To this end, we design two experiments: one based on the APD approach and the other using cross-validation. The APD-based approach does not require the selection of hyperparameters, allowing us to evaluate the entire test set. In contrast, the clustering-based approach involves several hyperparameters; therefore, we use cross-validation to select the optimal values for these hyperparameters for each of our models individually.

It is important to note that these two approaches enable us to address the Ranking LSCD task from different perspectives. APD offers a direct quantification of a word’s semantic change by calculating the mean of the scores associated with usage pairs from different time periods for each word. In contrast, the cross-validation evaluation method and clustering

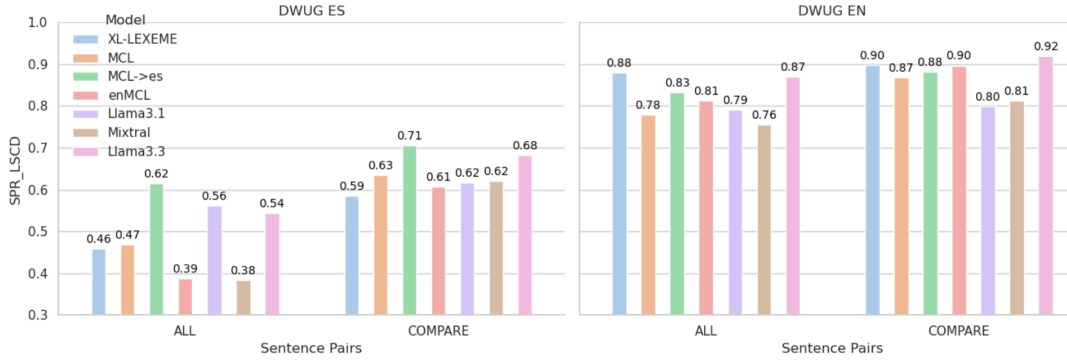


Figure 5.2: APD-based approach with various WiC models and LLMs, Spearman correlation with the gold graded change word scores is reported. “COMPARE” calculates APD across usage pairs consisting of usages from different time periods, while “ALL” employs usage pairs regardless of their time periods.

methods focus on the underlying WSI problem.

5.6.1. APD

We utilize the APD aggregation method to calculate the graded semantic change for each word based on annotations provided by the LLMs, DeepMistake, and XL-LEXEME. Although APD is typically applied to usage pairs originating from different time periods (COMPARE pairs), we also evaluate the performance of the APD-based approach using usage pairs from any time period (ALL pairs) to assess its effectiveness in determining the graded semantic score.

Figure 5.2 presents the results of this experiment. An initial observation reveals that, for English, Llama3.3:70B achieves the best performance, closely followed by XL-LEXEME and enMCL. Although Mixtral:8x7B and Llama3.1:8B yield the lowest results, their Spearman correlations of 0.81 and 0.80, respectively, remain notably high. Finally, although the performance of all models declines when using ALL sentence pairs compared to COMPARE, Llama3.3:70B and XL-LEXEME are the models least impacted by this effect.

For Spanish, the best-performing model is MCL->es, followed by Llama3.3:70B as the second-best. Mixtral:8x7B and Llama3.1:8B demonstrate similar performance—lower than MCL and Llama3.3:70B but higher than XL-LEXEME. However, when using ALL sentence pairs, the performance of all models declines significantly. Notably, in this case, Llama3.1:8B is less affected by this drop compared to Llama3.3:70B and Mixtral:8x7B.

This comparatively weaker performance for Spanish may be explained by several factors. First, large language models are predominantly trained on English-centric corpora, which can result in less robust semantic representations for other languages. Second, the richer morphological structure of Spanish may increase the variability of contextual realizations, making it more challenging for models to produce consistent semantic proximity judgments. Finally, the diachronic nature of the data may exacerbate these issues, as pretrained models are typically optimized for contemporary language usage rather than historical variation.

These observations suggest that the effectiveness of LLM-based approaches is influenced not only by prompt design and model size, but also by language-specific factors.

We conclude that recent prompt optimization techniques are crucial for achieving bet-

ter results on the Graded Change LSCD task, as demonstrated by the performance of Llama3.3:70B. However, medium-sized LLMs such as Mixtral:8x7B and Llama:3.1:8B still underperform compared to smaller and faster specialized LSCD models in both the DWUG EN and DWUG ES datasets. This suggests that, in addition to optimization techniques, the size of the model also significantly influences the results. Furthermore, our findings indicate that using COMPARE sentence pairs is more effective, as they more accurately capture the true relationships between sentence pairs from different time periods.

5.6.2. Cross validation with clustering

Table 5.5: Performance of various methods across cross-validation experiments on the DWUG ES and DWUG EN datasets. The table compares different clustering approaches: AC (Agglomerative Clustering), SC (Spectral Clustering), WSBM (Weighted Stochastic Block Model). APD is reported to evaluate the performance of the models in identifying the different senses of a word, as opposed to quantifying the semantic change of a word without discerning the senses of the target words. The random baseline assigns a score between 1 and 4 randomly.

Methods	Models	Spr_LSCD (ES)	ARI (ES)	Spr_LSCD (EN)	ARI (EN)
WSBM	Llama 3.1	.369 $\pm$.204	.356 $\pm$.116	.845 $\pm$.097	.152 $\pm$.067
SC		.271 $\pm$.461	.102 $\pm$.097	.014 $\pm$.411	-.03 $\pm$.01
AC		.478 $\pm$.286	.073 $\pm$.058	.205 $\pm$.540	-.01 $\pm$.03
APD		.636 $\pm$.236	-	.645 $\pm$.368	-
WSBM	Mixtral	.454 $\pm$.180	.380 $\pm$.104	.776 $\pm$.219	.161 $\pm$.07
SC		.565 $\pm$.141	.092 $\pm$.049	-.171 $\pm$.492	-.03 $\pm$.01
AC		.414 $\pm$.075	.068 $\pm$.027	-.04 $\pm$.525	-.003 $\pm$.02
APD		.567 $\pm$.332	-	.612 $\pm$.280	-
WSBM	Llama 3.3	.659 $\pm$.181	.502 $\pm$.09	.729 $\pm$.241	.183 $\pm$.08
SC		.507 $\pm$.231	.294 $\pm$.05	.302 $\pm$.436	.124 $\pm$.113
AC		.423 $\pm$.184	.228 $\pm$.05	.195 $\pm$.273	-.01 $\pm$.01
APD		.676 $\pm$.195	-	.752 $\pm$.227	-
WSBM	DeepMistake	.727 $\pm$.206	.397 $\pm$.074	.730 $\pm$.212	.231 $\pm$.212
SC		.561 $\pm$.140	.355 $\pm$.036	.520 $\pm$.436	.273 $\pm$.115
AC		.457 $\pm$.320	.341 $\pm$.054	.433 $\pm$.245	.215 $\pm$.128
APD		.653 $\pm$.250	-	.638 $\pm$.292	-
WSBM	XL-LEXEME	.630 $\pm$.377	.452 $\pm$.095	.686 $\pm$.200	.152 $\pm$.059
SC		.484 $\pm$.215	.318 $\pm$.043	.491 $\pm$.176	.137 $\pm$.063
AC		.426 $\pm$.255	.292 $\pm$.087	.143 $\pm$.367	.02 $\pm$.024
APD		.566 $\pm$.354	-	.814 $\pm$.199	-
WSBM	Random Baseline	-.199 $\pm$.310	-.02 $\pm$.184	-.111 $\pm$.254	-.05 $\pm$.147

We employ a cross-validation approach combined with hyperparameter optimization to assess the capacity of the LLMs to address the Graded Change LSCD task, as explained in Section 5.4. For the DWUG ES and DWUG EN datasets, we divide the sets of 60 and 37 target words into five folds. Subsequently, within each cross-validation iteration, we conduct an exhaustive hyperparameter grid search. This process encompasses hyperparameters for the WiC model, WUG, and clustering methods (see Section 5.4). The optimal configuration is identified in the training folds and subsequently evaluated in the testing fold. This procedure yields five performance scores (SPR_LSCD for LSCD and ARI for WSI) for each clustering method, from which we calculate the mean and standard deviations, following the procedure proposed by Schlechtweg et al. [151]. Finally, we incorporate APD with cross-validation across all models to facilitate a comparison between the performance of models that first identify the senses of each word and a method that directly quantifies semantic change for each word.

Table 5.5 presents the results of the experiments conducted on the DWUG ES and DWUG EN datasets. We compare the outcomes obtained from the WSI-based approach, where

predictions made by DeepMistake, XL-LEXEME, and various LLMs are used to construct WUGs.

Regarding the DWUG ES dataset, an initial observation reveals that the combination of DeepMistake and WSBM outperforms DeepMistake with APD, the latter previously considered the state-of-the-art for this dataset. Among the LLMs, Llama3.3:70B achieves the best performance, surpassing not only other LLMs but also XL-LEXEME, although it still trails behind DeepMistake combined with WSBM. Llama3.1:8B and Mixtral:8x7B perform less effectively than Llama3.3:70B, yet both still significantly outperform the random baseline.

While these findings suggest that LLMs can be valuable tools for tackling the LSCD task, they have not yet reached a level of performance that rivals the state-of-the-art model on the DWUG ES dataset. The results indicate that the annotations generated by the LLMs lack the necessary accuracy to effectively distinguish the different senses of the target words, thus limiting the interpretability of the outcomes.

The following experiment presents the results for the English benchmark. As shown in Table 5.5, the WSBM method combined with Llama3.1:8B emerges as the most effective approach to detect semantic change. This performance surpasses that of all other models, regardless of the clustering methods employed or APD, demonstrating Llama3.1:8B’s strong ability to generalize on the DWUG EN dataset and effectively address the WSI task.

Similarly, the results obtained by Mixtral:8x7B combined with WSBM are slightly lower than those of Llama3.1:8B with WSBM, yet they surpass those of DeepMistake, XL-LEXEME, and even Llama3.3:70B across all clustering methods. This underscores Mixtral:8x7B’s strong potential for the LSCD task, particularly when enhanced through techniques such as fine-tuning. Notably, both Llama3.1:8B and Mixtral:8x7B perform better with WSBM than with APD. In contrast, Llama3.3:70B combined with WSBM—its best-performing clustering method—shows a notable drop in performance compared to the APD approach, suggesting that its annotations may lack the quality needed to produce meaningful sense clusters.

The results of this study demonstrate that Llama3.1:8B is the most effective model for addressing the LSCD task using a WSI-based approach on the DWUG EN dataset. Its predictions, when combined with clustering methods, yield results that surpass those of state-of-the-art models employing similar clustering techniques, and even outperform Llama:3.3:70B, which achieved the best results using APD for English.

However, Mixtral:8x7B, Llama:3.1:8B, and Llama3.3:70B struggle to generalize effectively or produce accurate predictions on the 4-point scale for the Spanish dataset. These findings are consistent with the performance of all three models when using APD. A potential explanation for these contrasting results is the significantly larger volume of training data available for English compared to Spanish, which may have impacted the models’ ability to generalize and perform accurately.

5.7. Experiments at WiC annotation level

In this section, we analyze the semantic proximity judgments produced by both LLMs and specialized WiC-based models at the annotation level. Rather than evaluating the models through aggregated LSCD scores, we examine how closely their pairwise usage judgments align with human annotations. This analysis provides a more direct assessment of the quality of the semantic proximity scores generated by each model and complements the LSCD evaluation presented in Section 5.6.

The scores provided by the LLMs employed in this study correspond to annotations for

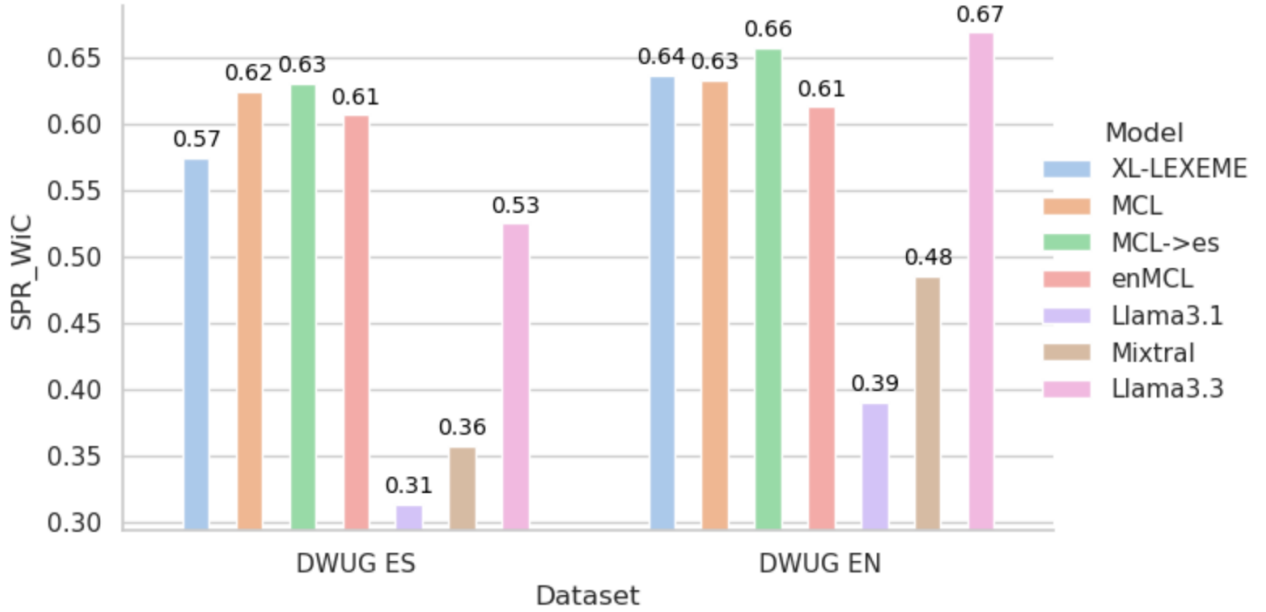


Figure 5.3: Spearman correlation between the annotations provided by human annotators, LLMs, DeepMistake models, and XL-LEXEME.

all usage pairs that were also annotated by human annotators. To ensure consistency, we first filter out any usage pairs that do not have a score within the 4-point scale. We then compute the intersection between the gold-standard usage pairs and the filtered annotations from the LLMs. The remaining gold usage pair annotations are aggregated by calculating the median score across all annotators for each pair. Finally, we report the Spearman correlation between the aggregated gold annotations and the scores produced by the LLMs and WiC models.

Figure 5.3 illustrates the Spearman correlation between the annotations provided by LLMs, DeepMistake models, and XL-LEXEME with the annotations provided by human annotators (SPR_WiC). The results across both datasets indicate that Mixtral:8x7B and Llama3.1:8B underperform compared to the other models. Notably, Llama3.3:70B achieves the highest SPR_WiC for the English dataset, which is consistent with its strong performance on the same dataset using the APD method. For Spanish, the MCL->es model achieves the highest correlation, reaffirming its effectiveness in the Spanish dataset.

The results achieved by Mixtral:8x7B and Llama3.1:8B appear somewhat contradictory: both models perform well in the graded change detection task for English, yet their performance in the annotation-level WiC task is notably low. Llama3.3:70B, on the other hand, presents the second-best performance for Spanish when using APD, but its performance in the annotation task remains low despite this. Overall, the performance of Mixtral:8x7B and Llama3.1:8B is inferior to that of smaller and faster specialized LSCD models, and Llama3.3:70B’s performance for Spanish also falls short of these specialized models.

5.8. Sensitive and Qualitative analysis

In this section, we present a qualitative analysis of the clustering methods used to induce semantic groupings of word usages, with a particular focus on the DWUG-ES dataset. While Chapter 4 evaluated the proposed approaches quantitatively through correlation-based LSCD

Table 5.6: Best-performing Spectral clustering configurations used for qualitative analysis on DWUG-ES dataset using semantic proximity annotations generated by Llama3.3:70B and MCL→es. The table reports the hyperparameter configurations achieving the highest Spearman correlation with the gold graded change scores.

Model	Spearman ρ	Fill Diagonal	Normalize	Quantile	Threshold	Distribution
Llama3.3:70B	0.793	False	False	–	–	discrete-binomial
MCL→es	0.761	True	False	7	0.512	real-normal

metrics, the objective here is to examine the linguistic coherence and interpretability of the induced clusters. The analysis centers on the Spanish dataset because it constitutes the primary resource developed and studied throughout this thesis, allowing for a more detailed examination of the semantic structures produced by the different methods. The goal of this section is therefore to investigate whether the clusters generated by the evaluated methods correspond to linguistically meaningful senses and to analyze how different methodological decisions influence the resulting semantic organization of word usages.

5.8.1. Experimental Configuration for Qualitative Analysis

Based on the quantitative results presented in Section 5.6.2 for the DWUG-ES dataset, we conduct a qualitative analysis of the clusters produced by the best-performing semantic proximity models: MCL→es and Llama3.3:70B. The objective of this analysis is to investigate whether high quantitative performance is also associated with linguistically meaningful sense induction and semantically coherent cluster structures.

For each model, we analyze the clusters generated by the evaluated clustering methods under the hyperparameter configurations that achieved the highest Spearman correlation with the gold graded change scores. In the case of Spectral Clustering, the number of clusters was selected dynamically for each target word using the Silhouette Score criterion, while WSBM inferred the community structure directly from the graph representation. The analysis focuses on representative target words exhibiting different types of polysemy and semantic variation in order to assess the semantic coherence, interpretability, and granularity of the resulting semantic groupings.

Since the goal of this chapter is interpretative rather than statistical, the qualitative analysis is conducted using the best-performing configurations obtained from experiments on the full dataset instead of cross-validation folds. This allows for a more comprehensive inspection of the semantic structures induced by the clustering methods.

5.8.2. Spectral Clustering Analysis

Table 5.6 presents the best Spearman correlation results obtained using Spectral Clustering over graphs constructed from semantic proximity annotations produced by the MCL→es variant of DeepMistake and by Llama3.3. Although the difference in Spearman correlation between both models is relatively small (approximately 0.011 points), the best overall performance was achieved using the probability estimates generated by MCL→es.

The results further reveal that the performance of Spectral Clustering is highly sensitive to the graph construction strategy and its associated hyperparameters. In the best-performing MCL→es configuration, the graph was sparsified by retaining only edges with normalized

semantic proximity scores greater than 0.516, corresponding to the eighth quantile of the semantic proximity distribution. Additionally, the **discretize** assignment strategy was used within Spectral Clustering. These observations suggest that, when WUGs are constructed from probabilistic predictions estimating whether two usages express the same sense, hyperparameter selection plays a central role in shaping the resulting graph structure and, consequently, the semantic partitions produced by the clustering algorithm.

In contrast, the annotations produced by Llama3.3:70B, which emulate ordinal human proximity judgments on a discrete scale from 1 to 4, achieved strong performance without requiring normalization or thresholding procedures. Preliminary experiments applying normalization and threshold-based sparsification to these ordinal annotations resulted in substantially worse performance, suggesting that such transformations distort the semantic structure encoded by the original judgments rather than improving it.

These differences indicate that the effectiveness of graph construction strategies depends strongly on the nature of the semantic proximity signal produced by each model. Consequently, the following qualitative analysis examines whether these quantitative differences are also reflected in the semantic coherence, interpretability, and granularity of the induced clusters.

5.8.2.1. Qualitative Analysis of Induced Clusters

To illustrate the qualitative behavior of the induced clusters, we performed a stratified sampling procedure and selected representative target words from different semantic change categories, including highly polysemous words and semantically stable words.

Figure 5.4 presents the cluster distribution obtained for the word *planta* using semantic proximity annotations generated by Llama3.3:70B, while Figure 5.5 shows the corresponding distribution obtained from DeepMistake annotations. Although both models achieved competitive quantitative performance, the resulting semantic organizations differ substantially in terms of granularity. The clusters induced from Llama3.3:70B annotations form two major semantic groups, suggesting a broader partition of the semantic space. In contrast, the DeepMistake-based representation produces three clusters, yielding a more fine-grained semantic organization that more closely resembles the diversity of senses represented in the gold annotations.

These observations suggest that the type of semantic proximity signal used to construct the WUG strongly influences the resulting cluster structure. In particular, probabilistic semantic proximity estimates appear to preserve more detailed distinctions between usages, whereas ordinal semantic proximity judgments tend to induce broader semantic groupings.

Figure 5.6 presents the cluster distribution obtained for the semantically stable word *historia* (*history*). In contrast to the highly polysemous word *planta*, the gold annotations for *historia* contain a single semantic cluster, reflecting the absence of substantial semantic variation across time periods. Interestingly, both MCL→es and Llama3.3:70B produced the same predicted semantic partition, separating the usages into two distinct clusters despite the semantic stability represented in the gold standard.

These results suggest that the semantic proximity judgments generated by the models may be influenced by contextual, topical, or stylistic variation in addition to lexical meaning. Consequently, usages that remain semantically related in the gold annotations may still be separated into different induced semantic groups, producing a more fragmented semantic organization.

A representative example of this behavior can be observed in the usages shown in Exam-

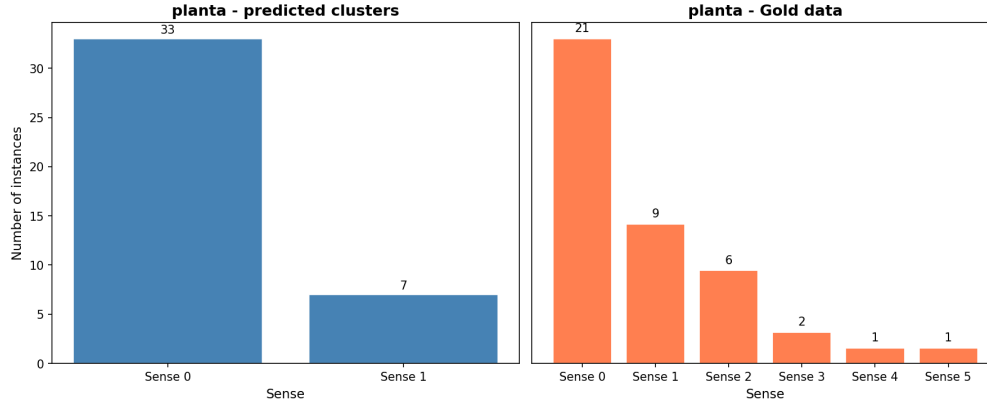


Figure 5.4: SFD for the word *planta* obtained from the annotations generated by Llama3.3:70B using Spectral Clustering. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.



Figure 5.5: SFD for the word *planta* obtained from the annotations generated by MCL→es using Spectral Clustering. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.

ples 5.1 and 5.2, which were assigned to different predicted clusters despite belonging to the same gold semantic partition.

- (5.1) Lo es. Pero la **historia** demuestra que no es un imperativo que siempre se pueda lograr de forma perfecta...
*‘It is. But **history** shows that this is not an imperative that can always be achieved perfectly...’*
- (5.2) No estoy muy cómodo con las emociones. Una de mis **historias** favoritas del cerebro describe a estos tipos de mediana edad...
*‘I am not very comfortable with emotions. One of my favorite brain **stories** describes these middle-aged men...’*

Although the usages occur in different discourse contexts, both express closely related meanings associated with narrative or historical accounts. From a linguistic perspective,

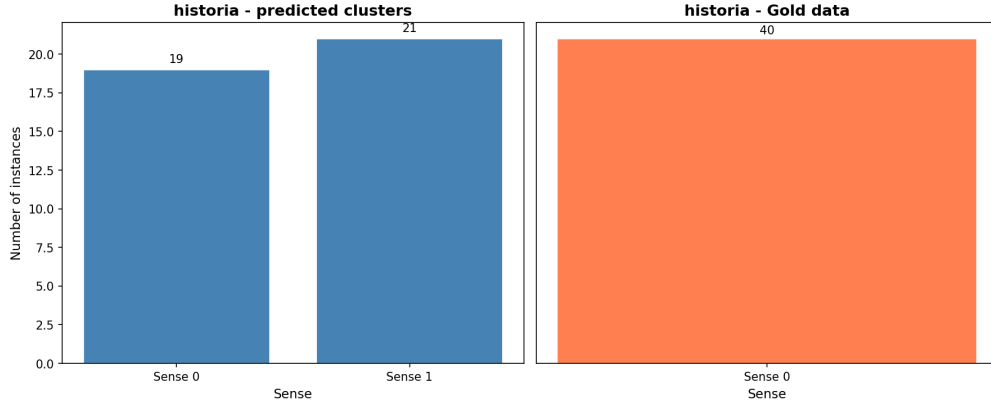


Figure 5.6: SFDs for the word *historia* obtained from the annotations generated by MCL→es and Llama3.3:70B using Spectral Clustering. Both models produced the same cluster distribution, where the bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.

Table 5.7: Best-performing WSBM configurations for the DWUG-ES dataset using semantic proximity annotations generated by Llama3.3:70B and MCL→es. The table reports the hyperparameter configurations achieving the highest Spearman correlation with the gold graded change scores.

Model	Spearman ρ	Fill Diagonal	Normalize	Quantile	Threshold	Distribution
Llama3.3:70B	0.793	False	False	–	–	discrete-binomial
MCL→es	0.761	True	False	7	0.512	real-normal

these usages can still be interpreted as belonging to a coherent semantic category, since the underlying notion of *historia* as a form of narration or account remains preserved across contexts. However, the separation of these usages into different predicted groups suggests that contextual and topical variation may influence the semantic organization induced from the annotations.

Together with the analysis of *planta*, these observations highlight the difficulty of consistently recovering linguistically meaningful sense inventories from automatically induced semantic structures. More importantly, they suggest that strong quantitative performance does not necessarily guarantee that the induced clusters correspond to semantically coherent or linguistically interpretable sense distinctions. While the models capture broad semantic relationships effectively, the qualitative analysis reveals important differences in the level of semantic granularity represented by the induced clusters.

5.8.3. WSBM Analysis

Table 5.7 presents the best-performing WSBM configurations obtained for the DWUG-ES dataset using semantic proximity annotations generated by Llama3.3:70B and MCL→es. In contrast to the results obtained with Spectral Clustering, the highest Spearman correlation was achieved using the annotations produced by Llama3.3:70B, reaching a correlation of 0.793. This result suggests that ordinal semantic proximity judgments expressed on the original annotation scale may provide sufficient information for constructing meaningful Word Usage Graphs when using probabilistic graph-based models such as WSBM.

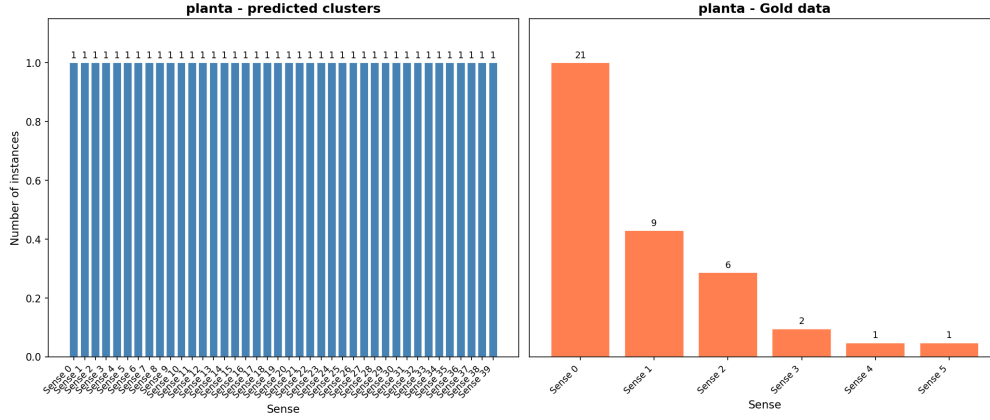


Figure 5.7: SFD for the word *planta* obtained from the annotations generated by Llama3.3:70B using WSBM. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.

These observations indicate that WSBM is capable of effectively exploiting the discrete semantic proximity structure encoded in the annotations generated by Llama3.3:70B. In particular, the strong performance achieved using the original ordinal annotation scale suggests that the semantic information captured by the model is sufficiently structured to support the inference of meaningful semantic distributions across time periods.

In contrast, the best-performing configuration using MCL→es required a more constrained graph construction strategy, including diagonal completion and a threshold over semantic proximity scores derived from normalized distributions. Although the resulting Spearman correlation remained competitive, the lower performance relative to Llama3.3:70B suggests that probabilistic semantic proximity estimates do not necessarily translate into better semantic change representations when used within WSBM. Overall, these results highlight that the interaction between the annotation source and the underlying graph model plays a crucial role in determining the quality of the induced semantic structures.

5.8.3.1. Qualitative Analysis of Induced Clusters

The SFDs for the word *planta* under WSBM, shown in Figures 5.7 and 5.8, further illustrate notable differences in the semantic granularity induced by the two annotation sources. Using Llama3.3:70B annotations (Figure 5.7), the model produces a markedly fine-grained partition, resulting in 40 distinct clusters. In contrast, DeepMistake (Figure 5.8) yields a more compact structure with 21 clusters.

This discrepancy suggests differences in the specificity of the underlying semantic proximity signals, with Llama3.3:70B inducing a more fragmented representation of the usage space. Despite this variation in granularity, both distributions preserve broadly similar usage patterns, indicating that the main semantic distinctions of the word are consistently captured across annotation sources, while the degree of fragmentation varies.

Figures 5.9 and 5.10 present the WSBM cluster distributions for the semantically stable word *historia* (*history*) under the two annotation sources. Using Llama3.3:70B annotations (Figure 5.9), the model induces a single cluster, which matches the gold standard and reflects the absence of semantic variation for this target word. In contrast, DeepMistake annotations (Figure 5.10) result in four distinct clusters, suggesting a more fragmented organization of

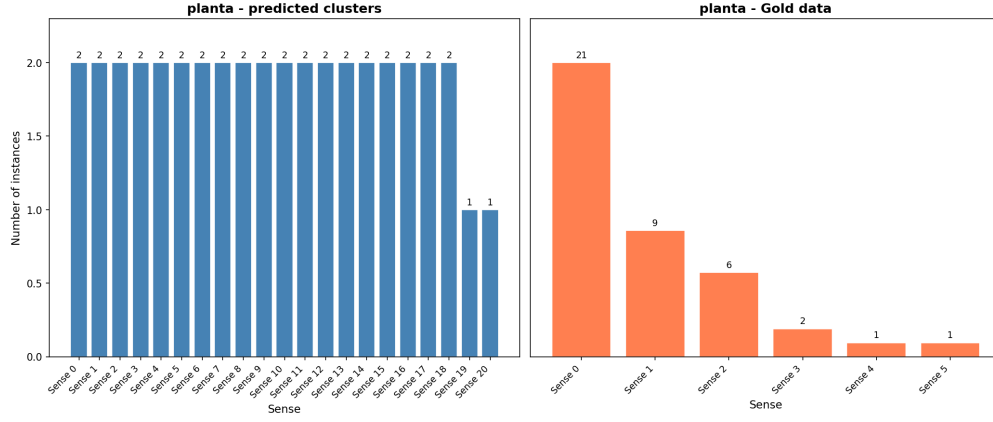


Figure 5.8: SFD for the word *planta* obtained from the annotations generated by MCL→es using WSBM. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.

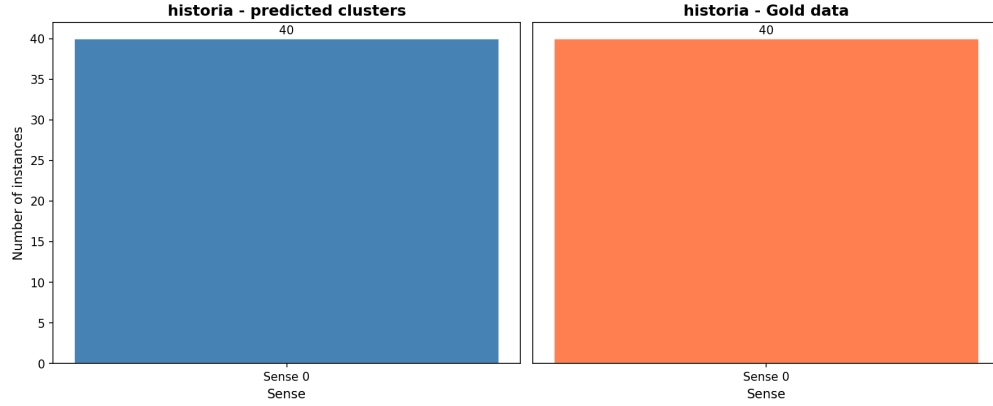


Figure 5.9: SFD for the word *historia* obtained from the annotations generated by Llama3.3:70B using WSBM. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.

the usage space.

This difference indicates that the inferred semantic structure is highly dependent on the nature of the underlying annotations. While Llama3.3:70B preserves the expected semantic stability of the word, DeepMistake introduces finer distinctions that lead to an over-segmentation of an otherwise stable lexical item.

Overall, these results suggest that WSBM is sensitive to the distributional characteristics encoded in the semantic proximity graphs, which can either preserve stable semantic structures or introduce additional subdivisions depending on the annotation source.

A further example illustrating this behavior is provided below, together with a contrasting usage that helps clarify its interpretation:

- (5.3) Probablemente lo sea. De partida, habría que crear salvaguardias políticas y jurídicas que aseguren flujos de energía seguros y fiables, tarea más bien difícil como lo muestra la historia de la región. El nacionalismo y la política de corto plazo han entrampado la razón una y otra vez en Sudamérica.

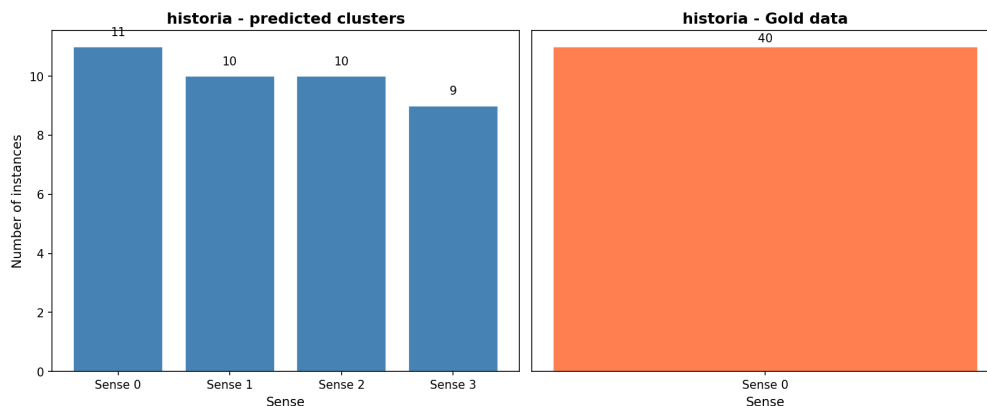


Figure 5.10: SFD for the word *historia* obtained from the annotations generated by MCL→es using WSBM. The bars represent the number of usages assigned to each predicted semantic cluster across historical and modern corpora.

‘It probably is. First of all, political and legal safeguards should be created to ensure safe and reliable energy flows, a rather difficult task as the history of the region shows. Nationalism and short-term politics have repeatedly trapped reason in South America.’

- (5.4) Su genio, su situación, su historia la reducían al papel secundario de levantar partidas. Obsequió al país con esta fruta nacional en tiempo de los Apostólicos (1827); durante la guerra de los siete años, en 1848, y en otras épocas de menos eco en la historia patria. Las partidas y los partidarios fueron siempre populares, circunstancia funesta que procedía de la guerra de la Independencia, una de esas cosas buenas que han sido origen de infinitas cosas detestables.

‘Her genius, her situation, her history reduced her to the secondary role of leading guerrilla groups. She offered the country this national fruit during the time of the Apostolics (1827); during the Seven Years’ War, in 1848, and in other periods with less resonance in national history. Bands and supporters were always popular, a disastrous circumstance that came from the War of Independence, one of those good things that have been the origin of countless detestable things.’

In the first usage, *historia* is employed in an evidential and argumentative sense, where historical experience is invoked as support for a claim about present political and institutional challenges. Here, the term does not function as a narrative object, but rather as a source of inference drawn from past regional events.

In contrast, the second usage refers to a more descriptive and historiographical sense of *historia*, related to the recording and narration of past political and military events. It functions as part of a structured historical account embedded in a broader narrative discourse.

Although these usages differ in their discourse roles—one being argumentative and the other descriptive—they remain semantically related through the overarching concept of *historia* as structured knowledge of past events. This suggests that the induced clustering may be influenced not only by sense distinctions in a strict lexical-semantic sense, but also by differences in discourse function and contextual framing.

Overall, the qualitative analysis reveals clear differences between Spectral Clustering and WSBM in terms of their sensitivity to the underlying annotation structure and hyperparameter choices. As observed in the previous experiments, Spectral Clustering relies on an

explicitly predefined number of clusters, which constrains the resulting partition and leads to more stable and controlled segmentations across different words.

In contrast, WSBM appears to be more sensitive to variations in the WUG and the annotation source. This behavior is observed both for polysemous words such as *planta* and for semantically stable words such as *historia*. In the case of polysemous words, WSBM tends to produce a wider range of cluster granularities depending on the annotation model, while for stable words it may still introduce additional subdivisions that are not present in the gold standard.

These observations suggest that WSBM is more responsive to subtle changes in the semantic proximity signals encoded in the graph, whereas Spectral Clustering enforces a more constrained structure due to its requirement of specifying the number of clusters in advance. As a result, WSBM can capture more fine-grained variations, but it may also reflect contextual or topical variation in cases where the semantic structure is relatively stable.

Importantly, these results also indicate that good quantitative performance does not necessarily guarantee the production of linguistically meaningful or easily interpretable cluster structures. Even when high correlation values are obtained in the evaluation against gold standards, the qualitative inspection shows that the induced clusters may still differ in granularity and may not always align with intuitive sense distinctions.

5.9. Conclusion

This study evaluated the effectiveness of LLMs in addressing the Graded Change LSCD task, focusing on both Spanish and English datasets and comparing their performance against specifically designed and trained LSCD models that represent the state-of-the-art. In addition to evaluating model performance, we also explored the impact of automatic prompt optimization to better adapt LLMs to the task.

A key result of this work is that LLMs, when equipped with optimized prompts, can outperform task-specific LSCD models. In particular, Llama3.3:70B, a large model paired with a prompt selected through automatic optimization, achieves state-of-the-art performance on the English dataset using the APD evaluation. This finding highlights the potential of scaling and prompt engineering to unlock LLMs’ capabilities for semantic change detection. Furthermore, Llama3.3:70B achieves the best result at the annotation level for the English dataset. However, results across other evaluation settings are more mixed: While Llama3.1:8B and Mixtral:8x7B perform well under specific conditions (e.g., WSI-based clustering), they underperform at annotation level compared to the specialized LSCD models. Furthermore, for Spanish, even Llama3.3:70B, despite ranking second in APD-based evaluation, fails to surpass specialized LSCD models—especially in annotation-level performance—demonstrating that challenges remain in cross-lingual generalization and fine-grained semantic interpretation.

These findings underscore the limitations of Llama3.1:8B and Mixtral:8x7B as general-purpose LLMs for the LSCD task, while also highlighting the promising potential of optimized prompts and the use of larger models for improving model performance. We will further investigate factors that may be influencing these results, including the role of quantization—for instance, whether unquantized models yield better performance on fine-grained semantic tasks. Additionally, we plan to shift our optimization target from accuracy to Spearman correlation, which better reflects the nature of graded change detection. Finally, LLMs continue to struggle with the underlying WSI problem, which may be central to improving both performance and the interpretability of results in LSCD tasks.

Chapter 6

Conclusion

This thesis examined LSCD from a sense-oriented perspective, with the overarching goal of understanding how semantic change can be modeled, evaluated, and generalized across languages and modeling paradigms. To this end, it addressed several key research questions: whether optimization for WSI transfers to LSCD; whether LLMs constitute a viable alternative to specialized LSCD systems; whether automatic prompt optimization improves LLM performance and alignment with human semantic judgments; and whether quantitative LSCD performance aligns with the linguistic interpretability of the induced sense structures.

The findings of this thesis emerge from three complementary and interconnected studies. Chapter 3 introduced a Spanish LSCD dataset and established strong aggregation-based baselines for semantic change detection, showing that APD-based approaches remain highly effective for graded LSCD. Chapter 4 investigated clustering-based approaches and demonstrated that optimization for WSI transfers effectively to LSCD across multiple datasets, providing evidence for a close relationship between the two tasks. Finally, Chapter 5 evaluated LLM-based approaches, showing that automatic prompt optimization substantially improves semantic change detection performance and that LLMs can provide competitive semantic proximity judgments, particularly in English.

Together, these studies provide empirical and conceptual evidence that LSCD can be understood as a sense-driven task that directly depends on how effectively models recover or approximate underlying sense-level structure. By combining dataset creation, controlled experimentation, and large language model evaluation, this work advances both the methodological and theoretical foundations of LSCD research.

Contributions

This thesis makes the following main contributions to the field of LSCD:

- The creation of the first LSCD dataset and shared task for Spanish, extending LSCD research beyond predominantly studied languages and enabling systematic comparison of competing approaches.
- An empirical analysis of shared task submissions showing that clustering-based and WiC-based approaches are particularly effective, highlighting the central role of sense-related information in LSCD.
- The introduction of a diachronic sense-annotated dataset that enables controlled experimentation across WSI, WiC, and LSCD within a unified framework.

- A conceptual and practical finding demonstrating an asymmetric transfer between tasks: hyperparameters optimized for WSI consistently yield optimal LSCD performance, while LSCD-optimized configurations do not reliably transfer back to WSI, suggesting that WSI can serve as an effective proxy task for model selection in LSCD.
- A systematic evaluation of LLMs for LSCD using natural-language prompting, demonstrating that LLMs can produce competitive semantic proximity judgments, and showing that automatic prompt selection consistently outperforms manual prompt design across models and languages. Due to computational and budgetary constraints, LLM-based experiments are conducted on English and Spanish datasets, which are widely used LSCD benchmarks and allow direct comparison with prior work.
- A comparative analysis of annotation-level behavior, revealing that large LLMs can approximate human judgments in English but remain less reliable than specialized models in Spanish.

Conceptual reflections

Taken together, the findings of this thesis suggest that effective LSCD modeling fundamentally depends on capturing sense-level structure. The strong asymmetry observed between WSI-to-LSCD and LSCD-to-WSI transfer suggests that semantic change detection is closely tied to the quality of the underlying sense organization rather than constituting a completely independent modeling objective. Likewise, these results suggest that explicitly modeling sense distinctions provides a principled and interpretable framework for semantic change detection, while remaining competitive with approaches that operate directly on contextual representations.

The analysis of LLMs further refines this perspective. While LLMs can, under appropriate prompting conditions, replace WiC-based components in certain LSCD pipelines—most notably for English—their performance remains highly sensitive to prompt formulation and language-specific factors. This sensitivity challenges the assumption that large, general-purpose models can serve as universally reliable solutions for semantic change detection, especially in lower-resource or linguistically distinct settings such as Spanish.

The qualitative analysis further refines this interpretation by showing that strong quantitative performance does not necessarily guarantee that the induced semantic structures are linguistically coherent or easily interpretable. In particular, models with comparable LSCD performance may produce markedly different cluster granularities, especially depending on the type of semantic proximity signal and clustering method employed. These findings suggest that evaluation based solely on correlation-based metrics may obscure important structural differences in the induced sense representations.

Limitations

Despite its contributions, this thesis has several limitations. First, the empirical analyses are limited to English and Spanish in the LLM-based experiments, while German is only used in the clustering-based study. This restriction is due to computational and budgetary constraints associated with prompt optimization and repeated inference for large language models. Additionally, the pretrained models and LLMs employed throughout this work are primarily trained on contemporary text, which may limit their ability to accurately represent historical language usage, particularly for older corpora and longer diachronic time spans. Second, the annotation schemes adopted in this work—both at the sense level and at the graded annotation level—necessarily impose structural assumptions that may influence model

behavior as well as the evaluation outcomes for WSI and LSCD. Third, the experiments involving LLMs rely on a finite set of model architectures and prompting strategies. In particular, the evaluated models were constrained by the computational and financial costs associated with repeated inference during prompt optimization and multilingual evaluation, which limited the inclusion of substantially larger frontier-scale models. Moreover, due to the inherently non-deterministic nature of LLM inference, exact replication of results may be affected by stochastic decoding and model updates over time.

Future research directions

Building on these findings, several avenues for future research emerge. A first direction concerns meaning interpretability [158–161]: Developing methods to assess whether induced clusters and LLM representations correspond to linguistically meaningful senses would enable a deeper validation of both annotation-level predictions and model behavior. Such work could combine clustering analysis, probing techniques, and human-in-the-loop evaluation to bridge the gap between evaluation metrics and the extent to which model outputs correspond to linguistically meaningful sense distinctions.

A second direction concerns a closer examination of JSD as a metric for semantic change. Future work should assess whether this effectiveness arises from its ability to capture variation in sense distributions without requiring explicit correspondence between senses across time periods, and whether alternative divergence measures yield comparable or more interpretable results. Such investigations would help inform the design of future LSCD benchmarks and annotation schemes.

Finally, extending LSCD research to additional languages and resource conditions remains a crucial goal. Future work should also investigate whether substantially larger frontier LLMs exhibit similar or stronger transfer capabilities for LSCD and semantic proximity modeling.

Closing remarks

In conclusion, this thesis provides evidence that LSCD can be understood as a sense-driven task, whose successful modeling depends on representations that capture semantic structure across time. While LLMs offer promising capabilities, particularly when supported by automatic prompt optimization, they do not yet provide a universal solution. Instead, progress in LSCD will continue to rely on the careful integration of sense-aware modeling, principled evaluation, and interpretability-driven analysis.

Appendices

Appendix A. Lemmatization

Manual inspection showed that spaCy sometimes yielded erroneous lemmatization. This happened more frequently for sentences in the old corpus and for tokens at the beginning of sentences as shown in the example below:

Example:

"Decidióse ésta por Teresa la expósita, y así se vio a la vagamunda tomar bajo su amparo a la pobre desheredada como ella."

Lemmatization:

Decidióse este por Teresa el expósita , y así él ver a el vagamunda tomar bajo su amparo a el pobre desheredado como él .

As can be seen, the lemma of the word *Decidióse* was not found, nor was the word converted to lowercase. SpaCy version 3.1.1 with `es_core_news_md` (3.1.0) was used.

Appendix B. Target indices of annotated usages

In the first version of the extracted word usages which were uploaded to the DUREl interface for annotation there were frequent errors for the target word indices. As a result, the wrong target words were marked in these usages. However, annotators were instructed to search for the correct target words and to judge these instead. We corrected the indices for the data provided to participants during the shared task. However, we later noticed that some indices included punctuation immediately following the target word as shown below:

Example

lemma: sexo

context: 136. Los apellidos de familia no varían de terminación para los diferentes **sexos**; y así se dice «don Pablo Herrera», «doña Juana Hurtado», «doña Isabel Donoso». 137 (b).

indexes_target_token: 75:81

After the shared task we uploaded a data version with corrected indices.

Appendix C. Type-based models

Type-based models are designed to encapsulate the various usages of a word into a single, condensed vector representation. These models are characterized by the following features:

- A semantic representation mapping each word to a vector,
- An alignment method; and
- A change measure to quantify variations in word meanings over time.

Although type-based models assign a single vector to each word per time period, they still capture the predominant usage (sense and topical context) within each period. However, since these vectors are learned in separate spaces, they are not directly comparable. To address this, an alignment step is required to map the representations into a common space. Once aligned, a metric is applied to quantify the degree of semantic change over time.

C.1. Semantic representation

A foundational approach for modeling the contextual relationships between words in a corpus is through the construction of a co-occurrence matrix. This structure captures statistical patterns by counting how frequently words appear near one another within a specified context window k , k , which can span a sentence, paragraph, or a fixed number of tokens. The rows and columns of this matrix represent vocabulary items, and each cell $M_{i,j}$ contains a value reflecting the association between a target word w_i and a context word w_j .

Traditionally, these values correspond to raw co-occurrence counts. However, to better emphasize informative associations and suppress frequent but uninformative co-occurrences, these raw counts are often weighted or normalized. One of the most widely used weighting schemes is Positive Pointwise Mutual Information (PPMI), which captures how much more often two words co-occur than would be expected if they were statistically independent. Formally, the PPMI value for a word-context pair $(w|c)$ is defined as:

$$M_{i,j}^{PPMI}(w, c) = \max \left\{ \log \frac{P(w, c)}{P(w) P(c)} - \alpha, 0 \right\}$$

where $P(w, c)$ is the probability of the word and context co-occurring, $P(w)$, $P(c)$ are their marginal probabilities, and $\alpha > 0$ which provides a smoothing bias [12, 162, 163].

Despite its interpretability, the co-occurrence matrix is typically high-dimensional and sparse, which poses challenges for computational efficiency and generalization. To address this, dimensionality reduction techniques are applied. Among the most prominent are Singular Value Decomposition (SVD) [164] and Principal Component Analysis (PCA) [165], which reduce the matrix to a lower-dimensional space while preserving the most significant variance or structure. These methods were foundational in early distributional semantic models.

Alternatively, Random Indexing (RI) [166] offers a scalable, incremental approach to reduce dimensionality without the need for explicit matrix factorization. In this method, words and contexts are assigned sparse, randomly generated vectors, and word representations are built by summing context vectors encountered in a sliding window. Temporal Random Indexing (TRI) [167] extends this technique to diachronic corpora by creating time-specific random vectors, enabling the tracking of semantic change over time.

Another prominent type of semantic representation within type-based models is static word embeddings, which learn dense, fixed-length vector representations for each word based on its distributional context in large corpora. Among the most influential models in this category are Word2Vec’s two architectures: Skip-gram with Negative Sampling (SGNS) and Continuous Bag of Words (CBOW) [94, 95]. These models are prediction-based, learning word vectors not by counting co-occurrences directly, but by optimizing a task-specific objective that distinguishes real word-context pairs from noise.

In the Skip-gram architecture, the model is trained to predict the surrounding context words given a target word. At the time SGNS was introduced, computing the full softmax over very large vocabularies was computationally expensive, motivating the use of a binary classification objective based on negative sampling instead. Specifically, the model maximizes the probability of observed word-context pairs and minimizes it for randomly sampled (negative) pairs. Formally, for a target word w and a context word c , the SGNS objective is:

$$\log \sigma(\vec{w} \cdot \vec{c}) + \sum_{i=1}^k \log \sigma(-\vec{w} \cdot \vec{c}_i)$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function, k is the number of negative samples, and $P_n(c)$ is the noise distribution from which negative contexts are drawn. This objective encourages the dot product of actual word-context pairs to be high while penalizing randomly paired ones.

Conversely, the CBOW model operates in the opposite direction: it predicts a target word given the average of the embeddings of its surrounding context words. This makes it more efficient for smaller datasets and more suited for frequent words. The CBOW model seeks to maximize the log-likelihood of the target word w given its context $\{c_1, c_2, \dots, c_n\}$:

$$\log P(w \mid c_1, \dots, c_n) = \log \frac{\exp(\vec{w} \cdot \sum_{i=1}^n \vec{c}_i)}{\sum_{w' \in V} \exp(\vec{w}' \cdot \sum_{i=1}^n \vec{c}_i)}$$

C.2. Alignment methods

Since the vector representations for time periods A and B are learned independently, they are not directly comparable. Therefore, it is necessary to align them to make meaningful comparisons across time.

Vector Initialization This method relies on the SGNS model to generate word embeddings. The process begins by training the model on the first time period T^1 , and the resulting embeddings are then used to initialize the model for the subsequent period T^2 . This procedure is repeated sequentially up to the final time period T^n . The underlying assumption is that if a word undergoes semantic change over time, its vector representation will differ substantially between periods, reflecting shifts in contextual usage [11].

Temporal Referencing (TR) Temporal Referencing is an alternative approach to address the problem of incomparable word embeddings across different time periods. The method proposed by Dubossarsky et al. [168] involves appending a temporal marker to each target word, resulting in time-specific tokens of the form $w_i_t_i$, where t_i denotes the time period. This strategy ensures that all word representations are learned within a shared vector space, thereby making them directly comparable across time.

Orthogonal Procrustes (OP) The Orthogonal Procrustes method is a widely used

technique for aligning vector spaces. Since embeddings trained on corpora from different time periods reside in distinct vector spaces, direct comparison of word vectors across time is not meaningful without alignment. The OP method addresses this issue by finding the optimal orthogonal transformation—i.e., a rotation or reflection without scaling—that best aligns one embedding space to another. Given two embedding matrices X and Y of dimension $n \times d$, the goal is to find an orthogonal matrix $R \in \mathbb{R}^{d \times d}$ that minimizes the Frobenius norm of the difference between the transformed source matrix and the target matrix:

$$R = \arg \min_{R \in O(d)} \|XR - Y\|_F$$

The solution to this optimization problem is obtained through Singular Value Decomposition. Specifically, if $X^\top Y = U\Sigma V^\top$, then the optimal rotation matrix is given by $R = UV^\top$. This aligned space allows for meaningful comparison of word vectors across different time periods, facilitating the detection and analysis of lexical semantic change [12].

Column Intersection (CI) The Column Intersection method offers a straightforward strategy for comparing word embeddings across different time periods without applying complex alignment transformations. Let $\mathbf{E}^A \in \mathbb{R}^{|V^A| \times d}$ and $\mathbf{E}^B \in \mathbb{R}^{|V^B| \times d}$ be the word embedding matrices for time periods A and B , where V^A and V^B are the respective vocabularies and d is the dimensionality of the embeddings. Since V^A and V^B may differ, the method focuses on the intersection vocabulary $V^{A \cap B} = V^A \cap V^B$. From the original embedding matrices, submatrices $\mathbf{E}_\cap^A \in \mathbb{R}^{|V^{A \cap B}| \times d}$ and $\mathbf{E}_\cap^B \in \mathbb{R}^{|V^{A \cap B}| \times d}$ are extracted, containing only the embeddings of the shared words. These submatrices establish a common basis over the intersecting vocabulary, enabling direct comparison of embeddings from different time periods [12].

C.3. Change measures

Once word embeddings from different time periods have been aligned into a shared vector space, it becomes possible to quantify how a word’s meaning has evolved over time. To this end, change measures are applied to compute the difference between a word’s representations across periods, yielding a numerical estimate of semantic change.

Cosine Distance (CD) measures the angle between two vectors (e.g., embeddings in this case) from different time periods. CD is based on cosine similarity which is defined as follows:

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

Then, the cosine distance (CD) is defined as:

$$CD(\mathbf{x}, \mathbf{y}) = 1 - \cos(\mathbf{x}, \mathbf{y})$$

Euclidean Distance (ED) measures quantify the semantic change of a word across time by computing the straight-line distance between its vector representations in different time periods. Given two embedding vectors of a word v_{t_1} and v_{t_2} , corresponding to time periods t_1 and t_2 , the Euclidean distance is defined as:

$$d(\mathbf{v}_{t_1}, \mathbf{v}_{t_2}) = \|\mathbf{v}_{t_1} - \mathbf{v}_{t_2}\|_2 = \sqrt{\sum_{i=1}^n (v_{t_1}^{(i)} - v_{t_2}^{(i)})^2}$$

Jaccard Similarity (JS) measures the overlap between discrete sets that represents word meanings, such as context words or sense clusters. JS quantifies the degree of overlap between two sets by computing the ratio of their intersection to their union. Formally, given two sets of context features A and B , the Jaccard similarity is defined as:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

where $|A \cap B|$ denotes the cardinality of the intersection between sets A and B , and $|A \cup B|$ denotes the cardinality of their union. A and B represent the sets of significant contexts or nearest neighbors of a target word at different time periods. A lower Jaccard similarity indicates a greater degree of semantic change due to reduced contextual overlap [12, 87].

Appendix D. Experiments with models based on substitution

We also considered adding other architectures into our benchmark such as Amrami and Goldberg [169]. The author’s work uses a model to generate substitutes given a list of target words, these substitutes are assigned to a set of dynamic clusters that generate predictions for a test dataset. We use this model to generate a baseline on our data, but we do not report its results due to its poor performance on our data. These results may be due to several factors: this method was originally used for data in English, while our data is in German. Similarly, another factor contributing to the suboptimal results is the sensitivity of standard clustering algorithms with off-the-shelf BERT to historical spelling variations, which can also be compounded by grammatical variations in some words [14].⁵⁸

Appendix E. Cross-Validation Results and Hyperparameter Configurations

E.1. Results per fold

Tables E.1 and E.2 display the results of our experiments for each clustering method across individual test partitions as part of the 5-fold cross-validation. Both tables share common columns, including “fill-diag” representing the fill diagonal parameter, “ude” indicating the use of disconnected edges, and “mod-hyp” representing model hyperparameters (see Section 4.5). Notably, the “test-SPR” column in Table E.1 shows the best SPR results considering optimized parameters from the WSI task, whereas the “test-ARI” column in the Table E.2 displays the optimal results for the LSCD task, considering parameters optimized in the LSCD task.

In Table E.1, we observe that for Spectral Clustering with Silhouette, the selected model is the same for four out of five folds. This indicates that the method remains stable, consistently performing well under the same set of parameters; in contrast, in Table E.2 the method shows more variation. We observe a similar picture for WSBM showing stability when optimized for WSI, but less for LSCD.

⁵⁸ Ansell et al. [170] is another state-of-the-art model in the WSI task, we did not test it on our data because the model is trained on texts in English and it is too expensive to train it for other languages.

We observe for both tasks that setting the parameter “ude” (`use_disconnected_edges`) for Correlation Clustering and WSBM to True consistently yields superior results. Regarding the **distribution** parameter (“mod-hyp”) of the WSBM, it becomes evident that when set to “poisson” it obtains optimal performance across all five folds for WSI. For LSCD, the geometric distribution dominates in four out of five folds. Furthermore, we observe that for the WiC path parameter, the value “ALL” dominates across all methods for WSI. This pattern, however, is not consistently observed when the models are optimized for the LSCD task.

Tables E.3 and E.4 provide the detailed hyperparameter configurations selected during the 5-fold cross-validation experiments on the Spanish dataset. As in the German experiments, both tables report the parameters selected for each clustering method and fold.

Compared to the German dataset, the Spanish experiments exhibit greater variability in the selected configurations across folds. No clustering method consistently converges to a single hyperparameter configuration, suggesting that the optimization process is more sensitive to the composition of the training partitions. This observation is consistent with the larger number of target words and the greater diversity of semantic change patterns represented in the Spanish dataset.

A notable result is the stability of the semantic proximity model. Across virtually all methods and folds, the optimization procedure selects the same WiC model (`MCL→es`), indicating that the quality of the semantic proximity predictions plays a more important role than variations in the clustering algorithms themselves. This contrasts with the German experiments, where different WiC models were selected depending on the optimization criterion and clustering method.

Regarding clustering-specific parameters, the use of disconnected edges remains strongly preferred for most methods, particularly for Chinese Whispers, WSBM, and the Spectral Clustering variants. Correlation Clustering constitutes a partial exception, where both settings appear among the selected configurations depending on the fold and optimization criterion. This result provides further evidence that retaining disconnected components is generally beneficial for sense induction and semantic change detection.

For WSBM, no single probability distribution dominates across all folds. While the Poisson distribution is preferred when optimizing for WSI, LSCD optimization frequently selects Binomial and Geometric distributions. This contrasts with the clearer pattern observed in the German dataset and suggests that the optimal probabilistic assumptions of WSBM are more dataset-dependent than previously anticipated.

Finally, unlike the German dataset, where WSI optimization produced notably more stable configurations than LSCD optimization, the Spanish dataset exhibits substantial variability under both optimization criteria. Nevertheless, the cross-validation results reported in Chapter 4 show that competitive LSCD performance can still be obtained from configurations selected through WSI optimization, supporting the overall hypothesis that WSI provides useful guidance for model selection in semantic change detection.

Tables E.5 and E.6 present the detailed hyperparameter configurations selected during the 5-fold cross-validation experiments on the English dataset. As in the previous analyses, Table E.5 reports the configurations selected through WSI optimization together with their corresponding LSCD performance, whereas Table E.6 reports the configurations selected through LSCD optimization together with their resulting WSI performance.

A striking observation is the remarkable consistency of the selected semantic proximity model. Across all clustering methods, folds, and optimization criteria, the same WiC model

(**enMCL**) is selected. This suggests that the quality of the semantic similarity predictions constitutes a dominant factor in the English dataset and that the optimization process consistently identifies the same underlying representation as the most informative for both WSI and LSCD.

Compared to the Spanish dataset, several clustering methods exhibit greater stability across folds. In particular, Chinese Whispers and WSBM repeatedly select very similar hyperparameter configurations. For Chinese Whispers, the use of weighted graphs together with logarithmic or linear edge weighting dominates both optimization settings. Likewise, WSBM consistently selects `use_disconnected_edges=True` and most frequently chooses the Binomial distribution, indicating a strong preference for a specific probabilistic model of graph structure. Only occasional selections of the Geometric distribution appear when optimizing directly for LSCD.

The parameter `use_disconnected_edges` is selected for nearly all methods and folds. The only consistent exception is Correlation Clustering, where disconnected edges are generally disabled. This pattern mirrors the observations made for the German and Spanish datasets and suggests that preserving disconnected components is beneficial for most clustering approaches in semantic change detection.

Another noteworthy difference with respect to the German and Spanish datasets is the widespread use of graph binarization. Whereas binarization played a limited role in the previous experiments, it is selected in the vast majority of folds for the English dataset, regardless of the optimization criterion. This finding indicates that transforming semantic similarity graphs into binary structures may be particularly advantageous in this setting.

Finally, the English dataset reveals less separation between the hyperparameter configurations selected through WSI optimization and those selected through LSCD optimization than was observed for the German dataset. Although some variation remains, many of the selected configurations share common characteristics across both optimization criteria. This suggests that, despite the lower overall performance levels observed in the English experiments, the optimization procedures consistently identify similar graph structures and clustering behaviors. The results therefore provide additional evidence that WSI and LSCD rely on closely related information, even when the optimal clustering configurations vary across methods.

Appendix F. Annotation Guidelines

Einführung.

Ihre Aufgabe ist es, Wortverwendungen einer Bedeutungsbeschreibung zuzuordnen. Ihnen werden Sätze wie in (1) vorgelegt, für die Sie die Bedeutung des markierten Zielworts, hier *Affentheater*, einer Bedeutungsbeschreibung wie in (2a) und (2b) zuweisen sollen.

(F.1) Da ringt selbst Elisa Kirschbaum um ihre Fassung, Meyer übrigens auch um seine, bittet ihn zu sich, der ganze Ton hier, dieses **Affentheater**.

(F.2) a. Theater mit dressierten Affen
b. übertriebenes, unsinnig Getue

Aufgabenstruktur.

Sie bekommen ein ODS-Tabellendokument wie in Tabelle F.1 illustriert. Eine Zeile der Tabelle entspricht einem Satz. Die Spalten entsprechen Bedeutungsbeschreibungen für das

jeweilige Zielwort. Das Zielwort ist in jedem Satz fett markiert. Ihre Aufgabe ist es, für jeden Satz die Bedeutungsbeschreibung zu markieren, welche am besten die Bedeutung des Zielworts im jeweiligen Satz beschreibt. Falls keine der Beschreibungen zutrifft, haben Sie die Möglichkeit, die Spalte “andere” zu markieren. Wenn Sie keine Entscheidung treffen können, markieren Sie bitte keine der Bedeutungsbeschreibungen und tragen nur eine Begründung in die Spalte “Kommentar” ein. Bitte wählen Sie nicht mehr als eine Bedeutungsbeschreibung aus und markieren Sie diese mit einem “x” in der jeweiligen Spalte.

Historische Sprachdaten.

Die Sätze für diese Annotationsaufgabe wurden aus historischen Korpora ausgelesen.

Da sich Sprache mit der Zeit verändert, kann es sein, dass Worte anders benutzt werden, als Sie es gewohnt sind. Wenn Sie sich unsicher über die Bedeutung eines Wortes oder einer Konstruktion in einem Satz sind, versuchen Sie sie aus der Bedeutung des Kontexts zu erschließen.

Die Sätze können sehr kurz oder sehr lang sein und ungrammatisch erscheinen. Außerdem können Worte aufgrund älterer Orthographie anders geschrieben sein, als Sie es gewohnt sind. Zudem wurden einige Buchstaben bei der automatischen Texterkennung eingescannter Dokumente falsch erkannt. Es wurde versucht, die Leserlichkeit durch Normalisierung spezieller Buchstaben zu moderner Orthographie zu verbessern.

Versuchen Sie, diese Umstände zu ignorieren; konzentrieren Sie sich nur auf die Bedeutung des Zielwortes in seinem Kontext. Wenn Sie einen Satz zu fehlerhaft finden, um ihn zu verstehen, die Verwendung des Zielwortes mehrdeutig ist, oder die beiden Verwendungen des Zielwortes nicht zusammenpassen (d. h., nicht dasselbe Lemma haben), notieren Sie dies bitte auch in der Kommentarspalte.

Durchführung.

Während der Annotation der Sätze können Sie immer zu vorherigen Bewertungen zurückgehen und diese ändern; z. B. falls Sie Ihre Meinung ändern, nachdem Sie mehr Informationen bekommen haben.

Sie müssen nicht die ganze Datei in einer Sitzung annotieren. Wenn Sie einen Kommentar hinterlassen wollen, können Sie diesen in das Kommentarfeld eintragen.

Es kann hilfreich sein, die Rechtschreibprüfung zu deaktivieren, um nicht durch zusätzliche Hervorhebungen gestört zu werden.

Abschluss.

Bitte stellen Sie sicher, dass Sie nichts in der Datei ändern außer Spaltenbreite, Schriftgröße, Ihren Bewertungen und Kommentaren. Schicken Sie das annotierte Dokument an schlecdk@ims.uni-stuttgart.de. Wenn Sie noch Fragen zur Aufgabe haben, zögern Sie nicht, diese zu stellen.

Table E.1: Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the WSI task for the DWUG DE dataset. Each row corresponding to a method signifies a distinct fold.

Method: Chinese Whispers								
quantile	WiC-path	binarize	ude	mod-hyp	fill-diag	dev-ARI	test-ARI	test-SPR
6	ALL	True	True	w: lin	True	0.745	0.683	0.975
6	ALL	True	True	w: log	True	0.667	0.809	0.900
6	MCL→RU	True	True	w: log	True	0.714	0.637	0.400
5	MCL→RU	True	True	w: log	True	0.778	0.403	0.564
6	ALL	False	True	w: lin	False	0.670	0.580	1.000
Method: Correlation Clustering								
3	ALL	False	False	-	True	0.757	0.623	0.975
1	ALL	False	False	-	True	0.735	0.746	0.990
3	ALL	False	False	-	True	0.765	0.604	0.900
4	MCL→RU	True	True	-	True	0.757	0.387	0.700
5	enMCL	True	True	-	True	0.729	0.749	0.800
Method: WSBM								
0	ALL	True	True	poisson	False	0.718	0.679	0.975
0	ALL	True	True	poisson	False	0.691	0.781	0.999
0	ALL	True	True	poisson	True	0.727	0.679	0.600
5	ALL	True	True	poisson	True	0.737	0.615	0.999
0	ALL	True	True	poisson	False	0.705	0.794	1.000
Method: Spectral Clustering/s								
4	ALL	False	True	-	True	0.838	0.753	0.975
4	ALL	False	True	-	True	0.818	0.818	0.999
4	ALL	False	True	-	True	0.839	0.754	0.700
3	enMCL	False	True	-	False	0.811	0.629	0.700
4	ALL	False	True	-	True	0.811	0.902	1.000
Method: Spectral Clustering/c								
3	enMCL	False	True	-	False	0.788	0.838	0.872
4	ALL	True	True	-	False	0.794	0.748	0.900
3	enMCL	False	True	-	True	0.818	0.598	0.900
3	enMCL	False	True	-	False	0.811	0.753	0.700
3	enMCL	False	True	-	False	0.791	0.837	0.800
Method: Spectral Clustering/e								
5	ALL	True	True	-	False	0.680	0.652	0.975
0	ALL	True	True	-	True	0.629	0.807	0.900
0	ALL	True	True	-	True	0.643	0.738	0.700
0	ALL	False	True	-	True	0.711	0.390	0.462
5	ALL	False	True	-	True	0.682	0.580	1.000

Table E.2: Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the LSCD task for the DWUG DE dataset. Each row corresponding to a method signifies a distinct fold.

Method: Chinese Whispers								
quantile	WiC-path	binarize	ude	mod-hyp	fill-diag	dev-SPR	test-SPR	test-ARI
6	ALL	True	True	w: lin	True	0.900	0.975	0.683
8	ALL	True	True	w: log	False	0.862	0.900	0.690
6	MCL→ru	False	True	w: top	False	0.896	0.400	0.600
6	ALL	True	True	w: top	False	0.955	0.564	0.488
5	MCL→ru	False	True	w: log	False	0.837	1.000	0.547
Method: Correlation Clustering								
4	MCL→ru	True	True	-	False	0.928	0.975	0.692
1	ALL	False	False	-	True	0.914	0.900	0.793
5	MCL→ru	True	True	-	False	0.901	0.700	0.572
5	MCL→ru	True	True	-	True	0.934	0.600	0.540
1	ALL	False	False	-	True	0.936	0.400	0.735
Method: WSBM								
6	ALL	True	True	geometric	True	0.926	0.975	0.673
5	ALL	True	True	poisson	False	0.892	0.900	0.726
6	ALL	True	True	geometric	False	0.939	0.700	0.517
0	enMCL	True	True	geometric	True	0.955	0.564	0.356
6	ALL	True	True	geometric	True	0.939	1.000	0.794
Method: Spectral Clustering/s								
4	ALL	False	True	-	True	0.940	0.975	0.753
4	ALL	False	True	-	True	0.939	0.999	0.818
3	MCL→ru	False	True	-	True	0.973	0.700	0.686
1	enMCL	False	True	-	True	0.962	0.700	0.619
2	MCL→es	False	True	-	True	0.946	0.800	0.701
Method: Spectral Clustering/c								
3	enMCL	True	True	-	False	0.928	0.872	0.758
0	ALL	True	True	-	True	0.944	0.900	0.807
0	ALL	True	True	-	True	0.936	0.700	0.651
4	enMCL	False	True	-	True	0.961	0.600	0.737
2	MCL→ru	False	True	-	True	0.958	0.800	0.560
Method: Spectral Clustering/e								
4	MCL→ru	True	True	-	True	0.833	0.516	0.317
4	MCL	False	True	-	True	0.840	0.999	0.833
7	enMCL	True	True	-	False	0.899	0.600	0.405
0	enMCL	True	True	-	True	0.944	0.462	0.299
7	MCL→es	False	True	-	False	0.897	0.400	0.278

Table E.3: Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the WSI task for the DWUG ES dataset. Each row corresponding to a method signifies a distinct fold.

Method: Chinese Whispers								
quantile	WiC-path	binarize	ude	mod-hyp	fill-diag	dev-ARI	test-ARI	test-SPR
7	MCL→es	False	True	w: log	False	.489	.530	.678
6	MCL→es	False	True	w: lin	False	.499	.483	.649
6	MCL→es	False	True	w: lin	False	.530	.359	.111
7	MCL→es	False	True	w: log	False	.492	.520	.629
6	MCL→es	False	True	w: lin	False	.499	.484	.885
Method: Correlation Clustering								
6	MCL→es	False	False	-	True	.358	.632	.757
1	MCL→es	False	False	-	True	.415	.407	.637
3	MCL→es	False	False	-	True	.436	.327	.486
4	MCL→es	False	False	-	True	.431	.342	.641
5	MCL→es	False	False	-	True	.431	.351	.783
Method: WSBM								
9	MCL→es	False	True	poisson	False	.501	.439	.556
4	MCL→es	False	True	poisson	True	.545	.472	.559
4	MCL→es	False	True	poisson	True	.562	.413	.211
4	MCL→es	False	True	poisson	True	.549	.465	.734
4	MCL→es	False	True	binomial	True	.530	.431	.705
Method: Spectral Clustering/s								
1	MCL→es	False	True	-	True	.354	.501	.838
4	MCL→es	False	True	-	True	.369	.424	.559
9	MCL→es	False	True	-	True	.428	.181	.493
3	MCL→es	False	True	-	True	.401	.314	.324
4	MCL→es	False	True	-	False	.378	.336	.674
Method: Spectral Clustering/c								
2	MCL→es	False	True	-	True	.329	.513	.691
3	MCL→es	True	True	-	False	.362	.390	.563
3	MCL→es	False	True	-	True	.405	.219	.167
3	MCL→es	False	True	-	True	.381	.314	.099
3	MCL→es	False	True	-	True	.365	.380	.580
Method: Spectral Clustering/e								
1	MCL→es	True	True	-	True	.317	.413	.698
4	MCL→es	True	True	-	True	.334	.364	.584
4	MCL→es	True	True	-	True	.377	.176	.479
4	MCL→es	False	True	-	True	.360	.257	.394
3	MCL→es	False	True	-	True	.334	.339	.819

Table E.4: Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the LSCD task for the DWUG ES dataset. Each row corresponding to a method signifies a distinct fold.

Method: Chinese Whispers								
quantile	WiC-path	binarize	ude	mod-hyp	fill-diag	dev-SPR	test-SPR	test-ARI
5	MCL→es	False	True	w: log	True	.667	.636	.500
8	MCL→es	False	True	w: lin	False	.733	.630	.467
5	MCL→es	False	True	w: lin	False	.753	.591	.353
6	MCL→es	False	True	w: lin	False	.732	.381	.538
6	MCL→es	False	True	w: lin	True	.691	.785	.406
Method: Correlation Clustering								
2	MCL→es	False	True	-	False	.615	.757	.632
2	MCL→es	False	False	-	False	.721	.630	.410
2	MCL→es	False	True	-	False	.717	.442	.320
8	MCL→es	False	True	-	False	.716	.641	.341
1	MCL→es	False	False	-	True	.681	.783	.349
Method: WSBM								
5	MCL→es	True	True	binomial	False	.653	.933	.617
6	MCL→es	True	True	binomial	False	.805	.478	.507
4	MCL→es	True	True	geometric	True	.750	.211	.413
6	MCL→es	True	True	geometric	False	.738	.713	.401
6	MCL→es	True	True	binomial	True	.729	.676	.462
Method: Spectral Clustering/s								
4	MCL→es	False	True	-	True	.604	.694	.506
8	MCL→es	False	True	-	True	.678	.577	.438
3	MCL→es	False	True	-	True	.678	.479	.183
1	MCL→es	False	True	-	True	.711	.324	.314
1	MCL→es	False	True	-	True	.618	.783	.414
Method: Spectral Clustering/c								
3	MCL→es	False	True	-	True	.457	.690	.490
9	MCL→es	False	True	-	False	.533	.253	.255
7	MCL→es	False	True	-	True	.597	.000	.157
1	MCL→es	False	True	-	True	.608	.035	.296
2	MCL→es	False	True	-	True	.525	.529	.381
Method: Spectral Clustering/e								
4	MCL→es	False	True	-	True	.613	.677	.404
4	MCL→es	False	True	-	True	.690	.577	.343
4	MCL→es	False	True	-	True	.708	.457	.146
4	MCL→es	False	True	-	True	.730	.289	.244
7	MCL→es	False	True	-	True	.608	.843	.368

Table E.5: Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the WSI task for the DWUG EN dataset. Each row corresponding to a method signifies a distinct fold.

Method: Chinese Whispers								
quantile	WiC-path	binarize	ude	mod-hyp	fill-diag	dev-ARI	test-ARI	test-SPR
7	enMCL	True	True	w: log	True	.336	.257	.762
4	enMCL	True	True	w: lin	False	.299	.385	.738
2	enMCL	True	True	w: log	True	.343	.305	.714
7	enMCL	True	True	w: log	True	.388	.113	.964
6	enMCL	True	True	w: lin	False	.355	.157	.321
Method: Correlation Clustering								
9	enMCL	True	False	-	True	.103	.067	.119
9	enMCL	True	False	-	True	.070	.187	.238
9	enMCL	False	False	-	True	.094	.102	.143
8	enMCL	True	False	-	True	.119	-.011	.393
5	enMCL	True	False	-	True	.090	.118	.643
Method: WSBM								
3	enMCL	True	True	binomial	False	.145	.143	.905
8	enMCL	True	True	binomial	False	.156	.106	.833
9	enMCL	True	True	binomial	False	.142	.132	.107
8	enMCL	True	True	binomial	False	.127	.219	.697
1	enMCL	True	True	binomial	False	.155	.093	.357
Method: Spectral Clustering/s								
1	enMCL	False	True	-	False	.110	.142	.238
1	enMCL	True	True	-	False	.112	.132	.810
2	enMCL	True	True	-	True	.117	-.008	.714
1	enMCL	True	True	-	False	.119	.104	.607
1	enMCL	True	True	-	False	.130	.062	.429
Method: Spectral Clustering/c								
1	enMCL	True	True	-	False	.097	.142	.000
1	enMCL	True	True	-	False	.111	.091	.875
1	enMCL	True	True	-	False	.102	.129	.214
1	enMCL	False	True	-	False	.107	.104	-.464
1	enMCL	True	True	-	False	.117	.065	.179
Method: Spectral Clustering/e								
2	enMCL	True	True	-	False	.106	.088	.691
1	enMCL	True	True	-	False	.084	.127	.286
3	enMCL	True	True	-	True	.116	.025	.464
9	enMCL	True	True	-	True	.107	.033	.464
3	enMCL	True	True	-	True	.111	.049	.071

Table E.6: Detailed hyperparameters employed in constructing the WUG and the models during cross-validation, outlined for each fold and model within the context of the LSCD task for the DWUG EN dataset. Each row corresponding to a method signifies a distinct fold.

Method: Chinese Whispers								
quantile	WiC-path	binarize	ude	mod-hyp	fill-diag	dev-SPR	test-SPR	test-ARI
1	enMCL	True	True	w: log	False	.834	.810	.378
9	enMCL	True	True	w: lin	True	.836	.619	.365
3	enMCL	True	True	w: lin	True	.867	.429	.336
2	enMCL	True	True	w: top	False	.795	.811	.071
6	enMCL	True	True	w: lin	False	.850	.321	.133
Method: Correlation Clustering								
2	enMCL	True	True	-	True	.323	.119	.067
9	enMCL	True	False	-	True	.293	.238	.187
8	enMCL	True	False	-	True	.335	.143	.102
7	enMCL	True	True	-	True	.305	.321	-.019
1	enMCL	True	False	-	True	.173	.643	.060
Method: WSBM								
2	enMCL	True	True	geometric	True	.708	.714	.054
1	enMCL	True	True	binomial	True	.749	.238	.032
3	enMCL	True	True	binomial	True	.646	.750	.029
2	enMCL	True	True	binomial	True	.715	.714	.088
9	enMCL	True	True	binomial	True	.788	.357	.020
Method: Spectral Clustering/s								
6	enMCL	True	True	-	True	.675	.619	.097
6	enMCL	True	True	-	True	.754	.048	.150
6	enMCL	True	True	-	True	.664	.679	.058
8	enMCL	True	True	-	True	.570	.643	.067
6	enMCL	True	True	-	True	.652	.536	.070
Method: Spectral Clustering/c								
2	enMCL	True	True	-	True	.554	.310	.036
4	enMCL	True	True	-	True	.599	.286	.134
7	enMCL	True	True	-	True	.612	.306	.048
8	enMCL	True	True	-	True	.586	.607	.009
7	enMCL	True	True	-	True	.700	-.071	.067
Method: Spectral Clustering/e								
4	enMCL	True	True	-	True	.583	.191	.004
6	enMCL	True	True	-	True	.618	.048	.147
2	enMCL	True	True	-	True	.569	.464	.059
1	enMCL	True	True	-	False	.603	.179	.020
1	enMCL	True	True	-	True	.634	-.025	.041

Table F.1: Annotationstabelle.

example_table.ods - OpenOffice Calc					
Ganzer Bildschirm					
	A	B	C	D	E
1		Theater mit dressierten Affen	übertriebenes, usinniges Getue	andere	Kommentar
2	Da ringt selbst Elisa Kirschbaum um ihre Fassung, Meyer übrigens auch um seine, bittet ihn zu sich, der ganze Ton hier, dieses Affentheater .		x		
3	In Europa waren Affentheater , die meist als reisende Schaustellungen geführt wurden, vor allem in der zweiten Hälfte des 19. Jahrhunderts recht verbreitet.	x			
4	Aber auch diese Erzählung, die man auch aus dem Affentheater mitteilen kann, wird nichts nützen.				
5	Das Affentheater mit den Damen und Herren Stars, die einfliegen, auftreten, kassieren, abheben, macht er nicht mit.				
6	Die Frau schweigt, wortlos wiederholt sie das Affentheater von eben.				
7	Die Deutschen sollten daher dankbar sein, das "Affentheater" nicht mitmachen zu müssen.				

Bibliography

- [1] Blank, A., “Prinzipien des lexikalischen bedeutungswandels am beispiel der romanischen sprachen. tbingen”, 1997.
- [2] Online Etymology Dictionary, “cell, adj. and n.”, 2025, <https://www.etymonline.com/search?q=cell>. Accessed: May 3, 2026.
- [3] Schlechtweg, D., Human and Computational Measurement of Lexical Semantic Change. Tesis PhD, University of Stuttgart, Stuttgart, Germany, 2023.
- [4] Schlechtweg, D., McGillivray, B., Hengchen, S., Dubossarsky, H., y Tahmasebi, N., “SemEval-2020 Task 1: Unsupervised Lexical Semantic Change Detection”, in Proceedings of the 14th International Workshop on Semantic Evaluation, (Barcelona, Spain), Association for Computational Linguistics, 2020, <https://www.aclweb.org/anthology/2020.semeval-1.1/>.
- [5] Weaver, W., “Translation”, in Machine Translation of Languages (Locke, W. N. y Boothe, A. D., eds.), pp. 15–23, Cambridge, MA: MIT Press, 1949/1955. Reprinted from a memorandum written by Weaver in 1949.
- [6] Navigli, R., “Word sense disambiguation: A survey”, ACM computing surveys (CSUR), vol. 41, no. 2, pp. 1–69, 2009.
- [7] Karjus, A. y Cuskley, C., “Evolving linguistic divergence on polarizing social media”, 09 2023, [doi:10.48550/arXiv.2309.01659](https://arxiv.org/abs/2309.01659).
- [8] Kutuzov, A., Velldal, E., y Øvreliid, L., “One-to-X analogical reasoning on word embeddings: a case for diachronic armed conflict prediction from news texts”, in Proceedings of the 1st International Workshop on Computational Approaches to Historical Language Change (Tahmasebi, N., Borin, L., Jatowt, A., y Xu, Y., eds.), (Florence, Italy), pp. 196–201, Association for Computational Linguistics, 2019, [doi:10.18653/v1/W19-4724](https://arxiv.org/abs/1908.04724).
- [9] Pilehvar, M. T. y Camacho-Collados, J., “WiC: the word-in-context dataset for evaluating context-sensitive meaning representations”, in Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers) (Burstein, J., Doran, C., y Solorio, T., eds.), (Minneapolis, Minnesota), pp. 1267–1273, Association for Computational Linguistics, 2019, [doi:10.18653/v1/N19-1128](https://arxiv.org/abs/1908.04724).
- [10] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Nee-lakantan, A., Shyam, P., Sastry, G., Askell, A., *et al.*, “Language models are few-shot learners”, Advances in neural information processing systems, vol. 33, pp. 1877–1901, 2020.

- [11] Kim, Y., Chiu, Y.-I., Hanaki, K., Hegde, D., y Petrov, S., “Temporal analysis of language through neural language models”, in Proceedings of the ACL 2014 Workshop on Language Technologies and Computational Social Science (Danescu-Niculescu-Mizil, C., Eisenstein, J., McKeown, K., y Smith, N. A., eds.), (Baltimore, MD, USA), pp. 61–65, Association for Computational Linguistics, 2014, [doi:10.3115/v1/W14-2517](https://doi.org/10.3115/v1/W14-2517).
- [12] Hamilton, W. L., Leskovec, J., y Jurafsky, D., “Diachronic word embeddings reveal statistical laws of semantic change”, in Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (Erk, K. y Smith, N. A., eds.), (Berlin, Germany), pp. 1489–1501, Association for Computational Linguistics, 2016, [doi:10.18653/v1/P16-1141](https://doi.org/10.18653/v1/P16-1141).
- [13] Camacho-Collados, J. y Pilehvar, M. T., “From word to sense embeddings: A survey on vector representations of meaning”, *Journal of Artificial Intelligence Research*, vol. 63, pp. 743–788, 2018.
- [14] Laicher, S., Kurtyigit, S., Schlechtweg, D., Kuhn, J., y Schulte im Walde, S., “Explaining and improving BERT performance on lexical semantic change detection”, in Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Student Research Workshop, (Online), pp. 192–202, Association for Computational Linguistics, apr 2021, [doi:10.18653/v1/2021.eacl-srw.25](https://doi.org/10.18653/v1/2021.eacl-srw.25).
- [15] Yadav, S. y Schlechtweg, D., “XL-DUREl: Finetuning sentence transformers for ordinal word-in-context classification”, in Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (Inui, K., Sakti, S., Wang, H., Wong, D. F., Bhattacharyya, P., Banerjee, B., Ekbil, A., Chakraborty, T., y Singh, D. P., eds.), (Mumbai, India), pp. 338–351, The Asian Federation of Natural Language Processing and The Association for Computational Linguistics, 2025, <https://aclanthology.org/2025.findings-ijcnlp.19/>.
- [16] Devlin, J., Chang, M.-W., Lee, K., y Toutanova, K., “Bert: Pre-training of deep bidirectional transformers for language understanding”, in Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers), pp. 4171–4186, 2019.
- [17] Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., y Stoyanov, V., “Unsupervised cross-lingual representation learning at scale”, arXiv preprint arXiv:1911.02116, 2019.
- [18] Kutuzov, A. y Pivovarov, L., “Rushifteval: a shared task on semantic shift detection for russian”, *Komp’yuternaya Lingvistika i Intellektual’nye Tekhnologii: Dialog conference*, 2021.
- [19] Arefyev, N., Fedoseev, M., Protasov, V., Homskiy, D., Davletov, A., y Panchenko, A., “Deepmistake: Which senses are hard to distinguish for a word-in-context model”, vol. 2021-June, pp. 16–30, 2021.
- [20] Cassotti, P., Siciliani, L., DeGemmis, M., Semeraro, G., y Basile, P., “XL-LEXEME: WiC pretrained model for cross-lingual LEXical sEMantic changE”, in Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers) (Rogers, A., Boyd-Graber, J., y Okazaki, N., eds.), (Toronto, Canada), pp. 1577–1585, Association for Computational Linguistics, 2023, [doi:10.18653/v1/2023](https://doi.org/10.18653/v1/2023).

- [21] Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., *et al.*, “Language models are unsupervised multitask learners”, OpenAI blog, vol. 1, no. 8, p. 9, 2019.
- [22] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., y Amodei, D., “Language models are few-shot learners”, in Advances in Neural Information Processing Systems (Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., y Lin, H., eds.), vol. 33, pp. 1877–1901, Curran Associates, Inc., 2020, https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf.
- [23] Stone, M., “Cross-validators choice and assessment of statistical predictions”, Journal of the royal statistical society: Series B (Methodological), vol. 36, no. 2, pp. 111–133, 1974.
- [24] Loya, M., Sinha, D., y Futrell, R., “Exploring the sensitivity of llms’ decision-making capabilities: Insights from prompt variations and hyperparameters”, in Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 3711–3716, 2023.
- [25] Schlechtweg, D., Schulte im Walde, S., y Eckmann, S., “Diachronic usage relatedness (DUREL): A framework for the annotation of lexical semantic change”, in Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers) (Walker, M., Ji, H., y Stent, A., eds.), (New Orleans, Louisiana), pp. 169–174, Association for Computational Linguistics, 2018, [doi:10.18653/v1/N18-2027](https://doi.org/10.18653/v1/N18-2027).
- [26] Bolboaca, S.-D. y Jäntschi, L., “Pearson versus spearman, kendall’s tau correlation analysis on structure-activity relationships of biologic active compounds”, Leonardo Journal of Sciences, vol. 5, no. 9, pp. 179–200, 2006.
- [27] Kilgarriff, A., “I don’t believe in word senses”, Computers and the Humanities, vol. 31, pp. 91–113, 1997.
- [28] Kilgarriff, A., Word Senses, pp. 29–46. Dordrecht: Springer Netherlands, 2006, [doi:10.1007/978-1-4020-4809-8_2](https://doi.org/10.1007/978-1-4020-4809-8_2).
- [29] McCarthy, D. y Navigli, R., “The english lexical substitution task”, Language resources and evaluation, vol. 43, pp. 139–159, 2009.
- [30] Erk, K., McCarthy, D., y Gaylord, N., “Measuring word meaning in context”, Computational Linguistics, vol. 39, no. 3, pp. 511–554, 2013.
- [31] Brown, S. W., “Choosing sense distinctions for wsd: Psycholinguistic evidence”, in Proceedings of ACL-08: HLT, Short Papers, pp. 249–252, 2008.
- [32] Miller, G. A., “Wordnet: a lexical database for english”, Communications of the ACM, vol. 38, no. 11, pp. 39–41, 1995.
- [33] McCarthy, D., Apidianaki, M., y Erk, K., “Word sense clustering and clusterability”, Computational Linguistics, vol. 42, no. 2, pp. 245–275, 2016.
- [34] Schlechtweg, D., Tahmasebi, N., Hengchen, S., Dubossarsky, H., y McGillivray, B.,

- “DWUG: A large resource of diachronic word usage graphs in four languages”, in Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (Moens, M.-F., Huang, X., Specia, L., y Yih, S. W.-t., eds.), (Online and Punta Cana, Dominican Republic), pp. 7079–7091, Association for Computational Linguistics, 2021, [doi:10.18653/v1/2021.emnlp-main.567](https://doi.org/10.18653/v1/2021.emnlp-main.567).
- [35] Langone, H., Haskell, B. R., y Miller, G. A., “Annotating wordnet”, in Proceedings of the Workshop Frontiers in Corpus Annotation at HLT-NAACL, (Boston, MA, USA), 2004.
- [36] Hovy, E., Marcus, M., Palmer, M., Ramshaw, L., y Weischedel, R., “Ontonotes: The 90% solution”, in Proceedings of the Human Language Technology Conference of the NAACL, Companion Volume: Short Papers, NAACL-Short ’06, (USA), pp. 57—60, Association for Computational Linguistics, 2006.
- [37] Schütze, H., “Automatic word sense discrimination”, Computational linguistics, vol. 24, no. 1, pp. 97–123, 1998.
- [38] Tuggy, D., “Ambiguity, polysemy, and vagueness”, Cognitive Linguistics, vol. 4, no. 3, pp. 273–290, 1993.
- [39] Cruse, D. A., Polysemy and related phenomena from a cognitive linguistic viewpoint, cap. 2, pp. 33—49. Studies in Natural Language Processing, Cambridge: Cambridge University Press, 1995, [doi:10.1017/CBO9780511527227.004](https://doi.org/10.1017/CBO9780511527227.004).
- [40] Hanks, P., “Do word meanings exist?”, Computers and the Humanities, vol. 34, no. 1/2, pp. 205–215, 2000.
- [41] Palmer, M., Dang, H., y Fellbaum, C., “Making fine-grained and coarse-grained sense distinctions, both manually and automatically”, Natural Language Engineering, vol. 13, pp. 137–163, 06 2007, [doi:10.1017/S135132490500402X](https://doi.org/10.1017/S135132490500402X).
- [42] McCarthy, D. y Navigli, R., “SemEval-2007 task 10: English lexical substitution task”, in Proceedings of the Fourth International Workshop on Semantic Evaluations (SemEval-2007), (Prague, Czech Republic), pp. 48–53, Association for Computational Linguistics, 2007, <https://aclanthology.org/S07-1009>.
- [43] Schlechtweg, D., Castaneda, E., Kuhn, J., y Schulte im Walde, S., “Modeling sense structure in word usage graphs with the weighted stochastic block model”, in Proceedings of *SEM 2021: The Tenth Joint Conference on Lexical and Computational Semantics, (Online), pp. 241–251, Association for Computational Linguistics, aug 2021, [doi:10.18653/v1/2021.starsem-1.23](https://doi.org/10.18653/v1/2021.starsem-1.23).
- [44] Tahmasebi, N. y Risse, T., “Finding individual word sense changes and their delay in appearance”, in Proceedings of the International Conference Recent Advances in Natural Language Processing, (Varna, Bulgaria), pp. 741–749, 2017.
- [45] Kutuzov, A., Øvrelid, L., Szymanski, T., y Velldal, E., “Diachronic word embeddings and semantic shifts: a survey”, in Proceedings of the 27th International Conference on Computational Linguistics (Bender, E. M., Derczynski, L., y Isabelle, P., eds.), (Santa Fe, New Mexico, USA), pp. 1384–1397, Association for Computational Linguistics, 2018, <https://aclanthology.org/C18-1117/>.
- [46] Tahmasebi, N. y Dubossarsky, H., “Computational modeling of semantic change”, 2023.
- [47] Cook, P., Lau, J. H., McCarthy, D., y Baldwin, T., “Novel word-sense identification”,

in 25th International Conference on Computational Linguistics, Proceedings of the Conference: Technical Papers, (Dublin, Ireland), pp. 1624–1635, 2014.

- [48] Giulianelli, M., del Tredici, M., y Fernández, R., “Analysing lexical semantic change with contextualised word representations”, in Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, (Online), pp. 3960–3973, Association for Computational Linguistics, 2020.
- [49] Rodina, J. y Kutuzov, A., “RuSemShift: a dataset of historical lexical semantic change in Russian”, in Proceedings of the 28th International Conference on Computational Linguistics (Scott, D., Bel, N., y Zong, C., eds.), (Barcelona, Spain (Online)), pp. 1037–1047, International Committee on Computational Linguistics, 2020, [doi:10.18653/v1/2020.coling-main.90](https://doi.org/10.18653/v1/2020.coling-main.90).
- [50] Zamora-Reina, F. D., Bravo-Marquez, F., y Schlechtweg, D., “LSCDiscovery: A shared task on semantic change discovery and detection in Spanish”, in Proceedings of the 3rd International Workshop on Computational Approaches to Historical Language Change, (Dublin, Ireland), Association for Computational Linguistics, 2022, <https://aclanthology.org/2022.lchange-1.16/>.
- [51] Kutuzov, A., Touileb, S., Mæhlum, P., Enstad, T., y Wittemann, A., “NorDiaChange: Diachronic semantic change dataset for Norwegian”, in Proceedings of the Thirteenth Language Resources and Evaluation Conference (Calzolari, N., Béchet, F., Blache, P., Choukri, K., Cieri, C., Declerck, T., Goggi, S., Isahara, H., Maegaard, B., Mariani, J., Mazo, H., Odijk, J., y Piperidis, S., eds.), (Marseille, France), pp. 2563–2572, European Language Resources Association, 2022, <https://aclanthology.org/2022.lrec-1.274/>.
- [52] Chen, J., Chersoni, E., Schlechtweg, D., Prokic, J., y Huang, C.-R., “ChiWUG: A graph-based evaluation dataset for Chinese lexical semantic change detection”, in Proceedings of the 4th International Workshop on Computational Approaches to Historical Language Change, (Singapore), Association for Computational Linguistics, 2023, <https://aclanthology.org/2023.lchange-1.10/>.
- [53] Perrone, V., Palma, M., Hengchen, S., Vatri, A., Smith, J. Q., y McGillivray, B., “GASC: Genre-aware semantic change for ancient Greek”, in Proceedings of the 1st International Workshop on Computational Approaches to Historical Language Change, (Florence, Italy), pp. 56–66, Association for Computational Linguistics, 2019.
- [54] Basile, P., Caputo, A., Caselli, T., Cassotti, P., y Varvara, R., “Diacr-ita@ evalita2020: Overview of the evalita2020 diachronic lexical semantics (diacr-ita) task”, Evaluation Campaign of Natural Language Processing and Speech Tools for Italian, 2020.
- [55] McGillivray, B., “Latin lexical semantic annotation”, 2021, <http://hdl.handle.net/20.500.14106/2568>. Literary and Linguistic Data Service.
- [56] Asahara, M., Ikegami, N., Suzuki, T., Ichimura, T., Kondo, A., Kato, S., y Yamazaki, M., “CHJ-WLSP: Annotation of ‘word list by semantic principles’ labels for the corpus of historical Japanese”, in Proceedings of the Second Workshop on Language Technologies for Historical and Ancient Languages, (Marseille, France), pp. 31–37, European Language Resources Association, 2022, <https://aclanthology.org/2022.lt4hala-1.5>.
- [57] Paul, H., Deutsches Wörterbuch: Bedeutungsgeschichte und Aufbau unseres Wortschatzes. Tübingen: Niemeyer, 10. ed., 2002.

- [58] DWDS, “Digitales Wörterbuch der deutschen Sprache. Das Wortauskunftssystem zur deutschen Sprache in Geschichte und Gegenwart, hrsg. v. d. Berlin-Brandenburgischen Akademie der Wissenschaften”. <https://www.dwds.de/>, 2021. Accessed: 02.02.2021.
- [59] OED, Oxford English Dictionary. Online: Oxford University Press, 2009.
- [60] Manjavacas Arevalo, E. y Fonteyn, L., “Non-parametric word sense disambiguation for historical languages”, in Proceedings of the 2nd International Workshop on Natural Language Processing for Digital Humanities, (Taipei, Taiwan), pp. 123–134, Association for Computational Linguistics, 2022, <https://aclanthology.org/2022.nlp4dh-1.16>.
- [61] Jurgens, D. y Klapaftis, I., “Semeval-2013 task 13: Word sense induction for graded and non-graded senses”, in Second Joint Conference on Lexical and Computational Semantics (* SEM), Volume 2: Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013), pp. 290–299, 2013.
- [62] Navigli, R., “A quick tour of word sense disambiguation, induction and related approaches”, in SOFSEM 2012: Theory and Practice of Computer Science: 38th Conference on Current Trends in Theory and Practice of Computer Science, Špindlerův Mlýn, Czech Republic, January 21–27, 2012. Proceedings 38, pp. 115–129, Springer, 2012.
- [63] Lin, J., “Divergence measures based on the shannon entropy”, IEEE Transactions on Information theory, vol. 37, no. 1, pp. 145–151, 1991.
- [64] Donoso, G. y Sanchez, D., “Dialectometric analysis of language variation in twitter”, in Proceedings of the Fourth Workshop on NLP for Similar Languages, Varieties and Dialects, (Valencia, Spain), pp. 16–25, 2017.
- [65] Schlechtweg, D., Virk, S. M., y Arefyev, N., “The lscd benchmark: a testbed for diachronic word meaning tasks”, 2024.
- [66] Burtseva, T., Levashov, E., y Denisenko, Y., Novye slova i znacheniya. Slovar’-spravochnik po materialam pressy i literatury 90-h godov XX veka: V 2 tomah [New words and meanings. Reference dictionary by the materials of the press and literature in the 90’s of the XX century: in 2 volumes]. Institute for Linguistic Studies of the Russian Academy of Sciences.
- [67] Davies, M., “Expanding horizons in historical linguistics with the 400-million word corpus of historical american english”, Corpora, vol. 7, no. 2, pp. 121–157, 2012.
- [68] Textarchiv, D., “Grundlage für ein referenzkorpus der neuhochdeutschen sprache”, Herausgegeben von der Berlin-Brandenburgischen Akademie der Wissenschaften, 2017.
- [69] Språkbanken Text, “The Kubhist corpus, v2”. Department of Swedish, Multilingualism, Language Technology, University of Gothenburg, 2019, <https://spraakbanken.gu.se/en/resources/kubhist2>. Accessed: 2019.
- [70] McGillivray, B. y Kilgariff, A., “Tools for historical corpus research, and a corpus of latin”, New methods in historical corpus linguistics, vol. 1, no. 3, pp. 247–257, 2013.
- [71] Chen, J., Chersoni, E., Schlechtweg, D., Prokic, J., y Huang, C.-R., “ChiWUG: A graph-based evaluation dataset for Chinese lexical semantic change detection”, in Proceedings of the 4th Workshop on Computational Approaches to Historical Language Change (Tahmasebi, N., Montariol, S., Dubossarsky, H., Kutuzov, A., Hengchen, S., Alfter, D., Periti, F., y Cassotti, P., eds.), (Singapore), pp. 93–99, Association for Computational Linguistics, 2023, [doi:10.18653/v1/2023.lchange-1.10](https://doi.org/10.18653/v1/2023.lchange-1.10).

- [72] Artstein, R. y Poesio, M., “Inter-coder agreement for computational linguistics”, *Comput. Linguist.*, vol. 34, pp. 555—596, dec 2008, [doi:10.1162/coli.07-034-R2](https://doi.org/10.1162/coli.07-034-R2).
- [73] Harris, Z. S., “Distributional structure”, *Word*, vol. 10, no. 2-3, pp. 146–162, 1954.
- [74] Turney, P. D. y Pantel, P., “From frequency to meaning: Vector space models of semantics”, *Journal of artificial intelligence research*, vol. 37, pp. 141–188, 2010.
- [75] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., y Polosukhin, I., “Attention is all you need”, *Advances in neural information processing systems*, vol. 30, 2017.
- [76] Peters, M. E., Ammar, W., Bhagavatula, C., y Power, R., “Semi-supervised sequence tagging with bidirectional language models”, *arXiv preprint arXiv:1705.00108*, 2017.
- [77] Hu, Y., Talmor, A., Herzig, J., Berant, J., Choi, E., y Liang, P., “Diachronic contextualized word embeddings from a transformer model”, in *Proceedings of EMNLP*, 2019.
- [78] MacQueen, J., “Some methods for classification and analysis of multivariate observations”, in *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, pp. 281–297, Oakland, CA, USA, 1967.
- [79] Jain, A. K. y Dubes, R. C., *Algorithms for clustering data*. Prentice-Hall, Inc., 1988.
- [80] Peixoto, T. P., “Bayesian stochastic blockmodeling”, in *Advances in Network Clustering and Blockmodeling*, cap. 11, pp. 289–332, John Wiley & Sons, Ltd, 2019, [doi:10.1002/9781119483298.ch11](https://doi.org/10.1002/9781119483298.ch11).
- [81] Dempster, A. P., Laird, N. M., y Rubin, D. B., “Maximum likelihood from incomplete data via the EM algorithm”, *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 39, no. 1, pp. 1–22, 1977.
- [82] von Luxburg, U., “A tutorial on spectral clustering”, 2007, [doi:10.48550/ARXIV.0711.0189](https://doi.org/10.48550/ARXIV.0711.0189).
- [83] Biemann, C., “Chinese whispers: An efficient graph clustering algorithm and its application to natural language processing problems”, in *Proceedings of the First Workshop on Graph Based Methods for Natural Language Processing, TextGraphs-1*, (USA), p. 73–80, Association for Computational Linguistics, 2006.
- [84] Kutuzov, A. y Giulianelli, M., “UiO-UvA at SemEval-2020 Task 1: Contextualised Embeddings for Lexical Semantic Change Detection”, in *Proceedings of the 14th International Workshop on Semantic Evaluation*, (Barcelona, Spain), Association for Computational Linguistics, 2020.
- [85] Kurtyigit, S., Park, M., Schlechtweg, D., Kuhn, J., y Schulte im Walde, S., “Lexical semantic change discovery”, in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, (Online), Association for Computational Linguistics, aug 2021, <https://aclanthology.org/2021.acl-long.543/>.
- [86] Schlechtweg, D. y Schulte im Walde, S., “Simulating Lexical Semantic Change from Sense-Annotated Data”, in *The Evolution of Language: Proceedings of the 13th International Conference (EvoLang13)* (Ravignani, A., Barbieri, C., Martins, M., Flaherty, M., Jadoul, Y., Lattenkamp, E., Little, H., Mudd, K., y Verhoef, T., eds.), 2020,

[doi:10.17617/2.3190925](https://doi.org/10.17617/2.3190925).

- [87] Tahmasebi, N., Borin, L., y Jatowt, A., “Survey of computational approaches to lexical semantic change detection”, *Computational approaches to semantic change*, vol. 6, no. 1, 2021.
- [88] Hengchen, S., Tahmasebi, N., Schlechtweg, D., y Dubossarsky, H., “Challenges for Computational Lexical Semantic Change”, in *Computational Approaches to Semantic Change* (Tahmasebi, N., Borin, L., Jatowt, A., Xu, Y., y Hengchen, S., eds.), vol. *Language Variation*, cap. 11, Berlin: Language Science Press, 2021, <https://arxiv.org/abs/2101.07668v1>.
- [89] Ahmad, A., Desta, K., Lang, F., y Schlechtweg, D., “Shared task: Lexical semantic change detection in german”, *CoRR*, vol. *abs/2001.07786*, 2020.
- [90] Tiedemann, J., “Parallel data, tools and interfaces in opus”, in *Proceedings of the Eight International Conference on Language Resources and Evaluation (LREC’12)* (Chair, N. C. C., Choukri, K., Declerck, T., Dogan, M. U., Maegaard, B., Mariani, J., Odijk, J., y Piperidis, S., eds.), (Istanbul, Turkey), European Language Resources Association (ELRA), may 2012.
- [91] Honnibal, M., Montani, I., Van Landeghem, S., y Boyd, A., “spaCy: Industrial-strength Natural Language Processing in Python”, 2020, [doi:10.5281/zenodo.1212303](https://doi.org/10.5281/zenodo.1212303).
- [92] Bansal, N., Blum, A., y Chawla, S., “Correlation clustering”, *Machine Learning*, vol. 56, no. 1-3, pp. 89–113, 2004, [doi:10.1023/B:MACH.0000033116.57574.95](https://doi.org/10.1023/B:MACH.0000033116.57574.95).
- [93] Baldissin, G., Schlechtweg, D., y Schulte im Walde, S., “DiaWUG: A Dataset for Diatopic Lexical Semantic Variation in Spanish”, in *Proceedings of the 13th Language Resources and Evaluation Conference*, (Marseille, France), European Language Resources Association, 2022.
- [94] Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., y Dean, J., “Distributed representations of words and phrases and their compositionality”, *Advances in neural information processing systems*, vol. 26, 2013.
- [95] Mikolov, T., Chen, K., Corrado, G., y Dean, J., “Efficient estimation of word representations in vector space”, *arXiv preprint arXiv:1301.3781*, 2013.
- [96] Kaiser, J., Schlechtweg, D., Papay, S., y Schulte im Walde, S., “IMS at SemEval-2020 Task 1: How low can you go? Dimensionality in Lexical Semantic Change Detection”, in *Proceedings of the 14th International Workshop on Semantic Evaluation*, (Barcelona, Spain), Association for Computational Linguistics, 2020, <https://aclanthology.org/2020.semeval-1.8/>.
- [97] Kaiser, J., Schlechtweg, D., y Schulte im Walde, S., “OP-IMS @ DIACR-Ita: Back to the Roots: SGNS+OP+CD still rocks Semantic Change Detection”, in *Proceedings of the 7th evaluation campaign of Natural Language Processing and Speech tools for Italian (EVALITA 2020)* (Basile, V., Croce, D., Di Maro, M., y Passaro, L. C., eds.), (Online), CEUR.org, 2020, <https://arxiv.org/abs/2011.03258>. Winning Submission!
- [98] Schlechtweg, D., Hättty, A., del Tredici, M., y Schulte im Walde, S., “A Wind of Change: Detecting and Evaluating Lexical Semantic Change across Times and Domains”, in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, (Florence, Italy), pp. 732–746, Association for Computational Linguistics, 2019,

- [99] Pömsl, M. y Lyapin, R., “CIRCE at SemEval-2020 Task 1: Ensembling Context-Free and Context-Dependent Word Representations”, in Proceedings of the 14th International Workshop on Semantic Evaluation, (Barcelona, Spain), Association for Computational Linguistics, 2020.
- [100] Pražák, O., Příbákň, P., Taylor, S., y Sido, J., “UWB at SemEval-2020 Task 1: Lexical Semantic Change Detection”, in Proceedings of the 14th International Workshop on Semantic Evaluation, (Barcelona, Spain), Association for Computational Linguistics, 2020.
- [101] Kutuzov, A., Pivovarova, L., y Giulianelli, M., “Grammatical profiling for semantic change detection”, in Proceedings of the 25th Conference on Computational Natural Language Learning, (Online), pp. 423–434, Association for Computational Linguistics, 2021, [doi:10.18653/v1/2021.conll-1.33](https://doi.org/10.18653/v1/2021.conll-1.33).
- [102] Kashleva, K., Shein, A., Tukhtina, E., y Vydrina, S., “Hse at lscdiscovery in spanish: Clustering and profiling for lexical semantic change discovery”, in Proceedings of the 3rd International Workshop on Computational Approaches to Historical Language Change, (Dublin, Ireland), Association for Computational Linguistics, 2022.
- [103] Straka, M. y Straková, J., “Tokenizing, POS tagging, lemmatizing and parsing UD 2.0 with UDPipe”, in Proceedings of the CoNLL 2017 Shared Task: Multilingual Parsing from Raw Text to Universal Dependencies, (Vancouver, Canada), pp. 88–99, Association for Computational Linguistics, 2017, [doi:10.18653/v1/K17-3009](https://doi.org/10.18653/v1/K17-3009).
- [104] Rachinskiy, M. y Arefyev, N., “Glossreader at lscdiscovery: Train to select a proper gloss in english – discover lexical semantic change in spanish”, in Proceedings of the 3rd International Workshop on Computational Approaches to Historical Language Change, (Dublin, Ireland), Association for Computational Linguistics, 2022.
- [105] Teodorescu, D., McIntosh von der Ohe, S., y Kondrak, G., “Ualberta at lscdiscovery: Lexical semantic change detection via word sense disambiguation”, in Proceedings of the 3rd International Workshop on Computational Approaches to Historical Language Change, (Dublin, Ireland), Association for Computational Linguistics, 2022.
- [106] Orlando, R., Conia, S., Brignone, F., Cecconi, F., y Navigli, R., “AMuSE-WSD: An all-in-one multilingual system for easy Word Sense Disambiguation”, in Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, (Online and Punta Cana, Dominican Republic), pp. 298–307, Association for Computational Linguistics, 2021, [doi:10.18653/v1/2021.emnlp-demo.34](https://doi.org/10.18653/v1/2021.emnlp-demo.34).
- [107] Sabina Uban, A., Maria Cristea, A., Daniela Dinu, A., Georgescu, S., y Zoicas, L., “Cotohili at lscdiscovery: the role of linguistic features in predicting semantic change”, in Proceedings of the 3rd International Workshop on Computational Approaches to Historical Language Change, (Dublin, Ireland), Association for Computational Linguistics, 2022.
- [108] Artetxe, M., Labaka, G., y Agirre, E., “Learning principled bilingual mappings of word embeddings while preserving monolingual invariance”, in Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing, (Austin, Texas), pp. 2289–2294, Association for Computational Linguistics, 2016.

- [109] Artetxe, M., Labaka, G., y Agirre, E., “Learning bilingual word embeddings with (almost) no bilingual data”, in Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics, pp. 451–462, 2017.
- [110] Artetxe, M., Labaka, G., y Agirre, E., “Generalizing and improving bilingual word embedding mappings with a multi-step framework of linear transformations”, in Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, pp. 5012–5019, February 2018.
- [111] Artetxe, M., Labaka, G., Lopez-Gazpio, I., y Agirre, E., “Uncovering divergent linguistic information in word embeddings with lessons for intrinsic and extrinsic evaluation”, in Proceedings of the 22nd Conference on Computational Natural Language Learning, (Brussels, Belgium), pp. 282–291, Association for Computational Linguistics, 2018.
- [112] Homskiy, D. y Arefyev, N., “Deepmistake at lscdiscovery: Can a multilingual word-in-context model replace human annotators?”, in Proceedings of the 3rd International Workshop on Computational Approaches to Historical Language Change, (Dublin, Ireland), Association for Computational Linguistics, 2022.
- [113] Martelli, F., Kalach, N., Tola, G., y Navigli, R., “SemEval-2021 task 2: Multilingual and cross-lingual word-in-context disambiguation (MCL-WiC)”, in Proceedings of the 15th International Workshop on Semantic Evaluation (SemEval-2021), (Online), pp. 24–36, Association for Computational Linguistics, 2021, [doi:10.18653/v1/2021.semeval-1.3](https://doi.org/10.18653/v1/2021.semeval-1.3).
- [114] Pasini, T., Raganato, A., Navigli, R., *et al.*, “Xl-wsd: An extra-large and cross-lingual evaluation framework for word sense disambiguation”, in Proceedings of the AAAI Conference on Artificial Intelligence, AAAI Press, 2021.
- [115] Kudisov, A. y Arefyev, N., “Bos at lscdiscovery: Lexical substitution for interpretable lexical semantic change detection”, in Proceedings of the 3rd International Workshop on Computational Approaches to Historical Language Change, (Dublin, Ireland), Association for Computational Linguistics, 2022.
- [116] Arefyev, N. y Bykov, D., “An interpretable approach to lexical semantic change detection with lexical substitution”, vol. 2021-June, pp. 31–46, ABBYY PRODUCTION LLC, 2021, [doi:10.28995/2075-7182-2021-20-31-46](https://doi.org/10.28995/2075-7182-2021-20-31-46).
- [117] Davletov, A., Arefyev, N., Gordeev, D., y Rey, A., “LIORI at SemEval-2021 task 2: Span prediction and binary classification approaches to word-in-context disambiguation”, in Proceedings of the 15th International Workshop on Semantic Evaluation (SemEval-2021), (Online), pp. 780–786, Association for Computational Linguistics, 2021, [doi:10.18653/v1/2021.semeval-1.103](https://doi.org/10.18653/v1/2021.semeval-1.103).
- [118] Hubert, L. y Arabie, P., “Comparing partitions”, Journal of Classification, vol. 2, pp. 193–218, 1985.
- [119] Spearman, C., “The proof and measurement of association between two things”, American Journal of Psychology, vol. 15, pp. 88–103, 1904.
- [120] Ustalov, D., Panchenko, A., Biemann, C., y Ponzetto, S. P., “Watset: Local-Global Graph Clustering with Applications in Sense and Frame Induction”, Computational Linguistics, vol. 45, no. 3, pp. 423–479, 2019, [doi:10.1162/COLI_a_00354](https://doi.org/10.1162/COLI_a_00354).
- [121] Aksenova, A., Gavrishina, E., Rykov, E., y Kutuzov, A., “RuDSI: Graph-based word sense induction dataset for Russian”, in Proceedings of TextGraphs-16: Graph-based

Methods for Natural Language Processing (Ustalov, D., Gao, Y., Panchenko, A., Valentino, M., Thayaparan, M., Nguyen, T. H., Penn, G., Ramesh, A., y Jana, A., eds.), (Gyeongju, Republic of Korea), pp. 77–88, Association for Computational Linguistics, 2022, <https://aclanthology.org/2022.textgraphs-1.9>.

- [122] Aicher, C., Jacobs, A. Z., y Clauset, A., “Learning latent block structure in weighted networks”, *Journal of Complex Networks*, vol. 3, no. 2, pp. 221–248, 2015.
- [123] Peixoto, T. P., “Nonparametric weighted stochastic block models”, *Physical Review E*, vol. 97, 08 2017, [doi:10.1103/PhysRevE.97.012306](https://doi.org/10.1103/PhysRevE.97.012306).
- [124] Kotchourko, S., “Optimizing human annotation of word usage graphs in a realistic simulation environment”, 2021. Bachelor thesis.
- [125] Tunc, B., “Optimierung von clustering von wortverwendungsgraphen”, 2021, <https://elib.uni-stuttgart.de/handle/11682/11923>.
- [126] Rousseeuw, P. J., “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis”, *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987, [doi:https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7).
- [127] Caliński, T. y Harabasz, J., “A dendrite method for cluster analysis”, *Communications in Statistics*, vol. 3, no. 1, pp. 1–27, 1974, [doi:10.1080/03610927408827101](https://doi.org/10.1080/03610927408827101).
- [128] Blei, D. M. y Lafferty, J. D., “Dynamic topic models”, in *Proceedings of the 23rd international conference on Machine learning*, pp. 113–120, 2006.
- [129] Tsitsulin, A., Palowitch, J., Perozzi, B., y Müller, E., “Graph clustering with graph neural networks”, *Journal of Machine Learning Research*, vol. 24, no. 127, pp. 1–21, 2023.
- [130] Fortunato, S. y Barthelemy, M., “Resolution limit in community detection”, *Proceedings of the national academy of sciences*, vol. 104, no. 1, pp. 36–41, 2007.
- [131] Cawley, G. C. y Talbot, N. L., “On over-fitting in model selection and subsequent selection bias in performance evaluation”, *The Journal of Machine Learning Research*, vol. 11, pp. 2079–2107, 2010.
- [132] Kutuzov, A. y Pivovarova, L., “Three-part diachronic semantic change dataset for russian”, 2021.
- [133] Tang, X., “A state-of-the-art of semantic change computation”, *Natural Language Engineering*, vol. 24, 01 2018, [doi:10.1017/S1351324918000220](https://doi.org/10.1017/S1351324918000220).
- [134] Periti, F. y Montanelli, S., “Lexical semantic change through large language models: a survey”, *ACM Comput. Surv.*, vol. 56, no. 11, 2024, [doi:10.1145/3672393](https://doi.org/10.1145/3672393).
- [135] Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., *et al.*, “A survey of large language models”, *arXiv preprint arXiv:2303.18223*, 2023.
- [136] Minaee, S., Mikolov, T., Nikzad, N., Chenaghlu, M., Socher, R., Amatriain, X., y Gao, J., “Large language models: A survey”, *arXiv preprint arXiv:2402.06196*, 2024.
- [137] Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., y Neubig, G., “Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing”, *ACM computing surveys*, vol. 55, no. 9, pp. 1–35, 2023.

- [138] Zhu, K., Wang, J., Zhou, J., Wang, Z., Chen, H., Wang, Y., Yang, L., Ye, W., Zhang, Y., Zhenqiang Gong, N., *et al.*, “Promptbench: Towards evaluating the robustness of large language models on adversarial prompts”, arXiv e-prints, pp. arXiv–2306, 2023.
- [139] Zhou, Y., Muresanu, A. I., Han, Z., Paster, K., Pitis, S., Chan, H., y Ba, J., “Large language models are human-level prompt engineers”, in The Eleventh International Conference on Learning Representations, 2022.
- [140] Guo, Q., Wang, R., Junliang, G., Li, B., Song, K., Tan, X., Liu, G., Bian, J., y Yang, Y., “Connecting large language models with evolutionary algorithms yields powerful prompt optimizers”, 09 2023, [doi:10.48550/arXiv.2309.08532](https://doi.org/10.48550/arXiv.2309.08532).
- [141] D’Oosterlinck, K., Khattab, O., Remy, F., Demeester, T., Develder, C., y Potts, C., “In-context learning for extreme multi-label classification”, arXiv preprint arXiv:2401.12178, 2024.
- [142] Wang, R. y Choi, M., “Large language models on lexical semantic change detection: An evaluation”, arXiv preprint arXiv:2312.06002, 2023.
- [143] Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., *et al.*, “Gpt-4 technical report”, arXiv preprint arXiv:2303.08774, 2023.
- [144] Loureiro, D., D’Souza, A., Muhajab, A. N., White, I. A., Wong, G., Anke, L. E., Neves, L., Barbieri, F., y Camacho-Collados, J., “Tempowic: An evaluation benchmark for detecting meaning shift in social media”, arXiv preprint arXiv:2209.07216, 2022.
- [145] Periti, F., Dubossarsky, H., y Tahmasebi, N., “(chat) gpt v bert dawn of justice for semantic change detection”, in Findings of the Association for Computational Linguistics: EACL 2024, pp. 420–436, 2024.
- [146] Yadav, S., Chopra, T., y Schlechtweg, D., “Towards automating text annotation: A case study on semantic proximity annotation using gpt-4”, arXiv preprint arXiv:2407.04130, 2024.
- [147] Krippendorff, K., Content analysis: An introduction to its methodology. Sage publications, 2018.
- [148] Periti, F. y Tahmasebi, N., “A systematic comparison of contextualized word embeddings for lexical semantic change”, in Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers) (Duh, K., Gomez, H., y Bethard, S., eds.), (Mexico City, Mexico), pp. 4262–4282, Association for Computational Linguistics, 2024, [doi:10.18653/v1/2024.naacl-long.240](https://doi.org/10.18653/v1/2024.naacl-long.240).
- [149] Martelli, F., Kalach, N., Tola, G., Navigli, R., *et al.*, “Semeval-2021 task 2: Multilingual and cross-lingual word-in-context disambiguation (mcl-wic)”, in Proceedings of the 15th International Workshop on Semantic Evaluation (SemEval-2021), pp. 24–36, 2021.
- [150] Raganato, A., Pasini, T., Camacho-Collados, J., y Pilehvar, M. T., “Xl-wic: A multilingual benchmark for evaluating semantic contextualization”, in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 7193–7206, 2020.
- [151] Schlechtweg, D., Zamora-Reina, F. D., Bravo-Marquez, F., y Arefyev, N., “Sense through time: Diachronic word sense annotations for word sense induction and lex-

ical semantic change detection”, Language Resources and Evaluation, 2024.

- [152] Jiang, A. Q., Sablayrolles, A., Roux, A., Mensch, A., Savary, B., Bamford, C., Chaplot, D. S., Casas, D. d. l., Hanna, E. B., Bressand, F., *et al.*, “Mixtral of experts”, arXiv preprint arXiv:2401.04088, 2024.
- [153] Jacob, B., Kligys, S., Chen, B., Zhu, M., Tang, M., Howard, A., Adam, H., y Kalenichenko, D., “Quantization and training of neural networks for efficient integer-arithmetic-only inference”, in Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 2704–2713, 2018.
- [154] von Luxburg, U., “A tutorial on spectral clustering”, 2007, <https://arxiv.org/abs/0711.0189>.
- [155] Rousseeuw, P. J., “Silhouettes: a graphical aid to the interpretation and validation of cluster analysis”, Journal of computational and applied mathematics, vol. 20, pp. 53–65, 1987.
- [156] Khattab, O., Singhvi, A., Maheshwari, P., Zhang, Z., Santhanam, K., Vardhamanan, S., Haq, S., Sharma, A., Joshi, T. T., Moazam, H., *et al.*, “Dspy: Compiling declarative language model calls into self-improving pipelines”, arXiv preprint arXiv:2310.03714, 2023.
- [157] Opsahl-Ong, K., Ryan, M. J., Purtell, J., Broman, D., Potts, C., Zaharia, M., y Khattab, O., “Optimizing instructions and demonstrations for multi-stage language model programs”, in Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (Al-Onaizan, Y., Bansal, M., y Chen, Y.-N., eds.), (Miami, Florida, USA), pp. 9340–9366, Association for Computational Linguistics, 2024, [doi:10.18653/v1/2024.emnlp-main.525](https://doi.org/10.18653/v1/2024.emnlp-main.525).
- [158] Giulianelli, M., Luden, I., Fernandez, R., y Kutuzov, A., “Interpretable word sense representations via definition generation: The case of semantic change analysis”, in Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (Rogers, A., Boyd-Graber, J., y Okazaki, N., eds.), (Toronto, Canada), pp. 3130–3148, Association for Computational Linguistics, 2023, [doi:10.18653/v1/2023.acl-long.176](https://doi.org/10.18653/v1/2023.acl-long.176).
- [159] Fedorova, M., Kutuzov, A., y Scherrer, Y., “Definition generation for lexical semantic change detection”, in Findings of the Association for Computational Linguistics: ACL 2024 (Ku, L.-W., Martins, A., y Srikumar, V., eds.), (Bangkok, Thailand), pp. 5712–5724, Association for Computational Linguistics, 2024, [doi:10.18653/v1/2024.findings-acl.339](https://doi.org/10.18653/v1/2024.findings-acl.339).
- [160] Fedorova, M., Mickus, T., Partanen, N., Siewert, J., Spaziani, E., y Kutuzov, A., “AXOLOTL’24 shared task on multilingual explainable semantic change modeling”, in Proceedings of the 5th Workshop on Computational Approaches to Historical Language Change (Tahmasebi, N., Montariol, S., Kutuzov, A., Alfter, D., Periti, F., Cassotti, P., y Huebscher, N., eds.), (Bangkok, Thailand), pp. 72–91, Association for Computational Linguistics, 2024, [doi:10.18653/v1/2024.lchange-1.8](https://doi.org/10.18653/v1/2024.lchange-1.8).
- [161] Fedorova, M., Kutuzov, A., Periti, F., y Scherrer, Y., “Explaining novel senses using definition generation with open language models”, in Findings of the Association for Computational Linguistics: EMNLP 2025 (Christodoulopoulos, C., Chakraborty, T., Rose, C., y Peng, V., eds.), (Suzhou, China), pp. 22294–22302, Association for

- [162] Levy, O. y Goldberg, Y., “Neural word embedding as implicit matrix factorization”, *Advances in neural information processing systems*, vol. 27, 2014.
- [163] Levy, O., Goldberg, Y., y Dagan, I., “Improving distributional similarity with lessons learned from word embeddings”, *Transactions of the association for computational linguistics*, vol. 3, pp. 211–225, 2015.
- [164] Stewart, G. W., “On the early history of the singular value decomposition”, *SIAM review*, vol. 35, no. 4, pp. 551–566, 1993.
- [165] Wall, M. E., Rechtsteiner, A., y Rocha, L. M., “Singular value decomposition and principal component analysis”, in *A practical approach to microarray data analysis*, pp. 91–109, Springer, 2003.
- [166] Kanerva, P., Kristoferson, J., y Holst, A., “Random indexing of text samples for latent semantic analysis”, in *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 22, 2000.
- [167] Basile, P., Caputo, A., y Semeraro, G., “Analysing word meaning over time by exploiting temporal random indexing”, in *Proceedings of the First Italian Conference on Computational Linguistics CLiC-it 2014 & and of the Fourth International Workshop EVALITA 2014: 9-11 December 2014, Pisa*, pp. 38–42, Pisa University Press, 2014.
- [168] Dubossarsky, H., Hengchen, S., Tahmasebi, N., y Schlechtweg, D., “Time-out: Temporal referencing for robust modeling of lexical semantic change”, in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (Korhonen, A., Traum, D., y Màrquez, L., eds.)*, (Florence, Italy), pp. 457–470, Association for Computational Linguistics, 2019, [doi:10.18653/v1/P19-1044](https://doi.org/10.18653/v1/P19-1044).
- [169] Amrami, A. y Goldberg, Y., “Towards better substitution-based word sense induction”, *arXiv preprint arXiv:1905.12598*, 2019.
- [170] Ansell, A., Bravo-Marquez, F., y Pfahringer, B., “Polym: Learning about polysemy through language modeling”, *arXiv preprint arXiv:2101.10448*, 2021.